Testing Conditional Moment Restrictions: A Partitioning Approach

Antonio Raiola*
Aarhus BSS

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Abstract

This paper proposes χ^2 tests for assessing the specification of regression models or general conditional moment restrictions. The data is partitioned according to the explanatory variables into several cells, and the tests evaluate whether the difference between the observed average of the dependent variable and its expected value under the model specification arises by chance. In contrast to existing omnibus procedures, χ^2 tests are asymptotically pivotal and fairly insensitive to the curse of dimensionality. The computation is straightforward and does not require bootstrapping or smoothing techniques. Importantly, the asymptotic properties of the test are invariant to sample-dependent partitions, which can be chosen to favor certain alternatives. A Monte Carlo study provides evidence of the good performance of the tests using samples of small or moderate size compared with existing omnibus alternatives, particularly with many explanatory variables. Empirical illustrations in difference-in-difference settings with continuous treatments and in static binary treatment designs with numerous covariates complement the finite sample study.

JEL classification: C12, C31, C52

Keywords: Goodness-of-fit tests; Regression specifications; Distribution-free tests; Regressogram; Neyman-Pearson cells.

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1 Introduction

The identification and estimation of causal relationships often rely on models defined by conditional moment restrictions (CMR), typically in the form of parametric regression models. Ensuring the correct model specification is crucial for drawing valid inferences and providing reliable interpretations. This article introduces χ^2 tests to check the goodness-of-fit of CMR model specifications.

Existing tests, rooted in the definition of the integrated regression function (IRF), expand upon classical goodness-of-fit tests designed for cumulative density function (CDF) specifications and apply them to the regression model checking. The IRF generalizes the one-to-one relationship between density and the CDF to the regression framework, enabling the extension of all the tests proposed for CDFs to the context of regression. The alternative test statistics are characterized by a functional of the standard empirical process (SEP), which is defined as the difference between the empirical IRF/CDF and its restricted counterpart under the null hypothesis, suitably scaled.

Omnibus tests are designed to assess whether any deviations of the SEP from zero arise by chance rather than due to model misspecification. Classical test statistics for CDF specifications, which are based on a norm of the SEP, such as the well-known Kolmogorov-Smirnov, Cramér-von Mises, or Anderson-Darling tests, have been extended to the context of regression model specifications by Bierens (1982), Stute (1997), and Andrews (1997), among others. Several tests are based on transformations of the SEP. For instance, test statistics based on Fourier transforms of the SEP, which in fact compare characteristic functions (e.g., Koutrouvelis and Kellermeier 1981), have been adapted to the regression case by Bierens (1982) and Bierens and Ploberger (1997). Tests based on martingale transforms of the SEP (Khmaladze 1982) have been extended by Khmaladze and Koul (2004). Additionally, specification tests for Lebesgue density specification, which rely on the convolution of the SEP and a kernel function (e.g., Bickel and Rosenblatt 1973), have been extended to regression models by Hardle and Mammen (1993). All of these tests are known as minimum-distance tests (Pollard 1980).

When the model parameters are estimated from the sample, the SEP converges in distribution to a case-dependent process, and the tests are implemented using bootstrap

techniques. Two exceptions exist: (i) test statistics based on the SEP martingale transform, which converge in distribution to a Brownian motion with tabulated critical values; and (ii) test statistics based on the convolution of the SEP and a kernel function, suitably centered and standardized, with a standard normal limit distribution under the null when the bandwidth vanishes at a suitable rate. However, the size properties of tests based on convolutions depend on the bandwidth choice, and in practice, the tests are implemented using bootstrap.

It is well known that tests based on the SEP exhibit poor power properties in the direction of high-frequency alternatives, as pointed out by Durbin and Knott (1972) based on the study of the SEP's principal components. Stute (1997) generalized these results to the regression case. Tests that rely on convolutions of the SEP and a kernel, with a vanishing bandwidth, are capable of detecting high-frequency alternatives but fail to detect the standard departures converging to the model in the null at the parametric rate. All of these tests are affected by the curse of dimensionality, showing limited power when the number of explanatory variables is large.

In practice, χ^2 tests, such as the Pearson (1900) primeval test and its subsequent versions, are among the most popular goodness-of-fit tests for CDF model specifications. These tests, after partitioning the data into a number of cells, say L, evaluates whether the difference between the observed and the expected frequencies within each cell occur by chance. The Pearson test statistics, for instance, can be represented as a quadratic form in the vector of differences between observed and expected frequencies. Note that the χ^2 tests are not omnibus: their goal is to detect alternatives where the true probability of one observation falling into a particular cell differs from the expected value specified by the model. However, they have important advantages with respect to tests based on SEP functionals. They are distribution-free and do not require the use of bootstrapping or smoothing techniques. Additionally, these tests are invariant to sample-dependent partitions, which can then be chosen to favor certain alternatives of interest. Also, they are computationally friendly.

Surprisingly, the χ^2 tests have not been extended to check the specification of CMR. This paper aims to fill this gap by proposing to group the data according to a partition of the explanatory variables into L classes, and considering χ^2 tests given by quadratic

forms in the vector of differences between observed and expected (according to the model) sample averages within each class. The Pearson test analog is a weighted sum of these squared differences, which is in fact a J test for the orthogonality conditions between regression errors and the explanatory variables in each of the L classes. The paper also considers a generalized Wald test statistic: a quadratic form in the vector of differences between observed and expected sample averages using any $\sqrt{n}$-consistent estimator of the parameters in the model.

Much like the classical χ^2 test statistic, these tests have a limiting null distribution which is invariant to sample-dependent partitions. I propose algorithms for the calculation of balanced cells, each designed to contain approximately the same number of observations, as well as unbalanced partitions intended to favor specific alternatives. I consider tests based on Neyman-Pearson (NP) partitions built upon deviations of the model fits under the null hypothesis from a predefined alternative, see Balakrishnan, Voinov, and Nikulin (2013). Under the correct specification of the alternative model, the NP partition maximizes the distance between the null and alternative models, thereby enhancing the power of the tests. In cases where there are no specific alternatives to prioritize, the recommendation is to utilize NP cells that compare the model under the null hypothesis with an auxiliary, flexible specification of the regression model.

Monte Carlo simulations demonstrate the strong performance of the χ^2 tests compared to omnibus tests, particularly in settings with many covariates. The finite-sample study is supplemented with two empirical applications of the tests in common quasi-experimental frameworks: one employing a difference-in-differences design with a continuous treatment and no never-treated units, and another with a binary treatment and numerous covariates under unconfoundedness. These revisit the studies by Pierce and Schott (2016) and Fryer and Greenstone (2010), examining the impact of the U.S. government's decision to grant Permanent Normal Trade Relations (PNTR) to China and the returns for African American students attending Historically Black Colleges and Universities (HBCUs), respectively.

The structure of the paper is as follows. In the next section, I introduce the tests focusing on regression specifications. This facilitates the motivation and presentation while reducing the notational burden required for the more general case. Section 3 discusses

the asymptotic properties of the tests using sample-dependent partitions. Section 4 introduces partitioning algorithms aimed at enhancing the tests' power. Section 5 examines the tests' local power under Pittman alternatives. In Section 6, I extend the discussion to general CMR. Section 7 presents a Monte Carlo study, and the final section provides the two empirical illustrations.

2 Chi-Squared Tests for CMR

Let $\{Z_i\}_{i=1}^n = \{(Y_i, X_i')'\}_{i=1}^n$ be an i.i.d sample from the $\mathbb{R}^{1+d_x}$-valued random vector $Z = (Y, X')'$ with distribution P, where Y is the response variable, and X is the d_x-dimensional vector of explanatory variables taking values in $\mathcal{X} \subset \mathbb{R}^{d_x}$. The correct specification hypothesis is stated as,

$$H_0 : \ m \in \mathcal{M} \tag{1}$$

with $m(\cdot) = \mathbb{E}[Y|X = \cdot]$ and $\mathcal{M} = \{m_\theta(\cdot) : \theta \in \Theta \subset \mathbb{R}^{d_\theta}\}$ being a family of parametric regression functions and a suitable parameter space, respectively. Thus, under H_0, there exists a $\theta_0 \in \Theta$ such that the IRF, $M(x) = \mathbb{E}\left[Y\mathbb{I}_{(-\infty,x]}(X)\right]$, with $\mathbb{I}_{(-\infty,x]}(X) = \prod_{j=1}^{d_x} \mathbb{I}_{(-\infty,x_j]}(X_j)$, and its version imposing the null hypothesis restrictions, $M_\theta(x) = \mathbb{E}\left[m_\theta(X)\mathbb{I}_{(-\infty,x]}(X)\right]$, are equal in all the Borel sets of $\mathcal{X}$; that is,

$$M\{A\} = M_{\theta_0}\{A\} \ \text{ for all } A \in \mathcal{B}^{d_x} \text{ and some } \theta_0 \in \Theta, \tag{2}$$

where $\mathcal{B}^{d_x}$ represent the Borel sigma-field of $\mathbb{R}^{d_x}$, $M\{A\} = \int_A M(dx)$ and $M_\theta\{A\} = \int_A M_\theta(dx)$ denote the Lebesgue-Stieltjes measure of M and M_θ over A, respectively. Recall that (2) is the definition of the regression function for the specification in H_0 (e.g., see definition 34.1 in Billingsley 2017), which is equivalent to the following orthogonality conditions,

$$\mathbb{E}\left[\varepsilon_{\theta_0}|X\right] = 0 \ \text{ a.s. for some } \theta_0 \in \Theta,$$

with $\varepsilon_\theta(z) = y - m_\theta(x)$, $z = (y, x')'$, denoting the regression error.

Intuitively, when X takes values in a finite set and θ_0 is known (i.e., under the simple hypothesis), a test for H_0, following Pearson (1900) proposal, consists of comparing the

conditional mean with the assumed regression model for all $x \in \mathcal{X}$. In this case, such a test is indeed omnibus (i.e. the test detect all the possible alternatives). For any type of covariates, once the data is partitioned into L cells, say, χ^2 tests assess whether the difference between the expected and observed averages in each cell arose by chance.

Consider the partition $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_L)$ of the covariates space $\mathcal{X}$ into L cells and let $\mathbf{I}_{\boldsymbol{\gamma}}(x) = (\mathbb{I}_{\gamma_1}(x), \ldots, \mathbb{I}_{\gamma_L}(x))'$ denote the vector of indicator functions over the sets within $\boldsymbol{\gamma}$. The building block of the χ^2 test statistics is the standardized vector of differences between the observed averages of Y and the corresponding averages under the specification of H_0 in each cell,

$$\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\theta) = \sqrt{n}\left(\hat{\boldsymbol{\mu}}^0_{\boldsymbol{\gamma}} - \hat{\boldsymbol{\mu}}_{\boldsymbol{\gamma}}(\theta)\right) = \sqrt{n}\left(\hat{\mu}^0_1 - \hat{\mu}_1(\theta), \ldots, \hat{\mu}^0_L - \hat{\mu}_L(\theta)\right)', \tag{3}$$

where $\hat{\mu}^0_l = \hat{M}\{\gamma_l\} = n^{-1}\sum_{i=1}^n Y_i \mathbb{I}_{\gamma_l}(X_i)$ is the empirical IRF evaluated at γ_l, with $\hat{M}(x) = n^{-1}\sum_{i=1}^n Y_i \mathbb{I}_{(-\infty,x]}(X_i)$, and $\hat{\mu}_l(\theta) = \hat{M}_\theta\{\gamma_l\} = n^{-1}\sum_{i=1}^n m_\theta(X_i)\mathbb{I}_{\gamma_l}(X_i)$ is its restricted version under H_0, with $\hat{M}_\theta(x) = n^{-1}\sum_{i=1}^n m_\theta(X_i)\mathbb{I}_{(-\infty,x]}(X_i)$.

Under a simple hypothesis, i.e. when θ_0 is known, by the central limit theorem, under the null,

$$\hat{\Sigma}_{\boldsymbol{\gamma}}(\theta_0)^{-1/2}\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\theta_0) \xrightarrow{d} N(0, I_L), \tag{4}$$

where $\hat{\Sigma}_{\boldsymbol{\gamma}}(\theta) = n^{-1}\sum_{i=1}^n \varepsilon^2_\theta(Z_i)\mathbf{I}_{\boldsymbol{\gamma}}(X_i)\mathbf{I}_{\boldsymbol{\gamma}}(X_i)'$ estimates,

$$\Sigma_{\boldsymbol{\gamma},0} = \text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\theta_0)\right) = \text{diag}\{\sigma^2_{0,1}, ..., \sigma^2_{0,L}\}, \tag{5}$$

under H_0, with $\sigma^2_{0,l} = \sigma^2_l(\theta_0)$, and $\sigma^2_l(\theta) = \mathbb{E}\left[\varepsilon^2_\theta(Z)\mathbb{I}_{\gamma_l}(X)\right]$. Thus, taking for granted that $\rho_l = \mathbb{E}\left[\mathbb{I}_{\gamma_l}(X)\right] > 0$ for all l, under H_0,

$$\hat{\chi}^2_{\boldsymbol{\gamma},0}(\theta_0) \xrightarrow{d} \chi^2_L, \tag{6}$$

where

$$\hat{\chi}^2_{\boldsymbol{\gamma},0}(\theta) = \hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\theta)'\hat{\Sigma}_{\boldsymbol{\gamma}}(\theta_0)^{-1}\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\theta) = n\sum_{l=1}^L \frac{\left(\hat{\mu}^0_l - \hat{\mu}_l(\theta)\right)^2}{\hat{\sigma}^2_{0,l}}, \tag{7}$$

and $\hat{\sigma}^2_{0,l} = n^{-1}\sum_{i=1}^n \varepsilon^2_{\theta_0}(Z_i)\mathbb{I}_{\gamma_l}(X_i)$. The test statistic $\hat{\chi}^2_{\boldsymbol{\gamma},0}(\theta_0)$ extends the classical Pearson's chi-squared test to check regression model specifications. It does so by replacing the

comparison of observed frequencies with expected frequencies under the null hypothesis with a comparison of averages.

Of course, tests based on (3) are not omnibus, but designed for detecting deviations from H_0 of the type,

$$H_1(\boldsymbol{\gamma}) : \boldsymbol{\mu}^0_{\boldsymbol{\gamma}} \neq \boldsymbol{\mu}_{\boldsymbol{\gamma}}(\theta) \text{ for all } \theta \in \Theta, \tag{8}$$

where $\boldsymbol{\mu}^0_{\boldsymbol{\gamma}} = (\mu^0_{\gamma_1}, ..., \mu^0_{\gamma_L})' = (M\{\gamma_1\}, \ldots, M\{\gamma_L\})'$ is the vector of expected averages of Y in each cell and $\boldsymbol{\mu}_{\boldsymbol{\gamma}}(\theta) = (\mu_{\gamma_1}(\theta), ..., \mu_{\gamma_L}(\theta))' = (M_\theta\{\gamma_1\}, \ldots, M_\theta\{\gamma_L\})'$ is its restricted version under H_0. The partitions cover a fundamental role in implementing the test and provide a flexible tool to exploit out-of-sample information on the possible alternatives. Neyman-Pearson (NP) cells (Balakrishnan, Voinov, and Nikulin 2013), for instance, consist of the points in $\mathcal{X}$ where the specification under the null and a pre-specified alternative coincide (see Example 1 below). Under the pre-specified alternative, NP classes maximize the L_2 norm of $\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}$.

Example 1 (NP Classes)

Consider testing the linear model $m_{\theta_0}(X_i) = \theta_{00} + \theta_{01}X_i$ against the alternative specification,

$$H_1 : m_{1,\theta_0^*}(X_i) = \theta^*_{00} + \theta^*_{01}X_i + \theta^*_{02}\sin\left(\frac{50X_i}{2\pi}\right)$$

where $(\theta^*_{00}, \theta^*_{01}) = (\theta_{00}, \theta_{01})$ and $\theta^*_0 = (\theta_0, \theta^*_{02})$ is a known vector. NP classes split $\mathcal{X}$ over the points where $\theta^*_{02}\sin\left(\frac{50x}{2\pi}\right) = 0$ and, hence, $m_{\theta_0}(\cdot) = m_{1,\theta_0^*}(\cdot)$. As a result, under H_1, $m_{\theta_0}(\cdot)$ is strictly bigger or strictly smaller than $m_{1,\theta_0^*}(\cdot)$ within each cell, and most cell-specific errors have the same sign, implying that the average error of a single cell is larger than the average error over the union of two contiguous cells (in absolute terms). See Figure 1 for a graphical representation of the partition.

When θ_0 is unknown, the criterion in (7) suggests the following minimum distance estimator, hereafter referred to as the grouped GMM estimator:

$$\hat{\theta}_{\boldsymbol{\gamma}} = \arg\min_{\theta\in\Theta} \chi^2_{\boldsymbol{\gamma}}(\theta) \tag{9}$$

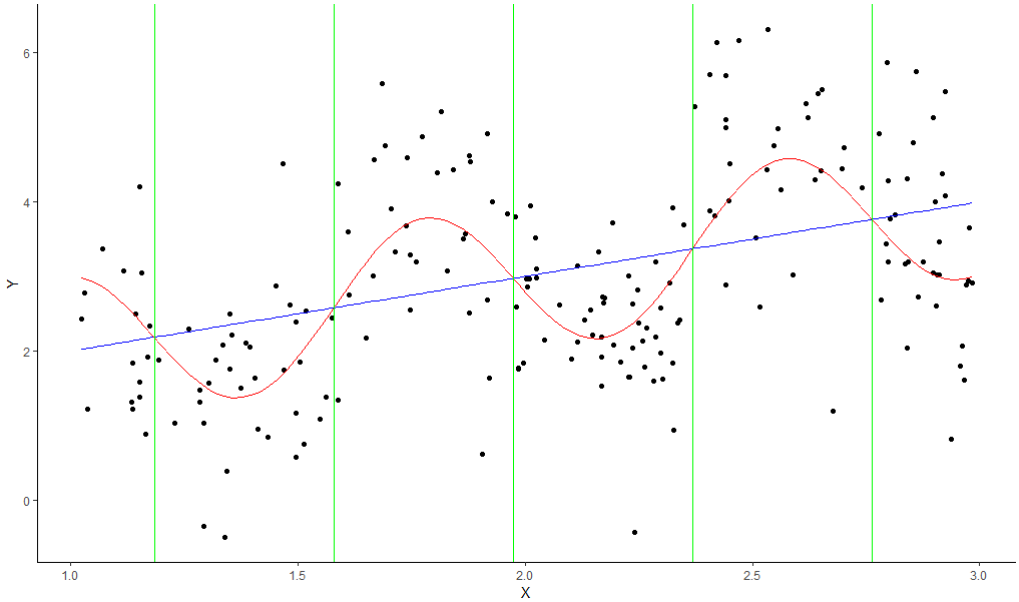


Figure 1: The graph depicts a random draw from the model under H_1 with $\theta_0^* = (1, 1, 1)$. The green lines depict the points where the model under the null (blue line) and under the alternative (red line) meet.

where

$$\hat{\chi}^2_{\boldsymbol{\gamma}}(\theta) = \hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\theta)'\hat{\Sigma}_{\boldsymbol{\gamma}}(\tilde{\theta})^{-1}\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\theta) = n\sum_{l=1}^{L}\frac{(\hat{\mu}_l^0 - \hat{\mu}_l(\theta))^2}{\hat{\sigma}_l^2(\tilde{\theta})},$$

$\hat{\sigma}_l^2(\theta) = n^{-1}\sum_{i=1}^n \varepsilon_\theta^2(Z_i)\mathbb{I}_{\gamma_l}(X_i)$, and $\tilde{\theta}$ is some initial $\sqrt{n}$-consistent estimator of θ_0. The estimator $\hat{\theta}_{\boldsymbol{\gamma}}$ is analogous to the multinomial maximum-likelihood estimator (or minimum χ^2 estimator) in the classical case (see Cramér 1946). Under linear null hypothesis, i.e. $m_\theta(x) = x'\theta$, this corresponds to the feasible GLS estimator based on the aggregated data $\{\bar{Y}_l, \bar{X}_l\}_{l=1}^L$,

$$\hat{\theta}_{\boldsymbol{\gamma}} = \left[\sum_{l=1}^{L}\frac{\bar{X}_l\bar{X}_l'}{\hat{\sigma}_l^2(\tilde{\theta})}\right]^{-1}\sum_{l=1}^{L}\frac{\bar{X}_l\bar{Y}_l}{\hat{\sigma}_l^2(\tilde{\theta})},$$

with $\bar{Y}_l = n^{-1}\sum_{i=1}^n Y_i\mathbb{I}_{\gamma_l}(X_i)$ and $\bar{X}_l = n^{-1}\sum_{i=1}^n X_i\mathbb{I}_{\gamma_l}(X_i)$. In the non-linear case, we can iterate a feasible asymptotically efficient Gauss-Newton estimator starting from any preliminary $\sqrt{n}$-consistent estimator $\tilde{\theta}$,

$$\hat{\theta}_{\boldsymbol{\gamma}}^{(1)} = \tilde{\theta} + \left[\sum_{l=1}^{L}\frac{\hat{\mu}_l^*(\tilde{\theta})\hat{\mu}_l^*(\tilde{\theta})'}{\hat{\sigma}_l^2(\tilde{\theta})}\right]^{-1}\sum_{l=1}^{L}\frac{\hat{\mu}_l^*(\tilde{\theta})(\hat{\mu}_l^0 - \hat{\mu}_l(\tilde{\theta}))}{\hat{\sigma}_l^2(\tilde{\theta})} \tag{10}$$

with $\hat{\mu}_l^*(\theta) = n^{-1}\sum_{i=1}^n \nabla m_\theta(X_i)\mathbb{I}_{\gamma_l}(X_i)$ and $\nabla m_{\bar{\theta}} = d/d\theta' m_\theta|_{\theta=\bar{\theta}}$. The estimator belongs

to the class of minimum-distance estimators considered by Cristobal, Roca, and Manteiga (1987) and Koul and Ni (2004), with the main difference that the regressogram is used instead of kernels, and the weighting introduced to improve efficiency. Under suitable regularity conditions and under H_0,

$$\sqrt{n}(\hat{\theta}_{\boldsymbol{\gamma}} - \theta_0) \xrightarrow{d} N\left(0, \left[\boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma},0}(\Sigma_{\boldsymbol{\gamma},0})^{-1}\boldsymbol{\mu}^{*}_{\boldsymbol{\gamma},0}\right]^{-1}\right),$$

where $\boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} = \boldsymbol{\mu}^*_{\boldsymbol{\gamma}}(\theta_0)$, with $\boldsymbol{\mu}^*_{\boldsymbol{\gamma}}(\theta) = (\mu^*_1(\theta), ..., \mu^*_L(\theta))'$ and $\mu^*_l(\theta) = \mathbb{E}\left[\hat{\mu}^*_l(\theta)\right]$ denoting the matrix of partial derivatives of $\boldsymbol{\mu}_{\boldsymbol{\gamma}}(\theta)$.

The $\hat{\chi}^2$ test statistics is, for $L > d_\theta$,

$$\hat{\chi}^2_{\boldsymbol{\gamma}} = \min_{\theta\in\Theta} \hat{\chi}^2_{\boldsymbol{\gamma}}(\theta) = \hat{\chi}^2_{\boldsymbol{\gamma}}(\hat{\theta}_{\boldsymbol{\gamma}}), \tag{11}$$

which is in fact a J test on the set of the L, out of the (possibly) many, orthogonality conditions implied by the null,

$$\mathbb{E}\big[Y\mathbb{I}_{\gamma_l}(X)\big] = \mathbb{E}\big[m_{\theta_0}(X)\mathbb{I}_{\gamma_l}(X)\big] \text{ for all } l \in \{1, 2, .., L\}.$$

Thus, under H_0, and for $L > d_\theta$,

$$\hat{\chi}^2_{\boldsymbol{\gamma}} \xrightarrow{d} \chi^2_{L-d_\theta}. \tag{12}$$

Is also well motivated, as suggested in classical goodness-of-fit χ^2 tests (e.g. Nikulin 1973 and Rao and Robson 1974), using the Wald testing principle based on $\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})$, employing any $\sqrt{n}$-consistent estimator $\tilde{\theta}$,

$$\hat{\mathcal{W}}_{\boldsymbol{\gamma}}(\tilde{\theta}) = \hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\widehat{\text{Avar}^-}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta}), \tag{13}$$

where $\widehat{\text{Avar}^-}(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta}))$ is a consistent estimator of some generalized inverse of $\text{Avar}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$, $\text{Avar}^-\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ say. Assuming $\widehat{\text{Avar}^-}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right) \xrightarrow{p} \text{Avar}^-\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$, and suitable regularity conditions, under H_0,

$$\hat{\mathcal{W}}_{\boldsymbol{\gamma}}(\tilde{\theta}) \xrightarrow{d} \chi^2_{r(\text{Avar}(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})))}, \tag{14}$$

where for a given square matrix A, $r(A)$ denotes its rank. Taking for granted the asymptotic linearity of the estimator (see Assumption 3 in the next section), the covariance matrix of $\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})$ is characterized as,

$$\text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right) = \Sigma_{\boldsymbol{\gamma},0} - \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} C'_{\boldsymbol{\gamma},0} - C_{\boldsymbol{\gamma},0}\boldsymbol{\mu}^{*'}_{\boldsymbol{\gamma},0} + \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} L_0 \boldsymbol{\mu}^{*'}_{\boldsymbol{\gamma},0}, \tag{15}$$

where $C_{\boldsymbol{\gamma},0} = \mathbb{E}\left[\varepsilon_{\theta_0}(Z)\mathbf{I}_{\boldsymbol{\gamma}}(X) l_{\theta_0}(Z)'\right]$, $L_0 = \mathbb{E}\left[l_{\theta_0}(Z) l_{\theta_0}(Z)'\right]$, and $l_{\theta_0}(\cdot)$ is the influence function of $\tilde{\theta}$. When $\text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ is full rank (e.g., if $\varepsilon_{\theta_0}(\cdot)\mathbf{I}_{\boldsymbol{\gamma}}(\cdot)$ and $l_{\theta_0}(\cdot)$ have linearly independent components), the Wald test can be performed on any finite splitting of the data. In this case, a valid choice of $\widehat{\text{Avar}^-}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ is given by the inverse of,

$$\hat{W}_{\boldsymbol{\gamma}}(\tilde{\theta}) = \hat{\Sigma}_{\boldsymbol{\gamma}}(\tilde{\theta}) - \hat{\boldsymbol{\mu}}^*_{\boldsymbol{\gamma}}(\tilde{\theta})\hat{C}_{\boldsymbol{\gamma}}(\tilde{\theta})' - \hat{C}_{\boldsymbol{\gamma}}(\tilde{\theta})\hat{\boldsymbol{\mu}}^{*'}_{\boldsymbol{\gamma}}(\tilde{\theta}) + \hat{\boldsymbol{\mu}}^*_{\boldsymbol{\gamma}}(\tilde{\theta})\hat{L}(\tilde{\theta})\hat{\boldsymbol{\mu}}^{*'}_{\boldsymbol{\gamma}}(\tilde{\theta}), \tag{16}$$

where $\hat{\boldsymbol{\mu}}^*_{\boldsymbol{\gamma}}(\theta) = (\hat{\mu}^*_1(\theta), ..., \hat{\mu}^*_L(\theta))'$, $\hat{C}_{\boldsymbol{\gamma}}(\theta) = n^{-1}\sum_{i=1}^n \varepsilon_\theta(Z_i)\mathbf{I}_{\boldsymbol{\gamma}}(X_i) l_\theta(Z_i)'$, and $\hat{L}(\theta) = n^{-1}\sum_{i=1}^n l_\theta(Z_i) l_\theta(Z_i)'$. If the covariance matrix is rank deficient and the Moore-Penrose inverse of $\hat{W}_{\boldsymbol{\gamma}}(\tilde{\theta})$, denoted as $\hat{W}^+_{\boldsymbol{\gamma}}(\tilde{\theta})$, is such that $\limsup_{n\to\infty} ||\hat{W}^+_{\boldsymbol{\gamma}}(\tilde{\theta})||_2 < \infty$, with $||\cdot||_2$ being the euclidean norm, then $\hat{W}^+_{\boldsymbol{\gamma}}(\tilde{\theta}) \xrightarrow{p} \text{Avar}^+\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ (Theorem 2 of Andrews 1987). However, this need not be the case (Schott 2016, p. 222-224) and more complex methods might be required. Consider, for instance, the nonlinear least squares (NLLS),

$$\tilde{\theta} = \arg\min_{\theta\in\Theta} \sum_{i=1}^n \varepsilon_\theta(Z_i)^2.$$

Under homoskedasticity, i.e. $\mathbb{E}\left[\varepsilon^2_{\theta_0}(Z)|X\right] = \sigma^2_0 = \sum_{l=1}^L \sigma^2_{0,l}$, it holds that, $l_{\theta_0}(z) = \Psi_0^{-1}\nabla m_{\theta_0}(x)\varepsilon_{\theta_0}(z)$, where $\Psi_0 = \mathbb{E}\left[\nabla m_{\theta_0}(X)\nabla m_{\theta_0}(X)'\right]$. Under H_0, when $\tilde{\theta}$ is asymptotically more efficient than $\hat{\theta}_{\boldsymbol{\gamma}}$ (i.e., when $\text{Avar}(\hat{\theta}_\gamma) - \text{Avar}(\tilde{\theta})$ is p.d.), $\text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right) = \Sigma_{\boldsymbol{\gamma},0} - \sigma^2_0\boldsymbol{\mu}^*_{\boldsymbol{\gamma},0}\Psi_0^{-1}\boldsymbol{\mu}^{*'}_{\boldsymbol{\gamma},0}$ is also p.d., and $\hat{\mathcal{W}}(\tilde{\theta}) \xrightarrow{d} \chi_L$. However, if X includes a constant, $\sum_{i=1}^n \varepsilon_{\tilde{\theta}}(Z_i) = 0$ and $\text{Avar}(\hat{\theta}_\gamma)$ and $\text{Avar}(\tilde{\theta})$ have the same first row and column. Consequently, $r\left(\widehat{\text{Avar}^-}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)\right) = L-1$ in the homoskedastic case, and $r\left(\widehat{\text{Avar}^-}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)\right) \leq L-1$ under heteroskedasticity.

Similarly, in the probit/logit maximum likelihood estimator (MLE), $\text{Avar}(\hat{\theta}_\gamma) - \text{Avar}(\tilde{\theta})$ is p.d. (regardless of whether X includes a constant or whether errors are homoskedastic).

Thus, in this case, $r\left(\widehat{\text{Avar}}^{-}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)\right) = L$ under H_0, as discussed in Section 8.

The estimation of the covariance matrix, when $\text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ is full rank, can be avoided by using a random normalizing weighting matrix, as in Kuan and Lee (2006) (see also Kiefer, Vogelsang, and Bunzel 2000 for an early reference). Similarly, the over-identification test of Lee, Kuan, and Hsu (2014) provides a robust version of the $\hat{\chi}^2$ test which does not require estimating $\hat{\Sigma}_{\boldsymbol{\gamma}}(\tilde{\theta})$. In these cases, the limit null distribution is non-standard, but pivotal.

3 Data-dependent Cells

When studying the large sample behavior of the statistics, it is crucial to address the inherent influence of the data on the selection of cells (Watson 1959). Moore and Spruill (1975) were among the first to address this concern in the distribution model check literature providing a rigorous derivation of the null distribution of χ^2 tests with rectangular data-dependent cells. In a more general setting, Pollard (1979) established the result for cells of arbitrary form using uniform results for empirical processes indexed by sets, while Andrews (1988a,b) and Delgado and Vainora (2023) applied the methodology to conditional distribution testing. This section provides a similar result for the more general CMR testing framework, showing that the grouped GMM estimator and the tests keep standard limit distribution when the partition is built with data-dependent cells.

A minimal set of assumptions, consisting of smoothness conditions and restrictions on the partitioning algorithm complexity, allows to state the convergence results in a self-contained fashion.

Assumption 1 *(a) $\{Z_i = (Y_i, X_i')'\}_{i=1}^n$ is a sequence of i.i.d. random vectors with $\mathbb{E}|Y_i| < \infty$; (b) $\mathbb{E}\left[\varepsilon_{\theta_0}^2\right] < C$, with $C < \infty$; (c) Θ is a compact subset of $\mathbb{R}^{d_\theta}$ and θ_0 is an interior point of Θ.*

Assumption 2 *$m_\theta(\cdot)$ is twice continuously differentiable in a neighborhood $\mathcal{N}_{\theta_0}$ of θ_0, with $\mathcal{N}_{\theta_0} \subset \Theta$. The gradient, $\nabla m_\theta(\cdot) = d/d\theta m_\theta(\cdot)$, is bounded by a square-integrable function $R(\cdot)$ such that $\sup_{\theta\in\Theta_0} |\nabla^{(j)} m_\theta(\cdot)| \leq R(\cdot)$ for all $j \in \{1, .., d_\theta\}$, where $\nabla^{(j)}$ denotes the j-th partial derivative, and $\mathbb{E}\left[R(X)^2\right] < \infty$.*

Assumption 3

(a) The estimator $\tilde{\theta}$ satisfies the following asymptotic expansion under the null,

$$\sqrt{n}(\tilde{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} l_{\theta_0}(Z_i) + o_p(1)$$

where $\mathbb{E}\left[l_{\theta_0}(Z)\right] = 0$ and $L_0 = \mathbb{E}\left[l_{\theta_0}(Z) l_{\theta_0}(Z)'\right]$ is a finite and non-singular matrix.

(b) The vector-valued function $l_\theta(\cdot)$ is twice continuously differentiable in a neighborhood Θ_0 of θ_0 with first partial derivatives bounded by a square-integrable function $R_2(\cdot)$ such that $\sup_{\theta \in \Theta_0} |\nabla^{(j)} l_\theta(\cdot)| \leq R_2(\cdot)$ for all $j \in \{1, .., d_\theta\}$ and $\mathbb{E}\left[R_2(Z)^2\right] < \infty$.

Assumptions 1 and 2 are common in the model check literature (see Stute and Zhu 2002, for instance). Compared to papers developing χ^2 tests based on a probability model (e.g., Tauchen 1985), these require slightly higher smoothness conditions of the regression function, but leave completely unrestricted the data distribution. Assumption 3(a) holds for most of the estimators used in practice, such as least squares or GMM estimators, as well as for identification-robust minimum-distance estimators (see, e.g., Domínguez and Lobato 2004). While Assumption 3(b) is a technical requirement for the consistency of the plug-in estimator $\hat{W}_{\boldsymbol{\gamma}}(\tilde{\theta})$.

Let also state the necessary global identification and finite-variance conditions for the consistency and asymptotic normality of the grouped GMM estimator.

Assumption 2' *(a) $\Sigma_{\boldsymbol{\gamma},0}$ is positive definite (p.d.); (b) $\mathbb{E}\left[Y \mathbf{I}_{\boldsymbol{\gamma}}\right] = \mathbb{E}\left[m_\theta(X) \mathbf{I}_{\boldsymbol{\gamma}}\right]$ if and only if $\theta = \theta_0$; (c) $\boldsymbol{\mu}^*_{\boldsymbol{\gamma},0}$ is full rank.*

Following Pollard (1979) and Andrews (1988a), the data-dependent partitions are modeled as random functions over a class of properly restricted measurable sets. Specifically, let $\mathbb{C}$ be a class of measurable sets in $\mathcal{X}$ from which the cells of each partition are drawn, and denote as $\mathbb{D}$ the class of partitions of $\mathcal{X}$ comprised of L sets from $\mathbb{C}$ (L is fixed for all n); that is,

$$\mathbb{D} = \left\{ \boldsymbol{\gamma} = (\gamma_1, ..., \gamma_L) \in \mathbb{C}^L : \bigcup_{l=1}^{L} \gamma_l = \mathcal{X},\ \gamma_l \bigcap \gamma_f = \emptyset,\ \forall l \neq f \right\}, \tag{17}$$

where γ_l and γ_f denote sets of the partition $\boldsymbol{\gamma}$. We equip $\mathbb{C}$ with the topology generated by the $L_2(F_x)$ semi-norm, F_x being the distribution of X under P, and give $\mathbb{D}$ the corresponding product topology. This means that two set C_1, C_2 in $\mathcal{X}$ are close if $F_x\{C_1 \Delta C_2\}$ is small, where Δ denotes the symmetric difference operator, $C_1 \Delta C_2 = C_1 \cup C_2 \backslash C_1 \cap C_2$.

For each sample size n the corresponding partition is determined by a measurable mapping, denoted as $\hat{\boldsymbol{\gamma}} = (\hat{\gamma}_1, \ldots, \hat{\gamma}_L)$, from the underlying probability space to $\mathbb{D}$, which converges in probability to some fixed partition of cells in $\mathbb{D}$. In other words, for all $\epsilon > 0$,

$$P\left(F_x\{\hat{\gamma}_l \Delta \gamma_l\} > \epsilon\right) \to 0 \text{ as } n \to \infty, \text{ for all } l = 1, 2, ..., L.$$

Assumption 4 *Under H_0, $\hat{\boldsymbol{\gamma}} \xrightarrow{p} \boldsymbol{\gamma}$ for some fixed set of cells $\boldsymbol{\gamma} \in \mathbb{D}$*

Assumption 4 is a standard requirement in the classical χ^2 goodness-of-fit tests literature for probability distributions and is satisfied by a wide range of partitioning algorithms. For instance, this assumption holds when the cells depend continuously on some random function of the data which converges in probability to a non-random vector (Andrews, 1988b). Additionally, cells generated by comparing different estimators also meets this requirement, as discussed in the next section. It is possible to relax Assumption 4 to cases where $\hat{\boldsymbol{\gamma}}$ converges to a random element $\boldsymbol{\gamma}_0 \in \mathbb{D}$. This is achieved by replacing the convergence in probability with a uniform tightness condition and ensuring asymptotic independence between $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\theta_0)$. Refer to Section 6 of Pollard (1979) for further discussions.

Crucially, deriving the limit null distribution requires bounding the complexity of the partitions employed for test construction. This is achieved by assuming that the cells are drawn from a Vapnik-Cervonenkis (VC) class.

Assumption 5 *$\mathbb{C}$ is a VC class of sets.*

This assumption is convenient because it is independent of the data distribution P, yet general enough for our purposes. For example, algorithms that generate cells with a finite number of straight edges, as well as the class of hyper-ellipsoids, are VC classes.

Hyperplanes based on linear combinations of known functions of X, i.e.,

$$\mathbb{C} = \left\{ \left\{ \sum_{j=1}^{q} \beta_j f_j(x) \geq g(x) \right\} : (\beta_1, ..., \beta_q) \in \mathbb{R}^q \right\},$$

and the particular case

$$\mathbb{C} = \left\{ \{\beta' x \geq \delta\} : (\beta, \delta) \in \mathbb{R}^{d_x+1} \right\},$$

are VC classes with VC dimension equal to $q+1$ and d_x+2, respectively (Theorem 4.2.1 in Dudley 1999). Furthermore, unions, intersections, differences, and complements of VC classes are also VC classes (Andrews 1988a and Pollard 1984 provide a thorough discussion; see also Section 2.6 in Van Der Vaart 1996). A less stringent condition assumes that $\mathbb{C}$ is a Donsker class for the underlying probability measure (Pollard 1979). While this allows for a wider range of admissible partitions, verifying Donsker assumptions may prove more challenging in practice.

The following theorems show that the asymptotic distribution of the grouped GMM estimator and the Wald test depends only on the limit partition $\boldsymbol{\gamma}$, whereas the limit null distribution of the J test remains unaffected by data-dependent cells. All the proofs are relegated to the appendix.

Theorem 1 *Let Assumptions 1, 2, 2', 4, 5 hold. Then, under the null hypothesis H_0,*

$$\sqrt{n}(\hat{\theta}_{\hat{\gamma}} - \theta_0) \xrightarrow{d} N\left(0, \left[\boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma},0}(\Sigma_{\boldsymbol{\gamma},0})^{-1}\boldsymbol{\mu}^{*}_{\boldsymbol{\gamma},0}\right]^{-1}\right),$$

Theorem 2 *Let Assumptions 1, 2, 4, 5, and the null hypothesis H_0 hold. Then,*

(a) Under Assumption 2' and $L > d_\theta$,

$$\hat{\chi}^2_{\hat{\gamma}} \xrightarrow{d} \chi^2_{L-d_\theta}.$$

(b) Under Assumption 3, and $\widehat{\text{Avar}^-}\left(\hat{\boldsymbol{\Phi}}_{\hat{\gamma}}(\tilde{\theta})\right) \xrightarrow{p} \text{Avar}^-\left(\hat{\boldsymbol{\Phi}}_{\hat{\gamma}}(\tilde{\theta})\right)$,

$$\hat{\mathcal{W}}_{\hat{\gamma}}(\tilde{\theta}) \xrightarrow{d} \chi^2_{r(\text{Avar}(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})))}.$$

4 Partitioning Procedures

The partitioning algorithms can be divided into covariate-based methods and model-based methods, depending on the information used to split the sample. The first classifies the data exclusively based on the characteristics of the population and does not have a predictable effect on the power of the test. In contrast, model-based methods exploit information about the model under the null hypothesis and pre-specified alternatives to increase the power of the tests in those directions. The objective of these procedures is to construct partitions where the distance between the vectors of average predictions under the null and alternative hypotheses is maximized.

In this section, I first briefly explore the construction of partitions using covariate-based methods and the limitations related to the local identification of the $\hat{\chi}^2$ test. Next, the discussion turns to the construction of Neyman-Pearson (NP) classes. These partitioning algorithms aim to maximize the probability of rejecting the null hypothesis by comparing the parametric fits under the null and the pre-specified alternative. This method is particularly useful when the researcher's primary concern is rejecting a subset of deviations critical for the application at hand.

When the alternative is left unrestricted, the recommendation is to construct NP classes using an auxiliary flexible specification of the regression model (see Davidson, MacKinnon, et al. 2004 Ch. 15.2 for a review of tests based on artificial regressions). The procedure assumes a structured alternative model, which allows to capture deviations from the null hypothesis by exploiting the conditional mean dependence of the parametric residuals from the fitted values under the alternative.

Lastly, I discuss the conditions for convergence in probability of NP classes to fixed cells in $\mathbb{D}$. Conditions for convergence of covariate-based partitioning methods are already discussed in Andrews (1988a) and Andrews (1988b).

4.1 Covariate-based Partitioning Methods

Partitioning methods based on covariate encompass both straightforward approaches, such as dividing the data into cubic cells, and more sophisticated techniques like k-means clustering, hierarchical clustering, and nearest-neighbor clustering, among others (refer

to Chapter 14 of Hastie, Tibshirani, and Friedman 2009 for a review of unsupervised clustering methods).

In the testing context, statistical practice involves dividing data into equiprobable cells when testing against unknown probability distributions (see, for instance, Greenwood and Nikulin 1996). This can be achieved through various methods, including partitioning into unions of statistically equivalent blocks (SEB), as discussed by Gessaman (1970). The procedure, applied to an $n \times d_F$ design matrix, involves sorting observations by column values and grouping them into S blocks at each iteration. Here, S is a user-chosen parameter, resulting in a partition with S^{d_F} cells. To avoid generating excessively fine partitions, instead of applying the algorithm directly to the design matrix, $\mathbb{X}$, one can apply the algorithm to a lower-dimensional projection of the data, such as the first q principal components of $\mathbb{X}$ or the fitted values of the parametric model under the null hypothesis. In the latter case, the algorithm is equivalent to splitting the data based on the S quantile of $\{m_{\tilde{\theta}}(X_i)\}_{i=1}^n$.

In practice, as discussed in the next sub-section, these methods are implemented in conjunction with other model-based partitioning algorithms. Their primary objective is to generate additional splits for the preliminary two-cell partition created by the model-based methods, which are subsequently used to construct the final partition. However, it's important to note that the J test requires the L moment conditions generated by the partition to identify θ_0, as outlined in Assumption 2'. This imposes additional restrictions on the set of feasible partitioning algorithms. For example, consider a scenario where $X = (X_1, \ldots, X_{d_x})'$, with $d_x > 2$, and X_j being mutually independent and identically distributed random variables for each $j = 1, ..., d_x$. In the linear setting, $H_0 : m(x) = \theta_0' x$, Assumption 2'(c) - i.e., the matrix of partial derivatives $\mu^*_{\gamma,0}$ is of full rank - is equivalent to assuming that the vector of grouped covariates,

$$\mu^*_{\gamma,0} = \mathbb{E}\left[\mathbf{I}_{\boldsymbol{\gamma}}(X)X'\right],$$

is linearly independent. Then, it is easy to show, that if the partition is based only on one dimension of the data - e.g., $\gamma_l = \{x \in \mathcal{X} : x_1 \in [a_{l-1}, a_l]\}$ and $\cup_{l=1}^L [a_{l-1}, a_l] = \mathbb{R}$, for $l = 1, \ldots, L$ - the matrix $\mu^*_{\gamma,0}$ is rank deficient. This is because the partition cells are

uninformative about the values the other covariates can take and, by independence and identical distribution, it follows that $\mathbb{E}\left[X_j \mathbb{I}_{\gamma_l}(X_1)\right] = \mathbb{E}\left[X_k \mathbb{I}_{\gamma_l}(X_1)\right]$ for any $j, k \neq 1$ and all l.

In the Monte Carlo simulations of Section 7, the data is partitioned using the following three covariate-based methods: SEB on the first principal component of $\mathbb{X}$ (**PCS**), SEB on the vector of fitted values (**FS**), and k-means clustering on $\mathbb{X}$ (**KM**). The first two methods use a one-dimensional projections of the data, which might be problematic for the implementation of the $\hat{\chi}^2$ test. Simulations, however, show that the size is always under control in the considered settings. The k-means algorithm, on the other hand, partitions by maximizing the between-cluster distance, making it less likely to result in a rank-deficient $\mu^*_{\gamma,0}$.

4.2 Neyman-Pearson (NP) Classes

Frequently, the researcher's primary concern is rejecting a subset of deviations critical for the application at hand. For instance, when estimating Mincer's earning regression, it is common to compare it with alternative specifications of the log-income profile (e.g. Polachek et al. 2008). Similarly, in the Cox (1972) model, a parsimonious parametric specification of the baseline hazard is often compared to more flexible models (e.g. Seetharaman and Chintagunta 2003).

In this case, a procedure that splits the data where the deviations from the null toward the alternative hypothesis are the largest is preferable. To this end, consider the construction of Neyman-Pearson (NP) classes built over the intersection points between the null and the alternative specification. These classes, in the probability distribution model checking literature, increase (or even maximize) Pearson's measure of discrepancy between the null and the alternative hypothesis (see Remark 3.3 in Balakrishnan, Voinov, and Nikulin 2013).

Let H_1 be a given alternative parametric specification of the regression function,

$$H_1 : m \in \mathcal{M}_1$$

with $\mathcal{M}_1 = \{m_{1,\theta^*}(\cdot) : \theta^* \in \Theta^*\}$ being a family of parametric regression functions and

$\Theta^* \subset \mathbb{R}^{d_{\theta^*}}$ a suitable parameter space. Thus, under H_1, $m(X) = m_{1,\theta_0^*}(X)$ a.s. for some $\theta_0^* \in \Theta^*$. Neyman-Pearson classes partition the data over the points where the models under H_0 and under H_1 meet, i.e. where $m_{1,\tilde{\theta}^*}(\cdot) = m_{\tilde{\theta}}(\cdot)$ a.s., with $\tilde{\theta}^* = \theta_0^* + o_p(1)$ under H_1 and $\tilde{\theta}$ the corresponding consistent estimator of θ_0 under H_0. By doing so, the difference between the average prediction under the null and the alternative model in each class is the largest possible, making it easier to detect deviations from the null toward H_1.

If X is a univariate variable, finding the intersection set, $\{x \in \mathcal{X} : m_{1,\tilde{\theta}^*}(x) - m_{\tilde{\theta}}(x) = 0\}$, and dividing the data accordingly is a straightforward process, as we show in the first empirical application of Section 8. However, when X is multivariate, the intersection set includes surfaces, and splitting $\mathcal{X}$ across different dimensions becomes significantly more complex. In line with the original proposal of Balakrishnan, Voinov, and Nikulin (2013), rather than dividing the data at these intersections, $\mathcal{X}$ is partitioned into two classes based on whether the difference between the models is greater or less than zero.

$$\hat{\gamma}_1 = \{x \in \mathcal{X} : m_{1,\tilde{\theta}^*}(x) - m_{\tilde{\theta}}(x) > 0\},$$
$$\hat{\gamma}_2 = \{x \in \mathcal{X} : m_{1,\tilde{\theta}^*}(x) - m_{\tilde{\theta}}(x) \leq 0\},$$

Notice that, if the alternative model is correctly specified, i.e. $\mathbb{E}[Y|X] = m_{1,\theta_0^*}(X)$ a.s. for some $\theta_0^* \in \Theta^*$, then $\hat{\gamma}_1$ is the set of points in $\mathcal{X}$ where the null model underestimates the regression function, and $\hat{\gamma}_2$ is the set of points in $\mathcal{X}$ where the null model overestimates the regression function. In this case, χ^2 tests with NP classes detect deviations of the type,

$$H_1(\gamma): \ \mathbb{E}\left[(m_{1,\theta^*}(X) - m_\theta(X))\,\mathbb{I}_{\gamma_l}(X)\right] \neq 0 \text{ for some } l \in \{1,2\} \text{ and for all } \theta \in \Theta.$$

where $\gamma_1 = \{x \in \mathcal{X} : m_{1,\tilde{\theta}^*}(x) - m_\theta(x) > 0\}$ and $\gamma_2 = \{x \in \mathcal{X} : m_{1,\tilde{\theta}^*}(x) - m_\theta(x) \leq 0\}$. That is, if the alternative model is correct, χ^2 tests with Neyman-Pearson classes are capable of detecting any deviations from it, at least asymptotically.

It is convenient to further divide the data into additional cells to enhance the power of the tests and, ensuring $L > d_\theta$, perform the J test. This is done by employing one of the covariate-based procedures discussed above - **PCS**, **FS**, or **KM** - to split each of the

preliminary two-cell partitions into $L/2$ cells.

Algorithm 1 (Parametric Neyman-Pearson (PNP) Algorithm)

(0) Fix a number of cells, L, and a minimum number of observations in each cell, n_{min}.

(1) Compute the fitted values under the null $\{m_{\tilde{\theta}}(X_i)\}_{i=1}^n$.

(2) Compute the fitted values under the alternative $\{m_{1,\tilde{\theta}^*}(X_i)\}_{i=1}^n$.

(3) Split the data into two cells, $\hat{\gamma}_1$ and $\hat{\gamma}_2$, where $\hat{\gamma}_1 = \{x \in \mathcal{X} : m_{1,\tilde{\theta}^*} - m_{\tilde{\theta}}(x) > 0\}$ and $\hat{\gamma}_2 = \{x \in \mathcal{X} : m_{1,\tilde{\theta}^*} - m_{\tilde{\theta}}(x) \leq 0\}$.

(4) If $\sum_{i=1}^N \mathbb{I}_{\hat{\gamma}_1}(X_i) < n_{min}$ or $\sum_{i=1}^N \mathbb{I}_{\hat{\gamma}_2}(X_i) < n_{min}$, merge the two cells.

*(5) If $L > 2$, split the largest and smallest cells into $\lceil L/2 \rceil$ and $\lfloor L/2 \rfloor$ cells, respectively, by applying one of the following methods: **PCS**, **FS**, or **KM**.*

This is not the only solution to determine additional splits. The key point for the NP procedure is that, in each cell, the differences between the two models have the same sign. Once this is granted, the two-cell NP partition can be split with any other procedure.

Example 2 (PNP Classes)

Consider the same scenario as in Example 1, but now under a composite hypothesis, where the model parameters are unknown both under the null hypothesis and under the alternative. After obtaining the OLS estimates for both the null and alternative models, the Neyman-Pearson critical regions are defined based on the points $x \in \mathcal{X}$ where the equation $(\tilde{\theta}_{00} - \tilde{\theta}^*_{00}) + (\tilde{\theta}_{01} - \tilde{\theta}^*_{01})x - \tilde{\theta}^*_{02} \sin(50x/2\pi) = 0$ holds. The resulting partition is depicted in Figure 2.

4.3 Flexible Neyman-Pearson (FNP) Classes

Misspecification of the parametric regression model often arises in the form of a stochastic relationship between the parametric residuals and functions of the covariate vector, such as the model's fitted values. In some cases, indeed, the main focus is on detecting deviations from

$$H_0 : \mathbb{E}\left[\varepsilon_{\theta_0}(Z) | m_{\theta_0}(X)\right] = 0 \text{ a.s. for some } \theta_0 \in \Theta,$$

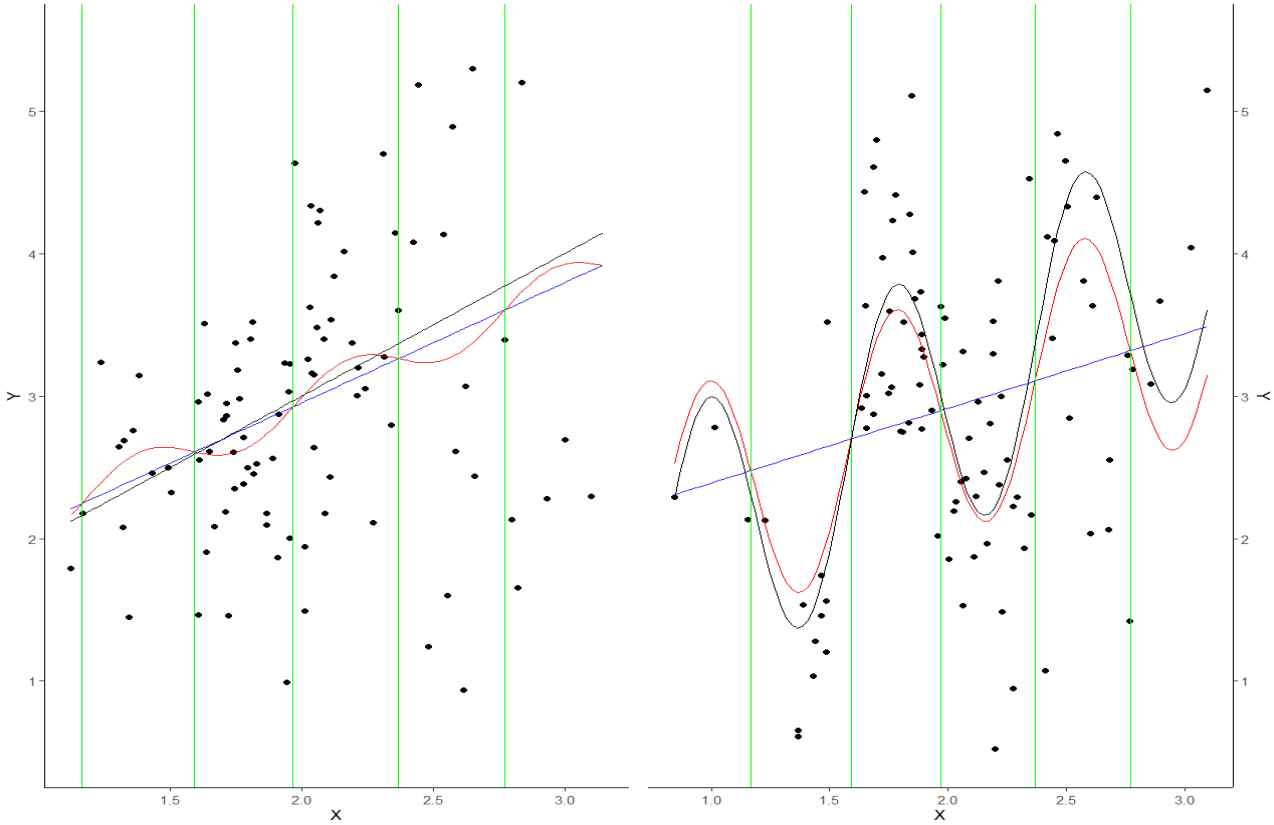


Figure 2: Neyman-Pearson classes with parametric estimate of the regression function. The black lines depict the true regression function under $H_0 : m(x) = 1 + x$ (left panel) and under $H_1 : m(x) = 1 + x + sin(50x/2\pi)$ (right panel). The green lines depict the points where the estimates under the null (blue line) and the estimates under the alternative (red line) meet.

such as when testing propensity score models (Sant'Anna and Song 2019) or generalized linear models (Stute and Zhu 2002). In these situations, it is useful to adopt a structured index model under the alternative hypothesis of the form,

$$H_{1,FNP} : \mathbb{E}\left[\varepsilon_{\theta_0}(Z)|m_{\theta_0}(X)\right] = w(\beta_0, m_{\theta_0}(X)) \text{ a.s. for some } \theta_0 \in \Theta \text{ and } \beta_0 \in B \subset \mathbb{R}^q,$$

where $w(\beta, \cdot)$ represents an unknown smooth function indexed by $\beta \in B$ (see, e.g., Han, Ma, Ren, and Wang 2023, for a similar approach). Consequently, under the null hypothesis, the regression error is independent in expectation from the assumed regression function, i.e. $\mathbb{E}\left[\varepsilon_{\theta_0}(Z)|m_{\theta_0}(X)\right] = 0$. In contrast, under the alternative hypothesis, a systematic dependence emerges. This motivates the adoption of an auxiliary regression

of the parametric residuals on a flexible specification of the fitted values,

$$Y - m_{\tilde{\theta}}(X_i) = w(\beta, m_{\tilde{\theta}}(X_i)) + \epsilon_i, \tag{18}$$

which is then used to split the data where the predicted fits under $H_{1,FNP}$ is greater or smaller than zero. The procedure consists of using the auxiliary model,

$$\mathcal{M}_{FNP} = \{m_{\tilde{\theta}}(X_i) + w(\beta, m_{\tilde{\theta}}(X_i)) : \beta \in \mathbb{B} \subset \mathbb{R}^q\},$$

to split the sample. Let $\tilde{\beta}$ be the estimates of β. Under the null hypothesis, $\tilde{\theta} = \theta_0 + o_p(1)$, and $w(\tilde{\beta}, m_{\tilde{\theta}}(X_i)) = o_p(1)$, implying that the fitted values have no predictive power over the residuals. Under the structured alternative, some of the estimated coefficients converge to a non-zero limit, and the fitted values exhibit predictive power over the residuals. The dependence is then used to split the sample into cells based on the predicted residuals, generating cells with residuals of the same sign and, thus, larger aggregate residuals.

Possible flexible specifications for the auxiliary model include a fixed-order polynomial expansion of the fitted values or a stepwise function,

$$w_{Poly}(\beta, m_{\tilde{\theta}}(X_i)) = \beta_0 + \sum_{j=1}^{q} \beta_j (m_{\tilde{\theta}}(X_i))^j, \tag{19}$$

$$w_{step}(\beta, m_{\tilde{\theta}}(X_i)) = \beta_0 + \sum_{j=1}^{q} \beta_j \mathbb{I}\{m_{\tilde{\theta}}(X_i) > \hat{p}_j\}, \tag{20}$$

where q is the order of the polynomial or the number of steps, respectively, and $\{\hat{p}_j\}_{j=1}^q$ can be fixed or random quantities, such as the quantiles of $\{m_{\tilde{\theta}}(X_i)\}_{i=1}^n$. The constant β_0 can be omitted when the residuals have mean zero to reduce estimation noise. Both $\{w_{poly}(\beta, m_\theta(x)) : \beta \in \mathbb{B} \subset \mathbb{R}^q, \theta \in \Theta\}$ and $\{w_{step}(\beta, m_\theta(x)) : \beta \in \mathbb{B} \subset \mathbb{R}^q, \theta \in \Theta\}$ have finite VC dimension when q is finite (see Example 19.7 in Van der Vaart 2000), and convergence to fixed cells is guaranteed under weak conditions, as discussed in the next subsection.

Tests built using these partitions can be seen as a formalization of the common statistical practice of examining the residual-fitted values scatter plot to detect model misspecification and heteroskedasticity (e.g., Cook 1994). Notably, when $m_\theta(x)$ is linear and

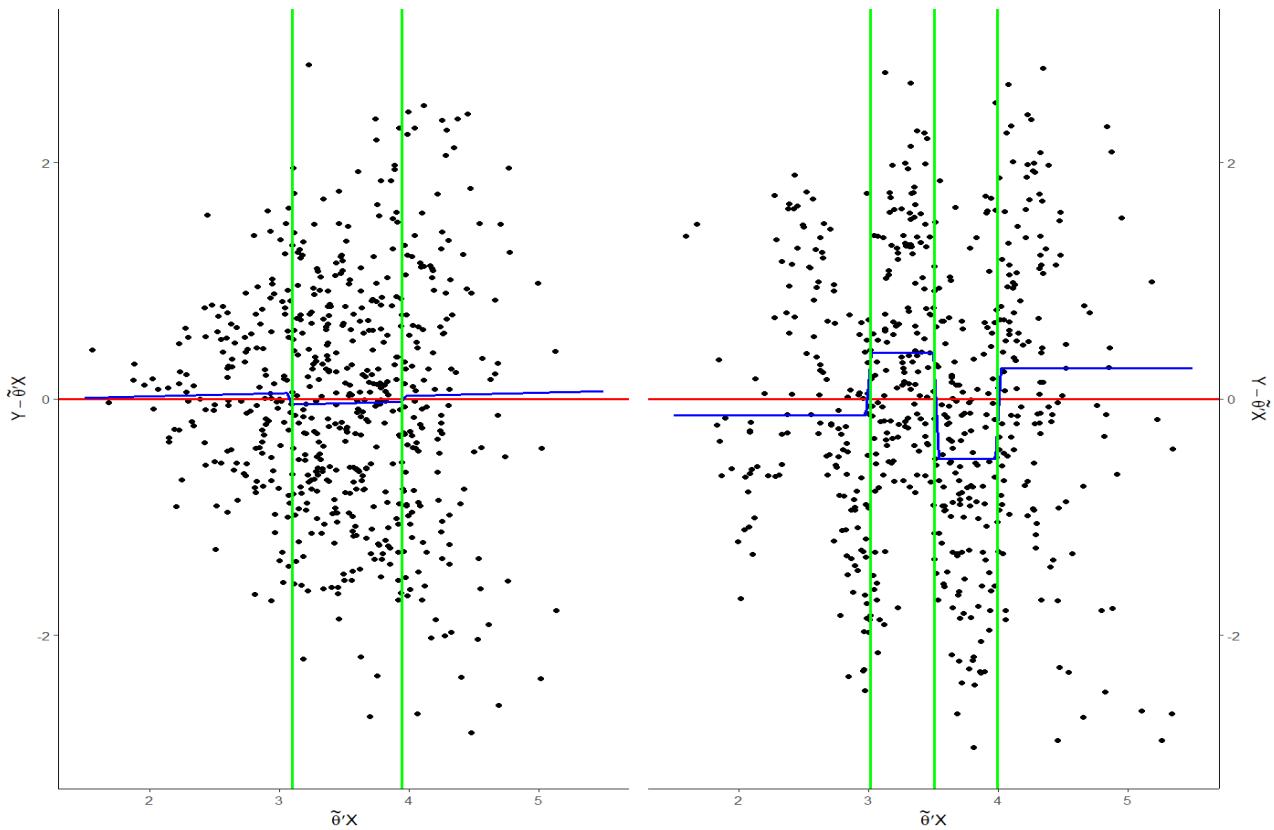


Figure 3: Residual-Fitted values scatter plot from a linear regression. Data generated under $H_0 : m(x) = 1 + \sum_{k=1}^{5} x$ (left panel) and under $H_1 : m(x) = 1 + \sum_{k=1}^{5} x + \sin(50 \sum_{k=1}^{5} x/2\pi)$ (right panel). The blue lines depict the predicted residuals $w_{step}(\tilde{\beta}, \tilde{\theta}'x)$, and the green line depict the points where the predicted residuals cross the zero line (red line).

$w(\beta, \cdot) = w_{poly}(\beta, \cdot)$, the auxiliary regression corresponds to the artificial regression used in the Ramsey (1969) RESET test for assessing the goodness-of-fit of the linear regression model.

Algorithm 2 (Flexible Neyman-Pearson (FNP) Algorithm)

(0) Fix a number of cells, L, and a minimum number of observations in each cell, n_{min}.

(1) Compute the fitted values under the null $\{m_{\tilde{\theta}}(X_i)\}_{i=1}^{n}$ and obtain the residuals, $\{\varepsilon_{\tilde{\theta}}(Z_i)\}_{i=1}^{n}$.

(2) Perform an auxiliary regression of $\varepsilon_{\tilde{\theta}}(Z_i)$ on a flexible specification of the fitted values and obtain the vector of predicted residuals, $\{\hat{\varepsilon}_{\tilde{\theta}}(Z_i)\}_{i=1}^{n}$.

(3) Split the data into two cells, $\hat{\gamma}_1$ and $\hat{\gamma}_2$, where $\hat{\gamma}_1 = \{x \in \mathcal{X} : \hat{\varepsilon}_{\tilde{\theta}}(x) > 0\}$ and

$\hat{\gamma}_2 = \{x \in \mathcal{X} : \hat{\varepsilon}_{\tilde{\theta}}(x) \leq 0\}$.

(4) If $\sum_{i=1}^{N} \mathbb{I}_{\hat{\gamma}_1}(X_i) < n_{min}$ *or* $\sum_{i=1}^{N} \mathbb{I}_{\hat{\gamma}_2}(X_i) < n_{min}$*, merge the two cells.*

(5) If $L > 2$*, split the largest and smallest cells into* $\lceil L/2 \rceil$ *and* $\lfloor L/2 \rfloor$ *cells, respectively, by applying one of the following methods:* ***PCS***, ***FS***, *or* ***KM***.

4.4 Convergence of NP Classes

I examine the conditions under which NP (PNP and FNP) partitions converge to fixed cells in $\mathbb{D}$. The validity of Assumption 4 for NP partitions may not be immediately apparent, considering that the splitting points in this case are determined through the comparison of different estimators of the regression function. However, as discussed below, a large class of NP partitions converges to fixed cells in $\mathbb{D}$.

First, consider the null hypothesis given by $H_0 : m(X) = \theta_0' X$ a.s., for some $\theta_0 \in \mathbb{R}^{d_x}$, whereas the alternative of interest is $H_1 : m(X) = \theta_0^{*\prime} X + \theta_1^* f(X)$ a.s., for some $(\theta_0^*, \theta_1^*) \in \mathbb{R}^{d_x+1}$, with $f : \mathbb{R}^{d_x} \to \mathbb{R}$ being a known function of the regressors. Parametric NP partitions are constructed around the points where the two models fit are equal; that is,

$$x \in \mathcal{X} : x'(\tilde{\theta}_0 - \tilde{\theta}_0^*) = f(x)\tilde{\theta}_1^*,$$

where $\tilde{\theta}_0$ is the OLS estimator of Y on X, and $(\tilde{\theta}_0^*, \tilde{\theta}_1^*)$ is the OLS estimator of Y on $\tilde{X} = (X, f(X))$. This is equivalent to stating that the set of solutions, x_0, satisfies

$$x_0'(\tilde{\theta}_0 - \tilde{\theta}_0^*) = f(x_0)\tilde{\theta}_1^*.$$

Working out the OLS algebra, we obtain the following representations under H_0,

$$\sqrt{n}(\tilde{\theta}_0 - \tilde{\theta}_0^*) = A_n \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \tilde{X}_i e_i,$$
$$\sqrt{n}\tilde{\theta}_1^* = B_n \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \tilde{X}_i e_i,$$

where $A_n = d_n^{-1}\begin{bmatrix} A_{1,n} & A_{2,n} \end{bmatrix}$, $B_n = d_n^{-1}\begin{bmatrix} 1 & A'_{1,n} \end{bmatrix}$, and,

$$A_{1,n} = \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} X_i f(X_i)\right),$$

$$A_{2,n} = -\left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i'\right)\left(\frac{1}{n}\sum_{i=1}^{n} f(X_i) X_i'\right)\left(\frac{1}{n}\sum_{i=1}^{n} X_i f(X_i)\right)\left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i'\right)^{-1},$$
$$d_n = \left(\frac{1}{n}\sum_{i=1}^{n} f(X_i)^2\right) - \left(\frac{1}{n}\sum_{i=1}^{n} f(X_i) X_i'\right)\left(\frac{1}{n}\sum_{i=1}^{n} f(X_i) X_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} X_i f(X_i)\right).$$

Thus, the set of splitting points is given by the solutions of the following equation,

$$(x' A_n - f(x) B_n)\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \tilde{X}_i e_i = 0,$$

which holds for any $x_0 \in \mathcal{X}$ such that $x_0' A_n = f(x_0) B_n$. That is, x_0 converge in probability to the fixed solutions of $x_0' A = f(x_0) B$, where A and B are the probability limits of A_n and B_n, respectively.

The following proposition generalizes this result by incorporating specific conditions on the behavior of the parameter estimators under the null and alternative hypotheses to ensure the convergence of NP classes.

Proposition 1 *Let $\mathcal{M} = \{m_\theta(\cdot) : \theta \in \Theta\}$ and $\mathcal{M}_1 = \{m_{1,\theta^*}(\cdot) : \theta^* \in \Theta^*\}$ represent the models under H_0 and H_1, respectively. Further, denote with $\tilde{\theta}$ and $\tilde{\theta}^*$ the respective $\sqrt{n}$-consistent estimators of θ_0 and θ_0^* under H_0 and H_1, where $m = m_{\theta_0}$ a.s. under H_0, and $m = m_{1,\theta_0^*}$ a.s. under H_1. Then, if under H_0:*

(a) There exists a pseudo-parameter $\bar{\theta}^*$, such that $\tilde{\theta}^* \xrightarrow{p} \bar{\theta}^*$ under H_0.

(b) $m_{\theta_0} = m_{1,\bar{\theta}^*}$ a.s.

(c) $\sqrt{n}(\tilde{\theta} - \theta_0) = Cn^{-1/2}\sum_{i=1}^{n} k(X_i, \varepsilon_{\theta_0}) + o_p(1)$, $\sqrt{n}(\tilde{\theta}^* - \theta_1^*) = Dn^{-1/2}\sum_{i=1}^{n} k(X_i, \varepsilon_{\theta_0}) + o_p(1)$, where C and D are $c \times d_k$ and $d \times d_k$ constant matrices, and $k(\cdot,\cdot)$ is an $\mathbb{R}^{d_k}$-valued function of regressors and errors such that $n^{-1/2}\sum_{i=1}^{n} k(X_i, \varepsilon_{\theta_0}) = O_p(1)$.

Then, the NP partition splitting points converge in probability to fixed points in $\mathcal{X}$.

The most crucial conditions for the convergence to fixed cells are Conditions (a) and (b), which require that the model under the alternative encompasses the model under the null hypothesis. These conditions are typically satisfied in the context of both PNP partitions and FNP partitions. In the case of FNP partitions this is particularly easy to check, as the alternative regression model nests the model under the null. Section B of the appendix provide Monte Carlo evidence demonstrating NP partitions converging to fixed cells in $\mathbb{D}$.

5 Local Power

To analyze the local power of the χ^2 tests under the alternative hypothesis, let's consider the following sequence of local alternatives,

$$H_{1,n} : m(x) = m_{\theta_0}(x) + \frac{1}{\sqrt{n}} h(x) \text{ a.s.}, \tag{21}$$

where $h(X)$ is a random variable representing departures from the null hypothesis with $\mathbb{E}\left[h(X)^2\right] < \infty$ and $0 \leq P(h(X) = 0) < 1$.

Denote as $\delta_{\boldsymbol{\gamma}} = (\delta_1, \ldots, \delta_L)$, with $\delta_l = \mathbb{E}\left[h(X)\mathbb{I}_{\gamma_l}(X)\right]$ for $l = 1, \ldots, L$. Under $H_{1,n}$ and θ_0 known, the test statistics converge to a noncentral chi-squared distribution with L degrees of freedom,

$$\begin{aligned} \hat{\chi}^2_{\hat{\boldsymbol{\gamma}},0} &\xrightarrow{d} \chi^2_L(\lambda), \\ \hat{\mathcal{W}}_{\hat{\boldsymbol{\gamma}}}(\theta_0) &\xrightarrow{d} \chi^2_L(\lambda), \end{aligned} \tag{22}$$

where,

$$\lambda = \delta'_{\boldsymbol{\gamma}} (\Sigma_{\boldsymbol{\gamma},0})^{-1} \delta_{\boldsymbol{\gamma}} \tag{23}$$

is the non-centrality parameter.

Equations (22) and (23) highlight an important trade-off when choosing L. On one hand, the inequality,

$$\frac{\delta_l^2}{\sigma_{0,l}^2} + \frac{\delta_f^2}{\sigma_{0,f}^2} \geq \frac{(\delta_l + \delta_f)^2}{\sigma_{0,l}^2 + \sigma_{0,f}^2},$$

shows that λ is non-decreasing for nested partitions. That is, as the partition becomes

finer with more cells, the test becomes more capable of detecting smaller deviations from the null hypothesis, resulting in increased power. On the other hand, the (global) power of χ^2 tests under the alternative hypothesis $H_{1,n}$ decreases as the number of cells increases. This decline in power is due to the higher variability of the limit distribution in (22) associated with a larger number of cells. As the number of cells grows, the distribution becomes more spread out, leading to a higher chance of observing test statistics falling in less extreme regions and, thus, not rejecting. Similar trade-offs have already been noted in the classical goodness-of-fit testing literature (see, e.g., Kallenberg, Oosterhoff, and Schriever 1985).

When θ_0 is unknown, under $H_{1,n}$, estimators of θ_0 typically follow the asymptotic expansion in Assumption 3' below (see, e.g., Newey 1985).

Assumption 3' *Under $H_{1,n}$, the estimator $\tilde{\theta}$ satisfies,*

$$\sqrt{n}(\tilde{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} l_{\theta_0}(Z_i) + \delta + o_p(1)$$

where the function $l_\theta(\cdot)$ is as defined in Assumption 3 and $\delta \in \mathbb{R}^{d_\theta}$ is a non-zero vector.

It's easy to show, for instance, that Assumption 3' is satisfied by the grouped GMM estimator, with

$$\delta = -\left[\boldsymbol{\mu}^*_{\gamma,0}(\Sigma_{\gamma,0})^{-1}\boldsymbol{\mu}^{*\prime}_{\gamma,0}\right]^{-1}\boldsymbol{\mu}^*_{\gamma,0}(\Sigma_{\gamma,0})^{-1}\delta_\gamma.$$

It follows that under $H_{1,n}$ and the same assumption of Theorem 2,

$$\begin{aligned} \hat{\chi}^2_{\hat{\gamma}} &\xrightarrow{d} \chi^2_{L-d_\theta}(\lambda_1), \\ \hat{\mathcal{W}}_{\hat{\gamma}}(\tilde{\theta}) &\xrightarrow{d} \chi^2_{r(\mathrm{Avar}(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})))}(\lambda_2), \end{aligned} \tag{24}$$

where

$$\lambda_1 = \delta'_\gamma\left[(\Sigma_{\gamma,0})^{-1} - (\Sigma_{\gamma,0})^{-1}\boldsymbol{\mu}^*_{\gamma,0}\left(\boldsymbol{\mu}^{*\prime}_{\gamma,0}(\Sigma_{\gamma,0})^{-1}\boldsymbol{\mu}^*_{\gamma,0}\right)^{-1}\boldsymbol{\mu}^{*\prime}_{\gamma,0}(\Sigma_{\gamma,0})^{-1}\right]\delta_\gamma \tag{25}$$

$$\lambda_2 = \left(\delta_\gamma - \boldsymbol{\mu}^*_{\gamma,0}\delta\right)'\mathrm{Avar}^-(\hat{\boldsymbol{\Phi}}_\gamma(\tilde{\theta}))\left(\delta_\gamma - \boldsymbol{\mu}^*_{\gamma,0}\delta\right) \tag{26}$$

are the respective non-centrality parameters. Comparing the power of these two tests

is not a straightforward task, given the differences in both degrees of freedom and non-centrality parameters in their respective distributions. In fact, scenarios can be constructed where the $\hat{\chi}^2$ test exhibits greater power than the Wald test, and vice versa, reasoning as Moore and Spruill (1975) and Moore (1977), for the classical case.

Instead of the non-centrality parameters, consider now the squared L2 norm of the drift,

$$||\delta_{\boldsymbol{\gamma}}||_2^2 = \delta_{\boldsymbol{\gamma}}'\delta_{\boldsymbol{\gamma}} = \sum_{l=1}^{L} \delta_l^2 = \sum_{l=1}^{L} \mathbb{E}\left[h(X)\mathbb{I}_{\gamma_l}(X)\right]^2.$$

Then, for any pair of δ_l and δ_f such that $\text{sgn}(\delta_l) = \text{sgn}(\delta_f)$, the inequality $(\delta_l+\delta_f)^2 \geq \delta_l^2+\delta_f^2$ shows that the optimal partitioning for $||\delta_{\boldsymbol{\gamma}}||_2^2$ consists of two cells: one containing the points where $h(\cdot) = \sqrt{n}\left(m(\cdot) - m_{\theta_0}(\cdot)\right)$ takes only positive values, and the other where it takes only negative values. Essentially, the optimal partitioning of $||\delta_{\boldsymbol{\gamma}}||_2^2$ is achieved with two Neyman-Pearson classes.

Proposition 2 *The euclidean norm of the drifts, $||\delta_{\boldsymbol{\gamma}}||_2$, is maximized by two Neyman-Pearson classes, $\gamma^* = \{\gamma_i^*\}_{i=1}^2$,*

$$\gamma_1^* = \{x \in \mathcal{X} : h(x) \geq 0\} \quad \gamma_2^* = \{x \in \mathcal{X} : h(x) < 0\}$$

This suggests that, in some sense, Neyman-Pearson classes correspond to optimization criteria for the non-centrality parameter. Of course, a rigorous argument to justify efficiency should account for the dependence of the non-centrality parameter denominators on the cell boundaries.

6 General CMR

The analysis is extended to general moment restrictions with the introduction of a response variables vector, Y, taking values in $\mathcal{Y} \subset \mathbb{R}^{d_y}$, $d_y \geq 1$, and a generalized residual vector (Wooldridge 1990), $\boldsymbol{\varepsilon}_\theta : (\mathcal{Y}, \mathcal{X}) \to \mathbb{R}^{d_\varepsilon}$ with $\boldsymbol{\varepsilon}_\theta(\cdot) = (\varepsilon_{1,\theta}(\cdot), ..., \varepsilon_{d_\varepsilon,\theta}(\cdot))'$, defining parametric relationships between Y and X. The null hypothesis is defined as before, i.e. $H_0 : \mathbb{E}\left[\boldsymbol{\varepsilon}_{\theta_0}|X\right] = 0$ a.s. for some $\theta_0 \in \Theta$. The generality of this framework allows testing for a wide range of econometric models such as simultaneous equation model identified by

instrumental variables (Newey 1990) or nonlinear in parameters and endogenous variables models, e.g. Box-Cox transform.

When the dimension of the generalized residual is bigger than one, it might be optimal to consider a partition for each component of $\boldsymbol{\varepsilon}_\theta(\cdot)$. In particular, for each $j \in \{1, ..., d_\varepsilon\}$, let $\mathbb{D}_j$ be a class of partitions of $\mathcal{X}$ comprised of L_j sets from $\mathbb{C}$ (L_j is fixed for all n); that is,

$$\mathbb{D}_j = \left\{ \boldsymbol{\gamma}_j = (\gamma_{j,1}, ..., \gamma_{j,L_j})' \in \mathbb{C}^{L_j} : \cup_{l=1}^{L_j} \gamma_{j,l} = \mathcal{X},\ \gamma_{j,l} \cap \gamma_{j,f} = \emptyset,\ \forall l \neq f \right\}. \tag{27}$$

The partition corresponding to the j-th component of the generalized residual is an element of $\mathbb{D}_j$, $\boldsymbol{\gamma}_j \in \mathbb{D}_j$, or a random element $\hat{\boldsymbol{\gamma}}_j \in \mathbb{D}_j$, with probability limit $\boldsymbol{\gamma}_j \in \mathbb{D}_j$.

Denote as $\mathcal{E}_\theta(\cdot)$ the $\bar{L} \times \bar{L}$ block diagonal matrix of generalized residuals with main diagonal elements given by $\{\varepsilon_{j,\theta}(\cdot) I_{L_j}\}_{j=1}^{d_\varepsilon}$, where $\bar{L} = \sum_j L_j$. If $L_1 = L_2 = \cdots = L_{d_\varepsilon} = L$, then $\mathcal{E}_\theta(\cdot) = \text{diag}[\boldsymbol{\varepsilon}_\theta(\cdot)] \otimes I_L$, where $\text{diag}[\boldsymbol{\varepsilon}_\theta(\cdot)] = \text{diag}\{\varepsilon_{1,\theta}(\cdot), ..., \varepsilon_{d_\varepsilon,\theta}(\cdot)\}$ is the $d_\varepsilon \times d_\varepsilon$ diagonal matrix with the components of $\boldsymbol{\varepsilon}_\theta(\cdot)$ on the main diagonal and $\otimes$ denotes the Kronecker product. The χ^2 test statistics can be expressed as quadratic forms of

$$\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathcal{E}_\theta(Z_i) \mathbf{I}_{\boldsymbol{\gamma}}(X_i), \tag{28}$$

where $\mathbf{I}_{\boldsymbol{\gamma}}(\cdot) = (\mathbf{I}'_{\boldsymbol{\gamma}_1}, ..., \mathbf{I}'_{\boldsymbol{\gamma}_{d_\varepsilon}})'$ is the vector of indicator functions over all the partitions. While the covariance matrices of $\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\theta_0)$ and $\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})$ under the null are given by,

$$\Sigma_{\boldsymbol{\gamma},0} = \mathbb{E}\left[\mathcal{E}_{\theta_0}(Z) \mathbf{I}_{\boldsymbol{\gamma}}(X) \mathbf{I}_{\boldsymbol{\gamma}}(X)' \mathcal{E}_{\theta_0}(Z)'\right], \tag{29}$$

and,

$$\text{Avar}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right) = \Sigma_{\boldsymbol{\gamma},0} - \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} C'_{\boldsymbol{\gamma},0} - C_{\boldsymbol{\gamma},0} \boldsymbol{\mu}^{*'}_{\boldsymbol{\gamma},0} + \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} L_0 \boldsymbol{\mu}^{*'}_{\boldsymbol{\gamma},0}, \tag{30}$$

where $\boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} = \mathbb{E}\left[\nabla \mathcal{E}_{\theta_0}(Z) \mathbf{I}_{\boldsymbol{\gamma}}(X)\right]$ is the Jacobian matrix, $C'_{\boldsymbol{\gamma},0} = \mathbb{E}\left[\mathcal{E}_{\theta_0}(Z) \mathbf{I}_{\boldsymbol{\gamma}}(X) l_{\theta_0}(Z)'\right]$, and L_0 is defined as before. In general, $\Sigma_{\boldsymbol{\gamma},0}$ is not diagonal, unless $\boldsymbol{\varepsilon}_\theta(\cdot)$ consists of orthogonal components or $d_\varepsilon = 1$. However, when $\boldsymbol{\gamma}_1 = \boldsymbol{\gamma}_2 = \ldots = \boldsymbol{\gamma}_{d_\varepsilon}$, $\Sigma_{\boldsymbol{\gamma},0}$ is block diagonal with L blocks of size $d_\varepsilon \times d_\varepsilon$ corresponding to the covariance matrix of $\boldsymbol{\varepsilon}_\theta(\cdot)$ in each cell of the partition.

7 Monte Carlo Study

This section presents Monte Carlo studies to evaluate the finite sample performance of χ^2 tests for testing the null hypothesis of linearity within a regression model, i.e., $H_0 : m \in \{\theta' x : \theta \in \Theta\}$. The analysis is divided in two parts: the first replicates the experiments of Escanciano (2006) in low-dimensional settings, and the second explores the performance of the tests against high-frequency alternatives and with large covariate dimensions. To establish a benchmark for comparison with existing tests for H_0, the χ^2 tests are compared with two omnibus tests based on marked residual processes proposed by Stute (1997) and Escanciano (2006), and two directional tests, namely the Stute and Zhu (2002) test for generalized linear models and the unfeasible parametric t-test using the full information about the data generating process (DGP).

In the first block, the tests are evaluated in the context of the following DGPs:

$$
\begin{aligned}
&\textbf{DGP1: } Y_i = 1 + X_{1,i} + X_{2,i} + \epsilon_i \equiv X_i'\alpha_0 + \epsilon_i, \\
&\textbf{DGP1-E: } Y_i = X_i'\alpha_0 + v_i, \\
&\textbf{DGP2: } Y_i = X_i'\alpha_0 + 0.1(W_{1,i} - \pi)(W_{w,i} - \pi) + \epsilon_i, \\
&\textbf{DGP3: } Y_i = X_i'\alpha_0 + X_i'\alpha_0 \exp\{-0.01(X_i'\alpha_0)^2\} + \epsilon_i, \\
&\textbf{DGP4: } Y_i = X_i'\alpha_0 + \cos(0.6\pi X_i'\alpha_0) + \epsilon_i,
\end{aligned}
$$

where $X_{i,1} = (W_i + W_{1,i})/2$, $X_{i,2} = (W_i - W_{2,i})/2$, W_i, $W_{1,i}$, and $W_{2,i}$ are i.i.d. and distributed uniformly on $[0, 2\pi]$. The error terms ϵ_i and v_i are i.i.d. with $\epsilon_i \sim N(0,1)$, and $v_i \sim (Exp(1)-1)$, independent of X. The models under the null correspond to **DGP1** and **DGP1-E**, with the latter intended to demonstrate the robustness of the tests against fat-tailed error distributions. **DGP2** is as considered in Hong and White (1995), while **DGP3** is a slight modification of their DGP3. **DGP4** involves high-frequency deviations from the linear model, similar to those considered in Eubank and Hart (1992). Following Escanciano (2006), in the first block of models, I consider sample sizes $n \in \{50, 100, 300\}$.

In the second block, the DGP is defined as follows,

$$Y_i = \sum_{j=1}^{d_x} X_{j,i} + b \sin\left(\frac{c\sum_{j=1}^{d_x} X_{j,i}}{2\pi}\right) + \sigma(X_i)\epsilon_i \tag{31}$$

where $X_i = (X_{1,i}, ..., X_{d_x,i})$ is a vector of mutually independent covariates uniformly distributed over $[0,1]$, $X_{i,j} \sim U[0,1]$. The error term is normally distributed and independent of X, $\epsilon|X \sim N(0,1)$. The heteroskedasticity function is given by,

$$\sigma^2(X) = \frac{g(X)}{\mathbb{E}[g(X,a)]},$$

with $g(X,a) = e^{aX_1}$, and $\mathbb{E}[\sigma^2(X)] = 1$. The model under the null corresponds to $b = 0$, while for the alternative models $b = 0.5$. The parameter a controls the heteroskedasticity severity, with $a = 0$ corresponding to homoskedasticity, while c governs the extent of departures from linearity. I present evidence regarding the test size for $a \in \{0, 3\}$. Subsequently, with $a = 3$ fixed, I examine the tests power for $c \in \{10, 50\}$. Notably, when $c = 50$, deviations occurr at higher frequencies, making them hard to distinguish from sampling error. The generated samples have size $n \in \{100, 200, 500, 1000\}$ with the covariate vector dimension $d_x \in \{5, 10\}$.

The $\hat{\chi}^2$ test is built as described in Section 2 using the GMM estimator,

$$\hat{\theta}_{\boldsymbol{\gamma}} = \left[\sum_{l=1}^{L} \frac{\bar{X}_l \bar{X}_l'}{\hat{\sigma}_l^2(\tilde{\theta})}\right]^{-1} \sum_{l=1}^{L} \frac{\bar{X}_l \bar{Y}_l}{\hat{\sigma}_l^2(\tilde{\theta})}.$$

where $\tilde{\theta}$ are initial OLS estimates of θ_0. The Wald test is implemented using the Moore-Penrose inverse of the plug-in estimator $\hat{W}_{\boldsymbol{\gamma}}(\tilde{\theta})$ of Avar $\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$,

$$\begin{aligned}\hat{W}_{\boldsymbol{\gamma}}(\tilde{\theta}) = {} & \mathbb{E}_n\left[\varepsilon_{\tilde{\theta}}(Z_i)^2 \mathbf{I}_{\boldsymbol{\gamma}}(X_i)\mathbf{I}_{\boldsymbol{\gamma}}(X_i)'\right] - \mathbb{E}_n\left[\mathbf{I}_{\boldsymbol{\gamma}}(X_i)X_i'\right]\mathbb{E}_n\left[X_iX_i'\right]^{-1}\mathbb{E}_n\left[\varepsilon_{\tilde{\theta}}(Z_i)^2 X_i \mathbf{I}_{\boldsymbol{\gamma}}(X_i)'\right] - \\ & - \mathbb{E}_n\left[\varepsilon_{\tilde{\theta}}(Z_i)^2 \mathbf{I}_{\boldsymbol{\gamma}}(X_i)X_i'\right]\mathbb{E}_n\left[X_iX_i'\right]^{-1}\mathbb{E}_n\left[X_i\mathbf{I}_{\boldsymbol{\gamma}}(X_i)'\right] + \\ & + \mathbb{E}_n\left[\mathbf{I}_{\boldsymbol{\gamma}}(X_i)X_i'\right]\mathbb{E}_n\left[X_iX_i'\right]^{-1}\mathbb{E}_n\left[\varepsilon_{\tilde{\theta}}(Z_i)^2 X_iX_i'\right]\mathbb{E}_n\left[X_iX_i'\right]^{-1}\mathbb{E}_n\left[X_i\mathbf{I}_{\boldsymbol{\gamma}}(X_i)'\right],\end{aligned}$$

even when the regression error is homoskedastic, as in the first block, or in the second block with $a = 0$.

I assess the test's performance using the model-based partitioning methods described in Section 4. For both Parametric NP (**PNP**) and Flexible NP (**FNP**) classes, the division into L cells from the initial two-cell partition is performed using SEB on the vector of fitted values (**FS**), and K-means clustering (**KM**). Section C of the Appendix also reports results using SEB on the first principal component (**PCS**). In the first block of simulations, I consider partitions with $L \in \{2, 4, 6\}$ for the Wald test, and $L^* \in \{5, 7, 9\}$ for the J test. Notice that $L^* = L + d_\theta$ is selected to ensure that the limit null distributions of the two χ^2 tests are the same, enabling a fair comparison between them. In the second block, when $d_x = 5$, the partitioning is done using $L \in \{4, 6, 8\}$ cells for the Wald test, and $L^* \in \{9, 11, 13\}$ for the J test. For $d_x = 10$, $L \in \{8, 10, 12\}$ and $L^* \in \{18, 20, 22\}$. In all the partitioning algorithm, I set the minimum number of observations in each cell of the preliminary two-cell partition equal to $n_{min} = 5L$.

I build the **FNP** partition using predictions from a regression of parametric residuals on either a polynomial expansion of the fitted values, $w_{poly}(\beta, \cdot)$, or a stepwise function of the fitted values, $w_{step}(\beta, \cdot)$. The polynomial's order is set to two in the first block and three in the second, while the step function is estimated using $(\hat{p_1}, \hat{p_2})$ equal to the first and third quartile of the fitted values distribution.

The parametric **PNP** cells are constructed using the correctly specified alternative model up to linear coefficients. For example, under **DGP2** in the first block, the **PNP** cells are generated by comparing $X_i'\hat{\alpha}$ with $X_i'\tilde{\alpha}_1 + \tilde{\alpha}_2(W_{1,i} - \pi)(W_{2,i} - \pi)$, where $\hat{\alpha}$ is the OLS estimator of α_0, and $\tilde{\alpha}_1$ and $\tilde{\alpha}_2$ are the OLS estimates from a regression of Y on $(X_i, (W_{1,i} - \pi)(W_{2,i} - \pi))'$. Under the null hypothesis models, the **PNP** cells are constructed specifically against **DGP2** in the first block and against a trigonometric deviation with $c = 50$ in the second block.

In the first block, where X include a constant, I compare the Wald test with the quantile of a χ^2_{L-1} distribution. In the second block, where X does not include a constant, the rejection region of the Wald test is derived using a χ^2_L distribution. The rejection region for the J test is always evaluated using the quantile of a $\chi^2_{L-d_\theta}$ distribution.

The performance of the χ^2 tests is compared with three minimum-distance tests for regression specifications based on marked residuals processes indexed by real vectors, real numbers, and linear projections; cf. Stute (1997), Stute and Zhu (2002), and Escanciano

(2006), respectively.

$$R_{1,n}(x_1) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(Y_i - X_i'\tilde{\theta}\right) \mathbb{I}\{X_i \leq x_1\},\ x_1 \in \mathbb{R}^{d_x}$$
$$R_{2,n}(x_2) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(Y_i - X_i'\tilde{\theta}\right) \mathbb{I}\{X_i'\tilde{\theta} \leq x_2\},\ x_1 \in \mathbb{R},$$
$$R_{3,n}(\beta, x_3) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(Y_i - X_i'\tilde{\theta}\right) \mathbb{I}\{X_i'\beta \leq x_3\},\ (\beta, x_3) \in \Pi \equiv \mathbb{S}_{d_x} \times \mathbb{R},$$

with $\mathbb{S}_{d_x} = \{\beta = (\beta_1, ..., \beta_{d_x}) \in \mathbb{R}^{d_x} : \sum_{k=1}^{d_x} |\beta_k| = 1\}$. Incidentally, the marked residual process indexed by partitions generated with the **FS** method,

$$\hat{\mathbf{\Phi}}_{\hat{\gamma}_l}(\tilde{\theta}) = n^{-1/2} \sum_{i=1}^{n} \varepsilon_{\tilde{\theta}}(Z_i) \mathbb{I}\{X_i'\tilde{\theta} \in C_l\},$$

where $\hat{\gamma}_l = \{x \in \mathcal{X} : x'\tilde{\theta} \in C_l\}$ and $\bigcup_{l=1}^{L} C_l = \mathbb{R}$, is a finite cells version of the process used in the Stute and Zhu (2002) test. The test statistics consist of a functional of $R_{1,n}(\cdot)$, $R_{2,n}(\cdot)$, and $R_{3,n}(\cdot,\cdot)$. We consider the Kolmogorov-Smirnov functional of the Stute (1997) and Stute and Zhu (2002) tests, and the Cramer-von Mises functional of the Escanciano (2006) test:

$$KS1 = \sup_{x_1 \in \mathbb{R}^{d_x}} |R_{1,n}(x_1)|,$$
$$KS2 = \sup_{x_2 \in \mathbb{R}} |R_{2,n}(x_2)|,$$
$$CvM1 = \int_{\Pi} (R_{3,n}(\beta, x_3))^2 F_{n,\beta}(dx_3) d\beta,$$

with $F_{n,\beta}$ being the empirical distribution function of $\{X_i'\beta\}$ and d_β the uniform density on the unit sphere. The KS1 and CvM1 tests are omnibus tests, while the KS2 test is a directional test designed for generalized linear models, such as the ones considered in this simulations. Since these tests have non-pivotal limiting distribution, the critical values are obtained with a Wild bootstrap procedure (Stute, Manteiga, and Quindimil 1998). It is worth remarking that the Stute and Zhu (2002) test can be implemented by estimating an innovation process transformation of $R_{3,n}(\cdot)$, which is not considered here. In this case, the test is asymptotically pivotal, but the implementation of the test

statistics relies on Nadaraya-Watson estimates. The finite sample performance of the Stute and Zhu (2002) test with this implementation are already explored in Escanciano (2006) using **DGP1-DGP4** above, and in Stute and Zhu (2002) in nonlinear settings. Furthermore, I provide the rejection rates of a t-test for the null of a correctly specified linear model (e.g., $H_0 : b = 0$, in the second block of simulations). Both the t-test and the χ^2 tests with **PNP** partitions are infeasible in practice, as they require knowledge of the true DGP. Including them, however, allows for a comprehensive comparison with the other feasible testing methods.

I consider $R = 3000$ replications for the χ^2 tests and the t-test. The rejection rates of the integrated tests, KS1, KS2, and CvM1, are based on $R^* = 1000$ replications for each sample size and $B = 500$ bootstrap repetitions in the first block and $B = 999$ repetitions in the second. These values are chosen to ensure that the results of the first block of simulations are directly comparable with the Monte Carlo study in Escanciano (2006). The rejection rates are reported at the 5% nominal level, results at 1% and 10% are similar. The simulation results for the first block are presented in Table 1, while the results for the second block are shown in Tables 2 (size) and 3 (power).

In the first block, when $n = 50$ and with many cells, χ^2 tests tend to be undersized. However, with $n = 100$, the estimated rejection rates under the null hypothesis are fairly close to the nominal level, in both **DGP1** and **DGP1-E**. It is noteworthy that, in small samples, the rejection rates of χ^2 tests using w_{poly} and w_{step} are very similar, if not identical. This is because in most cases, the dimension of each cell in the preliminary two-cell partition falls below $n_{min} = 5L$, meaning the division into cells is predominantly driven by covariate-based partitioning. As expected, the unfeasible t-test and the χ^2 tests with the **PNP** partition show the highest rejection rates across the alternative models and for all sample sizes. In contrast, the comparison of feasible methods presents a mixed picture in terms of test performance. With large samples, the CvM1 test demonstrates the highest power under the alternative DGPs, followed by the J test using the **KM** and **PCS** partitioning (discussed in the appendix). The alternative model **DGP3** is detected by all tests in large samples, while the high-frequency alternative **DGP4** is captured by all tests except the Stute (1997) test, which exhibits low power against this type of alternative.

Table 1: First Block

				FS						**KM**									
				w_{poly}		w_{step}		**PNP**		w_{poly}		w_{step}		**PNP**					
n	DGP	L	L^*	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	KS1	KS2	CvM1	T
50	1	4	7	5.1	3.9	5.2	3.9	4.3	3.9	4.8	3.3	4.7	3.4	4.8	3.3	4.5	6.1	4.2	5.4
		6	9	3.6	2.7	3.6	2.7	3.6	2.7	3.2	2.5	3.3	2.4	3.3	2.5				
		8	11	2.4	1.5	2.4	1.5	2.4	1.5	2.4	1.7	2.5	1.7	2.5	1.7				
50	1-E	4	7	4	3.2	3.9	3.2	3.4	3.2	4.1	2.5	4.1	2.5	3.4	2.5	4.5	3.8	3	3.9
		6	9	2.3	1.6	2.3	1.6	2.3	1.6	2.5	1.6	2.5	1.7	2.5	1.6				
		8	11	1.5	1	1.5	1	1.5	1	1.8	1.3	1.8	1.3	1.7	1.2				
50	2	4	7	5.6	4.7	5.7	4.7	22.2	4.7	7.5	8	7.5	8.1	23	8	11.1	6.2	10.1	56.5
		6	9	4.3	3.5	4.3	3.5	4.3	3.5	6.5	6.4	6.5	6.3	6.5	6.2				
		8	11	2.6	1.9	2.6	1.9	2.6	1.9	5.9	3.6	5.9	3.5	5.8	3.4				
50	3	4	7	31.5	30.2	31.6	30.2	32.4	30.2	30.9	29.6	30.1	29.7	32.7	29.6	31.8	49.5	56.8	79.2
		6	9	28.4	20.5	28.4	20.5	28.4	20.5	22.6	18.4	22.6	18.5	22.6	18.6				
		8	11	18.8	14.1	18.8	14.1	18.8	14.1	15.1	11.5	15	11.8	15.1	11.5				
50	4	4	7	40.5	42.4	40.4	42.4	90.8	42.4	21.8	19.1	21.8	19.2	90.2	19.3	4.3	33.6	12.4	99.6
		6	9	44.5	38	44.5	38	44.5	38	19	22.3	19	22.4	18.8	22				
		8	11	42.2	29.7	42.2	29.7	42.2	29.7	20	23	20	23.3	19.9	23.1				
100	1	4	7	4.8	4	4.9	4.5	5.8	4.3	4.7	4.9	4.6	4.7	5	4.4	5.4	5.9	4.5	5.2
		6	9	4.9	3.9	4.7	4	4.7	4.3	5	3.3	4.9	3.4	4.7	4.1				
		8	11	3.8	3.3	3.8	3.3	3.8	3.3	3.5	3.4	3.5	3.7	3.4	3.7				
100	1-E	4	7	5.8	5.09	4.7	5.4	4.6	4.6	4.6	4.7	4.9	4.6	4.5	3.8	3.7	5.4	3.5	4.3
		6	9	4.8	4	4.3	4.1	4.1	4.4	3.5	4.2	4.1	4	4.3	4.2				
		8	11	3.8	3.1	3.8	3.1	2.7	3.1	2.6	2.7	2.7	3.1	2.8	2.7				
100	2	4	7	4.8	6.1	5.5	6.3	48.4	37.5	6.2	12.8	7	13.5	47.7	43.4	17.5	5.5	25.7	87.2
		6	9	4.5	5.7	4.8	5.7	35.8	23	9.5	19.2	11.3	18.9	36.2	28.3				
		8	11	5.3	4.9	5.3	4.9	28.2	4.9	14.4	17.2	14.2	16.8	28.9	17.3				
100	3	4	7	56.3	66	68.5	67.4	78.4	71.5	55.2	58.6	57.8	70.7	77.8	72	62.9	81	87.4	97.9
		6	9	66.2	61.5	69.6	61.5	74.5	61.6	50	59.8	62.7	60	68.5	59.9				
		8	11	61.8	55.8	62	55.8	62.2	55.8	55.1	51.8	55.2	52	55.3	52				
100	4	4	7	60	81.7	64.6	82.7	99.9	98.9	59.2	36.1	49.8	50.8	99.9	99.9	5.8	75.2	27	100
		6	9	92.2	88.7	88	88.9	99.7	98.2	40.2	52.7	35.5	53.3	99.7	92.3				
		8	11	94.7	91.8	94.7	91.8	99.7	91.8	47.7	63.5	47.6	62.8	99.4	63.1				
300	1	4	7	4.7	4.9	4.7	4.5	5.5	4.3	4.2	5.3	4.2	5.5	5.4	5.2	5.4	5.4	4.5	4.9
		6	9	4.5	4.5	4.1	4.8	4.4	4.9	4.4	4.9	5.2	4	5.6	5.2				
		8	11	4.1	4.2	4.4	4.1	4.3	4.6	5.09	4.8	5	4.3	4.3	5.2				
300	1-E	4	7	5.2	5.2	4.5	4.8	4.8	5.6	4.8	5.7	4.6	5.3	5.7	5.3	4.7	4.8	3.9	4.7
		6	9	6.4	6.3	5.2	5.5	6	5.6	4.8	5.9	5.5	5.7	5.3	5.6				
		8	11	6	5.3	5.9	5.8	5.7	5.6	5.7	5.6	5.9	6.4	5	6.6				
300	2	4	7	5.7	9.9	9.19	11.8	95.3	88.2	6.3	31.5	8	20.6	95	92.7	49.8	9.1	81.6	99.9
		6	9	8.6	10	11.1	11.9	91.2	87.2	27.5	45	29.6	41.9	90.9	90.5				
		8	11	8.6	10.2	9.7	11.3	87.2	82.5	32.8	52.9	38.4	54.1	88.5	85.7				
300	3	4	7	98.1	97.2	99.7	97.9	99.9	99.3	98.3	97.7	97.4	99.9	99.9	99.8	99.8	100	100	100
		6	9	99.7	99	99.8	98.9	99.9	99.8	97.2	98.9	99.8	99.5	99.6	99.5				
		8	11	99.7	99.1	99.9	99.2	99.9	99.7	95.9	99.5	99.3	99.7	99.4	99.5				
300	4	4	7	96.5	98.4	95.8	98.7	100	99.9	98.5	87.8	93.7	88.9	100	100	24.4	100	98.8	100
		6	9	100	99.8	99.9	99.7	100	100	98.2	96.2	70.8	94.2	100	100				
		8	11	100	99.9	100	99.9	100	100	98.2	97.7	75	96.7	100	100				

Percentage of rejections at the nominal level $\alpha = 0.05$ of the $\hat{\chi}^2$ and the Wald test under different partitioning methods. n, DGP, L, and L^* denote the sample size, the DGP, and the number of cells used for Wald and J tests, respectively. The last four columns report the rejection rates of the Kolmogorov-Smirnov tests of Stute (1997) (KS1) and Stute and Zhu (2002) (KS2), the Cramer-Von-Mises test of Escanciano (2006) (CvM1), and the T-test.

In the second block, χ^2 tests generally exhibit excellent size accuracy, with only minor distortions observed for small sample sizes and when there are many cells. The KS1 test, however, shows pronounced size distortions when $d_x = 10$, even for large samples. In contrast, the KS2 and CvM1 tests demonstrate greater robustness in higher-dimensional covariate settings. As in the first block, the χ^2 tests using **PNP** partitions and the

Table 2: Second Block: Size ($b = 0$)

					FS						KM									
					w_{step}		w_{poly}		PNP		w_{step}		w_{poly}		PNP					
n	d_x	a	L	L^*	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	KS1	KS2	CvM1	T
100	5	0	4	9	4.9	4.8	4.7	5.1	4.7	4.9	6.3	4.2	5.9	4.5	4.5	4.8	5.1	6.4	6.8	5.7
			6	11	4.7	4.3	4.8	4.3	4.8	4.3	5.1	4.0	5.4	3.9	4.3	4.0				
			8	13	4.0	4.3	4.3	4.3	4.3	4.3	4.8	3.5	5.0	3.9	4.6	3.8				
100	5	3	4	9	5.5	4.8	4.8	4.7	4.8	4.5	5.5	4.9	5.3	5.1	4.5	5.0	4.8	7.1	4.9	4.4
			6	11	4.1	4.2	4.2	4.2	4.2	4.2	4.7	4.3	4.5	4.1	4.2	4.3				
			8	13	3.5	2.8	3.7	2.8	3.7	2.8	4.3	3.5	4.3	3.2	4.5	3.5				
100	10	0	8	18	5.9	4.5	5.7	4.5	5.7	4.5	5.2	4.7	5.7	4.6	5.5	4.4	1.4	8.1	5.4	5.6
			10	20	5.4	4.5	5.4	4.5	5.4	4.5	4.4	4.6	4.8	4.1	4.7	4.1				
			12	22	4.2	4.3	4.2	4.3	4.2	4.3	4.2	3.6	4.0	3.7	4.3	3.8				
100	10	3	8	18	4.6	4.5	4.3	4.5	4.3	4.5	5.0	4.0	4.4	3.9	5.1	4.3	0.7	9.1	4.9	5.6
			10	20	4.5	4.3	4.5	4.3	4.5	4.3	4.0	3.6	4.5	3.5	4.5	3.3				
			12	22	3.5	2.8	3.5	2.8	3.5	2.8	3.7	2.3	4.6	2.6	3.8	2.8				
200	5	0	4	9	4.6	5.6	3.9	5.3	3.9	5.2	4.5	6.8	5.0	5.5	4.3	4.6	5.5	6.7	4.9	5.5
			6	11	4.6	5.0	4.5	5.1	4.5	5.1	5.7	5.9	5.3	4.9	5.3	4.3				
			8	13	4.8	4.8	4.3	4.6	4.3	4.7	4.4	5.0	5.1	4.5	4.5	4.6				
200	5	3	4	9	4.1	4.7	4.5	4.6	4.5	4.9	5.3	6.1	5.5	6.0	4.1	4.2	5.5	6.5	4.6	5.7
			6	11	4.8	4.4	4.3	4.5	4.3	4.6	4.8	5.1	4.6	5.3	3.9	4.1				
			8	13	4.9	4.4	3.6	5.0	3.6	4.1	4.3	4.7	4.5	4.6	3.7	4.1				
200	10	0	8	18	5.1	5.1	5.0	4.9	5.0	5.1	5.8	5.0	5.3	5.8	5.3	5.3	2.8	6.6	5.3	5.2
			10	20	4.5	4.8	4.9	4.8	4.9	4.8	5.8	5.0	6.0	4.2	5.5	4.6				
			12	22	5.0	4.3	4.7	4.3	4.7	4.3	5.4	4.3	5.5	3.9	4.9	4.5				
200	10	3	8	18	4.6	4.8	4.5	4.7	4.5	4.8	5.7	4.8	5.4	4.6	4.9	4.5	1.4	6.0	5.6	4.4
			10	20	4.3	4.3	4.5	4.3	4.5	4.3	5.1	5.0	4.8	3.8	4.8	4.0				
			12	22	4.0	4.3	4.3	4.3	4.3	4.3	5.1	4.8	5.0	4.4	3.7	4.6				
500	5	0	4	9	5.1	4.7	4.8	5.2	4.8	4.2	6.0	5.8	5.3	6.0	4.7	4.2	6.4	7.2	5.4	6.2
			6	11	4.7	5.3	5.3	5.7	5.3	6.1	5.5	6.8	6.0	5.1	4.7	4.8				
			8	13	5.2	5.0	4.7	4.9	4.7	4.8	5.7	5.6	5.4	5.4	4.8	5.2				
500	5	3	4	9	5.2	4.5	5.8	5.1	5.8	4.7	5.0	6.7	5.6	5.5	5.7	4.9	5.7	5.8	5.8	5.3
			6	11	4.5	5.1	5.4	5.3	5.4	4.8	5.7	6.0	5.7	4.9	5.0	4.5				
			8	13	3.7	5.3	4.2	4.7	4.2	4.8	4.9	5.0	4.3	5.0	4.7	4.3				
500	10	0	8	18	4.3	4.6	4.9	4.6	4.9	5.1	5.1	6.5	5.1	5.2	5.0	4.1	4.3	6.3	6.3	5.3
			10	20	5.1	4.9	5.7	5.0	5.7	4.7	5.1	6.0	5.6	5.6	5.9	4.9				
			12	22	4.6	4.8	4.6	4.7	4.6	4.7	4.8	5.8	5.2	5.2	4.5	4.5				
500	10	3	8	18	5.1	4.5	5.4	4.7	5.4	4.9	4.9	6.2	5.4	5.2	5.2	4.7	3.2	5.6	6.2	3.8
			10	20	4.3	4.6	4.7	4.4	4.7	4.3	5.6	5.2	5.0	5.2	5.3	4.5				
			12	22	4.4	4.4	4.5	4.7	4.5	4.7	4.8	5.2	4.7	5.4	4.6	4.2				
1000	5	0	4	9	4.9	5.1	5.4	5.5	5.4	5.1	5.9	7.1	5.6	6.4	5.6	5.0	3.4	5.9	4.7	4.0
			6	11	5.3	5.7	4.5	5.4	4.5	4.4	5.1	6.1	5.1	4.9	4.6	4.9				
			8	13	4.6	4.7	4.9	4.7	4.9	4.2	5.7	5.4	5.2	5.1	4.5	3.8				
1000	5	3	4	9	4.7	5.2	5.4	5.1	5.4	5.2	5.5	7.5	5.2	6.5	4.7	4.9	3.5	7.1	4.4	4.7
			6	11	4.5	5.4	5.0	5.6	5.0	4.6	5.3	6.0	5.4	5.5	4.5	4.5				
			8	13	5.3	4.9	5.0	4.7	5.0	4.9	5.2	6.0	5.4	5.0	5.0	4.7				
1000	10	0	8	18	5.0	5.2	4.5	5.2	4.5	4.6	6.0	6.9	5.9	5.5	5.8	4.4	6.3	4.8	4.9	5.9
			10	20	4.8	4.8	4.2	4.6	4.2	5.3	5.6	6.6	5.0	5.6	5.5	5.3				
			12	22	5.4	4.8	5.1	4.5	5.1	4.5	5.0	5.6	5.2	4.8	5.3	4.7				
1000	10	3	8	18	4.5	4.4	5.0	5.0	5.0	5.4	5.7	6.4	5.1	5.8	4.9	4.4	6.2	5.6	5.0	4.4
			10	20	4.9	4.5	4.8	5.0	4.8	5.3	5.2	5.6	5.2	6.0	5.2	4.2				
			12	22	4.3	4.3	5.0	4.9	5.0	3.7	4.9	5.5	5.2	5.5	4.1	4.3				

Percentage of rejections at the nominal level $\alpha = 0.05$ of the $\hat{\chi}^2$ and the Wald test under different partitioning methods. n, d_x, a, L, and L^* denote the sample size, the number of covariates, the degree of heteroskedasticity, and the number of cells of the Wald and J test, respectively. The last four columns report the rejection rates of the Kolmogorov-Smirnov tests of Stute (1997) (KS1) and Stute and Zhu (2002) (KS2), the Cramer-Von-Mises test of Escanciano (2006) (CvM1), and the T-test on $b = 0$.

Table 3: Second Block: Power ($b = 0.5$)

					FS						KM									
					w_{step}		w_{poly}		PNP		w_{step}		w_{poly}		PNP					
n	d_x	c	L	L^*	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	KS1	KS2	CvM1	T
100	5	10	4	9	39.6	19.6	47.0	19.6	47.0	24.8	39.3	20.6	43.9	25.3	41.9	26.2	31.5	35.8	34.1	47.4
			6	11	37.1	18.4	40.3	18.4	40.3	18.4	27.3	19.1	37.0	19.8	36.1	18.8				
			8	13	30.9	16.3	33.5	16.3	33.5	16.3	20.1	18.2	26.6	17.4	29.5	17.9				
100	5	50	4	9	7.1	7.2	73.1	7.1	73.1	46.4	6.5	4.1	7.3	4.5	73.6	61.6	5.2	8.0	4.10	93.7
			6	11	7.6	7.7	63.9	7.7	63.9	7.7	5.9	4.7	5.7	4.7	68.5	4.3				
			8	13	6.3	6.5	57.0	6.5	57.0	6.5	3.7	3.8	3.9	3.7	61.6	3.5				
100	10	10	8	18	28.7	15.6	35.2	15.6	35.2	15.6	8.5	7.7	9.0	7.3	24.5	7.6	1.6	28.2	10.1	85.8
			10	20	24.7	14.6	24.7	14.6	24.7	14.6	7.6	7.1	7.6	6.7	7.6	6.5				
			12	22	21.6	13.0	21.6	13.0	21.6	13.0	7.0	5.7	7.1	6.0	6.6	5.6				
100	10	50	8	18	4.5	4.5	54.0	4.5	54.0	4.5	5.3	4.6	6.1	4.7	57.3	4.3	0.7	8.6	5.30	93.1
			10	20	4.9	4.1	4.9	4.1	4.9	4.1	3.9	3.1	4.1	3.4	4.4	3.5				
			12	22	4.5	3.3	4.5	3.3	4.5	3.3	4.0	3.1	4.1	3.4	4.4	3.4				
200	5	10	4	9	75.1	36.9	80.6	36.9	80.6	63.4	72.5	69.8	75.0	68.5	75.1	73.5	59.8	65.1	56.4	73.0
			6	11	75.2	38.4	78.4	38.5	78.4	62.1	68.2	65.9	72.1	63.7	72.5	69.1				
			8	13	70.0	40.4	72.7	39.9	72.7	58.8	64.0	46.8	65.4	61.5	66.9	65.5				
200	5	50	4	9	12.3	17.0	96.6	18.8	96.6	83.9	9.1	7.0	8.9	7.3	97.1	96.9	5.0	10.2	4.90	99.8
			6	11	21.5	19.6	95.3	21.2	95.3	87.8	7.9	7.3	7.7	6.7	96.3	97.1				
			8	13	23.7	22.0	92.7	22.0	92.7	86.1	6.8	5.0	7.2	5.7	95.1	96.0				
200	10	10	8	18	73.0	41.7	82.1	41.7	82.1	41.8	54.6	12.1	64.9	12.2	81.8	11.8	4.0	50.4	15.2	99.5
			10	20	69.4	42.8	78.4	42.8	78.4	42.8	49.2	12.5	56.7	12.3	78.4	13.3				
			12	22	66.1	40.8	74.4	40.8	74.4	40.8	39.7	12.3	35.9	12.1	75.0	13.0				
200	10	50	8	18	7.7	7.6	92.4	7.7	92.4	71.3	5.4	4.5	5.6	4.3	93.8	88.6	1.4	7.4	6.50	99.8
			10	20	7.9	7.5	88.3	7.5	88.3	7.5	5.6	4.5	5.4	4.5	91.5	4.4				
			12	22	7.1	6.9	84.9	6.9	84.9	6.9	4.9	4.3	4.9	3.6	89.1	4.2				
500	5	10	4	9	99.2	72.8	99.7	69.5	99.7	95.0	99.1	98.8	99.5	97.6	99.3	98.9	94.7	98.2	92.5	98.4
			6	11	99.7	79.6	99.6	77.9	99.6	96.7	99.1	98.4	99.3	97.8	99.2	99.1				
			8	13	99.5	82.1	99.6	80.6	99.6	97.2	98.3	98.2	99.1	97.6	98.8	99.1				
500	5	50	4	9	36.7	62.4	100	64.7	100	98.6	22.7	14.5	20.0	16.3	100	100	5.4	19.6	5.60	100
			6	11	74.0	73.3	100	72.5	100	99.3	18.8	12.3	18.3	14.3	100	100				
			8	13	81.3	76.6	100	77.3	100	99.7	16.3	11.6	16.5	11.6	100	99.9				
500	10	10	8	18	99.9	87.4	100	87.3	100	97.2	97.3	96.1	99.0	98.9	99.8	99.8	17.5	94.4	32.7	100
			10	20	99.9	89.1	99.9	88.8	99.9	97.9	96.5	94.5	98.7	98.6	99.8	99.6				
			12	22	99.9	90.4	99.8	90.4	99.8	98.0	94.9	94.1	98.6	98.7	99.8	99.7				
500	10	50	8	18	42.3	35.8	100	33.8	100	98.2	8.2	7.7	7.7	6.2	100	99.9	4.4	10.0	4.50	100
			10	20	42.2	38.2	99.9	36.8	99.9	99.1	6.6	7.8	7.7	6.0	99.9	100				
			12	22	50.9	36.8	99.9	38.5	99.9	99.0	6.5	6.9	6.6	5.6	99.9	100				
1000	5	10	4	9	100	90.7	100	87.3	100	99.1	100	99.9	100	100	100	100	99.9	100	100	100
			6	11	100	95.6	100	93.8	100	99.7	100	99.9	100	100	100	100				
			8	13	100	97.3	100	96.8	100	99.9	100	100	100	100	100	100				
1000	5	50	4	9	71.8	89.6	100	91.7	100	99.7	47.4	26.9	40.1	28.9	100	100	4.6	45.0	4.90	100
			6	11	99.2	96.7	100	96.7	100	99.9	43.5	23.6	38.2	26.0	100	100				
			8	13	99.6	98.8	100	98.7	100	100	41.7	21.3	34.4	24.4	100	100				
1000	10	10	8	18	100	98.7	100	98.5	100	99.9	100	100	100	100	100	100	37.9	100	57.5	100
			10	20	100	99.4	100	99.3	100	99.9	100	99.9	100	100	100	100				
			12	22	100	99.5	100	99.5	100	100	100	99.9	100	100	100	100				
1000	10	50	8	18	94.6	84.2	100	84.2	100	99.9	16.0	12.7	14.7	10.4	100	100	4.3	20.0	6.10	100
			10	20	93.7	88.7	100	87.7	100	99.9	14.4	10.8	13.7	9.9	100	100				
			12	22	95.8	88.3	100	89.4	100	100	12.2	10.8	13.1	9.1	100	100				

Percentage of rejections at the nominal level $\alpha = 0.05$ of the $\hat{\chi}^2$ and the Wald test under different partitioning methods. n, d_x, a, L, and L^* denote the sample size, the number of covariates, the degree of heteroskedasticity, and the number of cells of the Wald and J test, respectively. The last four columns report the rejection rates of the Kolmogorov-Smirnov tests of Stute (1997) (KS1) and Stute and Zhu (2002) (KS2), the Cramer-Von-Mises test of Escanciano (2006) (CvM1), and the T-test on $b = 0$.

t-test consistently show the highest power across all scenarios, due to their ability to

leverage out-of-sample information. The $\hat{\mathcal{W}}$ test has higher power compared to the $\hat{\chi}^2$ test, particularly in high-frequency cases. When $c = 50$, the omnibus tests KS1 and CvM1 exhibit trivial power, while the directional KS2 test is able to detect alternatives in nearly half of the cases when $d_x = 5$, and once in every five cases when $d_x = 10$. On the other hand, the feasible χ^2 tests, where the data is split using w_{step} and w_{poly} for the initial two-cell partitioning and **FS** for additional splits, detect these alternatives in almost all cases, regardless of the number of covariates.

8 Empirical Applications

This section discusses the application of χ^2 testing procedures in two empirical contexts. First, I consider the estimation of the average treatment effect on treated in difference-in-difference (DID) settings with heterogeneous adoption designs (HAD) without stayers (terminology introduced by de Chaisemartin and D'Haultfoeuille (2020), see Assumption S1 in the corresponding web appendix). In this setting, units are untreated in the first period and receive a strictly positive dose of a non-binary treatment in the following period. The second context involves a classical binary treatment design, where ATE estimation relies on the unconfoundedness assumption and the correct specification of a parametric model, either for the regression function or the propensity score (i.e., the probability of treatment given observed characteristics).

The χ^2 tests are then applied to two real-data examples, one for each empirical setting. The first, following Pierce and Schott (2016), analyzes the effects of the U.S. government's 2001 decision to grant Permanent Normal Trade Relations (PNTR) status to China on employment. The second estimates the impact of attending Historically Black Colleges and Universities (HBCU) versus Traditional White Institutions on the labor and educational outcomes of African American students, based on Fryer and Greenstone (2010).

8.1 DID-HAD Setting

Consider a two-period DID setup where outcomes $Y_{i,t}$ and a continuously distributed treatment $D_{i,t}$ are observed for $i = 1, ..., N$ and $t = 1, 2$. Assume that $\{(Y_{i,1}, Y_{i,2}, D_{i,1}, D_{i,2})\}_{i=1}^N$

are i.i.d. draws from (Y_1, Y_2, D_1, D_2) and let $f_{D_2}(\cdot)$ denote the density of D_2. Define the support of D_2, $\mathcal{D}_2$ say, as the smallest set A such that $P(D_2 \in A) = 1$.

The design follows de Chaisemartin, Ciccia, D'Haultfœuille, and Knau (2024) (hereafter, CCDK) and de Chaisemartin, D'Haultfœuille, and Vazquez-Bare (2024), where units (e.g., firms or counties) receive no treatment in the first period, $D_1 = 0$, and a strictly positive dose of treatment in the second period, $D_2 > 0$, with $V(D_2) > 0$. It is assumed that $0 \in \mathcal{D}_2$ and $f_{D_2}(0) > 0$. Thus, although treatment values of zero are never observed, treatments arbitrarily close to zero occur with positive probability. Units experiencing treatments close to zero are termed *quasi-stayers*.

Let $Y_{i,t}(d)$ denote the potential outcome of unit i at time t under treatment d, i.e., the outcome that would have been observed if unit i at t received a treatment dose equal to d. The observed outcome is $Y_{i,t} = Y_{i,t}(D_{i,t})$. We aim to estimate the causal parameter $\tau_{\text{AS}} = \mathbb{E}[TE_2]$, where

$$TE_2 = \frac{Y_2(D_2) - Y_2(0)}{D_2}$$

is the slope of the individual potential outcome function between zero and the actual treatment. The τ_{AS} generalizes the average treatment effect on the treated to cases with a continuous treatment, offering a natural target for policy effect estimation.

Under the assumption below, the identification of the conditional AS (CAS), $\mathbb{E}[TE_2|D_2]$, follows from a parametric specification of $\mathbb{E}[\Delta Y|D_2]$, where $\Delta Y_i = Y_{i,2} - Y_{i,1}$. Throughout, $\mathbb{E}[TE_2|D_2 = 0] = 0/0 = 0$.

Assumption 6

(a) There is a real number μ_0 such that $\mathbb{E}[\Delta Y(0)|D_2] = \mu_0$.

(b) There exist a known integer K, with K known functions $(g_1(d), ..., g_K(d))$, such that,

$$\mathbb{E}[TE_2|D_2] = \sum_{k=1}^{K} \delta_k g_k(D_2) \text{ a.s. for some } (\delta_1, ..., \delta_K) \in \mathbb{R}^K,$$

$$g_k(0) = 0 \text{ for all } k, \text{ and } (1, D_2 g_1(D_2), ..., D_2 g_K(D_2))' \text{ is a linearly independent vector.}$$

Assumption 6 (a) generalizes the parallel trend assumption from classical DiD designs with binary treatments to the continuous case, while Assumption 6 (b) imposes a parametric

structure for the CAS. Toghether, they provide a sufficient condition for identifying τ_{AS} as $\mathbb{E}[\Delta Y|D_2] = \mu_0 + D_2 \sum_{k=1}^{K} \delta_k g_k(D_2)$.

Moreover, when $f_{D_2}(0) > 0$, any parametric specification of the CAS in Assumption 6 (b) is equivalent to a parametric model for $\mathbb{E}[\Delta Y|D_2]$, as stated in the next proposition.

Proposition 3 *Under Assumption 6 (a), if $f_{D_2}(0) > 0$, Assumption 6 (b) holds if and only if $\mathbb{E}[\Delta Y|D_2] = \beta_0 + D_2 \sum_{k=1}^{K} \beta_k g_k(D_2)$, where $(\beta_0, ..., \beta_K)$ are the linear projection coefficients of ΔY on $(1, D_2 g_1(D_2), ..., D_2 g_K(D_2))$, and $(\mu_0, \delta_1, ..., \delta_K) = (\beta_0, ..., \beta_K)$.*

In particular, $\mathbb{E}[TE_2|D_2] = \tau_{\text{AS}}$ if and only if $\mathbb{E}[\Delta Y|D_2]$ is linear in D_2. In this case, the two-way fixed effect (TWFE) estimator—the coefficient β_{fe} of $D_{i,2}$ in a regression of ΔY_i on a constant and $D_{i,2}$—is a consistent estimator of the τ_{AS}. This motivates pre-testing $H_0 : \mathbb{E}[\Delta Y|D_2] = \beta_0 + \beta_{fe} D_2$ to validate the causal interpretation of the TWFE estimator.

Building on CCDK, after conducting a placebo test for Assumption 6 (a) and for $f_{D_2}(0) > 0$, I propose testing $H_0 : \mathbb{E}[\Delta Y|D_2] = \beta_0 + \beta_{fe} D_2$ using χ^2 tests with **PNP** partitions, tailored for various choices of K and $(g_1(\cdot), ..., g_K(\cdot))$, .

Under the null hypothesis of linearity, and conditional on the pre-testing procedure, inference on β_{fe} is, at worst, conservative (Corollary 1 of de Chaisemartin and D'Haultfœuille 2024). A powerful testing procedure, however, is essential, for undetected deviations from linearity can lead to significantly biased inference. The χ^2 tests with **PNP** partitions exhibit strong power, especially against the alternative specifications used for partitioning. Furthermore, rejections indicate both misspecification of the linear model and that alternative models provide a better fit, which, according to Proposition 3, it is useful for the estimation of the τ_{AS} when linearity fails. χ^2 tests are also computationally efficient so that can be employed with large sample sizes.

The testing routine is applied to investigate the impact of removing uncertainty regarding China's Normal Trade Relations status on U.S. employment. The analysis draws on Pierce and Schott (2016), from whose replication package we obtain the data, and the revisitation of the paper by CCDK.

8.2 Effect on US employment of Permanent Normal Trade Relations (PNTR) to China

Before 2001, the U.S. Congress was required to renew China's Normal Trade Relations (NTR) status annually. Failure to renew this status would have resulted in a significant increase in trade tariffs, affecting firms across industries differently due to the varying non-NTR rates. The uncertainty surrounding potential tariff spikes influenced U.S. firms' incentives to incur sunk costs associated with shifting operations to China or establishing trade relationships with existing Chinese producers. Similarly, it affected Chinese firms' incentives to enter or expand into the U.S. market.

In 2001, Permanent Normal Trade Relations (PNTR) status was granted, eliminating tariff uncertainty for all sectors. This change influenced firm decisions and, consequently, employment levels in U.S. industrial sectors, which is the focus of the following analysis. Specifically, I estimate the effect of granting PNTR status to China on U.S. employment. The treatment in this case is the difference between NTR and non-NTR tariffs, which varies substantially across industries and is often close to zero. The unit of analysis is the industry level. The data are sourced from the replication package for Column (3) in Pierce and Schott (2016), along with publicly available UNIDO data. This dataset contains information on 103 industries for the periods 1997-2002 and 2004-2005.

Let $Y_{i,t}$ denote the natural logharithm of employment in sector i at time t, and $D_{i,t}$ represent the post-2000 difference between the NTR and non-NTR rate of sector i at time t, where $D_{i,t} = 0$ for $t < 2001$ and $D_{i,t} = D_{i,2001}$ for $t \geq 2001$. Following CCDK, we focus on unweighted regressions of the de-trended employment growth from 2000 to $t \in \{2001, 2002, 2004, 2005\}$,

$$\tilde{\Delta} Y_{i,t} = Y_{i,t} - Y_{i,2000} - (t - 2000)(Y_{i,2000} - Y_{i,1999}),$$

on $D_{i,2001}$, where $(Y_{i,2000} - Y_{i,19999})$ denotes the pre-treatment industry-specific linear trend. Controlling for the linear trend accounts for the fact that, in the absence of treatment, the growth rate of employment increases over the years, as suggested from placebo tests on regressions of $Y_{i,t} - Y_{i,2000}$ on $D_{i,2001}$ in $t \in \{1997, 1998, 1999\}$ (see Table 2 in CCDK). The

CCDK test for $H_0 : f_{D_{2001}}(0) > 0$ cannot reject the null hypothesis, so that we can focus on the parametric specification of $\mathbb{E}\left[\tilde{\Delta}Y_t|D_{2001}\right]$ to characterize the one of $\mathbb{E}\left[TE_t|D_{2001}\right]$.

Table 4 below reports the p-values of the χ^2 tests for

$$H_0 : \mathbb{E}\left[\tilde{\Delta}Y_t|D_{2001}\right] = \beta_0 + \beta_1 D_{2001}$$

using **PNP** partitions, specializing against the following alternative specifications,

$$H_{1,K} : \mathbb{E}\left[\tilde{\Delta}Y_t|D_{2001}\right] = \beta_0 + \beta_1 D_{2001} + \sum_{k=2}^{K} \beta_k D_{2001}^k,$$

for $K \in \{2, 3\}$ and for all $t \in \{2001, 2002, 2004, 2005\}$. In constructing the **PNP** partition, I do not use Algorithm 1 described in Section 4. Instead, I divide the data into several cells based on whether the fit under $H_{1,0}$ is greater or less than the fit under $H_{1,k}$. If a cell contains fewer than five observations, it is merged with the next smallest cell. Since D_{2001} is univariate, finding the intersection set where the fits under the specifications of H_0 and $H_{1,k}$ coincide is straightforward. Furthermore, because $X = (1, D_{2001})'$, we only require $L > 2$ cells to perform the J-test, a condition that holds for all t and k. The test statistics are compared with the quantiles of a χ^2_{L-1} distribution for the Wald test and a χ^2_{L-2} distribution for the J-test.

Table 5 reports estimates of τ_{AS}, $\hat{\tau}_{\text{AS}}$ say, under the parametric specifications of H_0 and $H_{1,k}$ for $k \in \{2, 3\}$, where

$$\hat{\tau}_{AS} = \sum_{k=1}^{K} \hat{\beta}_k \left(\frac{1}{n} \sum_{i=1}^{n} D_{i,2001}^{k-1} \right),$$

and $\{\hat{\beta}_k\}_{k=0}^{K}$ are OLS coefficients from a regression of $\tilde{\Delta}Y_t$ on $(1, D_2, ..., D_2^K)$. Standard errors are calculated using the delta method with HC2 robust standard errors.

Graphical representations of the partitions under $H_{1,2}$ (Figure 4) and $H_{1,3}$ (Figure 5) indicate that the quadratic specification offer a better fit than the linear model for $\mathbb{E}\left[\tilde{\Delta}Y_{2002}|D_{2001}\right]$, while a cubic fit is more suitable for $\mathbb{E}\left[\tilde{\Delta}Y_{2001}|D_{2001}\right]$. The test results reveal that the linear model is mispecified for $\mathbb{E}\left[\tilde{\Delta}Y_{2002}|D_{2001}\right]$, although linearity cannot be rejected for $\mathbb{E}\left[\tilde{\Delta}Y_t|D_{2001}\right]$ when $t \in \{2001, 2004, 2005\}$.Rejections under the

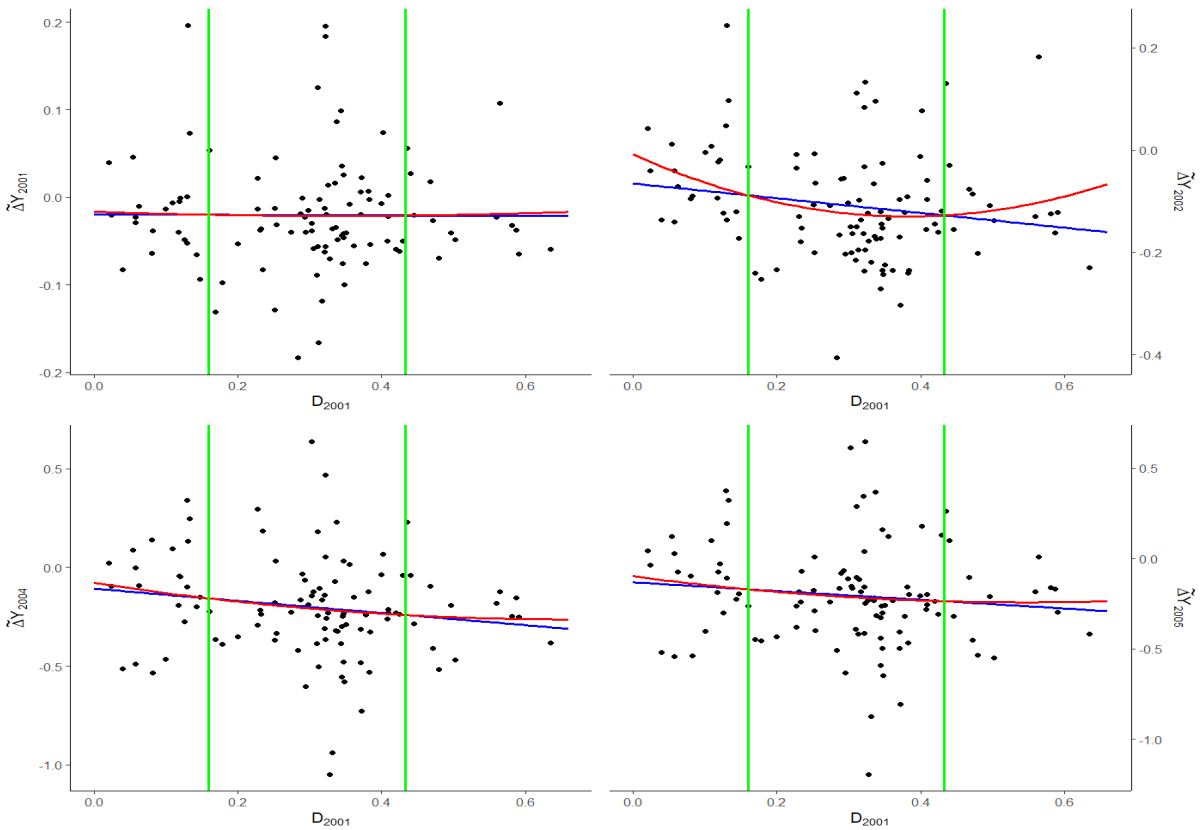


Figure 4: Neyman-Pearson classes comparing the linear (blue line) and quadratic (red line) fit of $\mathbb{E}\left[\tilde{\Delta}Y_t|D_{2001}\right]$ for $t \in \{2001, 2002, 2004, 2005\}$, shown respectively in the northeast, northwest, southeast, and southwest panels.

x

PNP partitions suggest that quadratic or cubic specifications better fits $\mathbb{E}\left[\tilde{\Delta}Y_{2002}|D_{2001}\right]$. All estimated average slopes are negative, consistent with Pierce and Schott (2016) and CCDK. For 2002, the quadratic and cubic models produce stronger, less noisy estimates with lower p-values compared to the linear model. However, effects in 2001, 2004, and 2005 remain statistically negligible across all model specifications.

	2001		2002		2004		2005	
$H_{1,K}$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$
$H_{1,2}$	0.610	0.386	0.015	0.004	0.444	0.223	0.218	0.210
$H_{1,3}$	0.876	0.909	0.003	0.004	0.694	0.451	0.231	0.139

Table 4: P-values from the χ^2 tests for H_0 using **PNP** partitions, with rejections favoring the alternative specifications in $H_{1,K}$, for $K \in \{2, 3\}$.

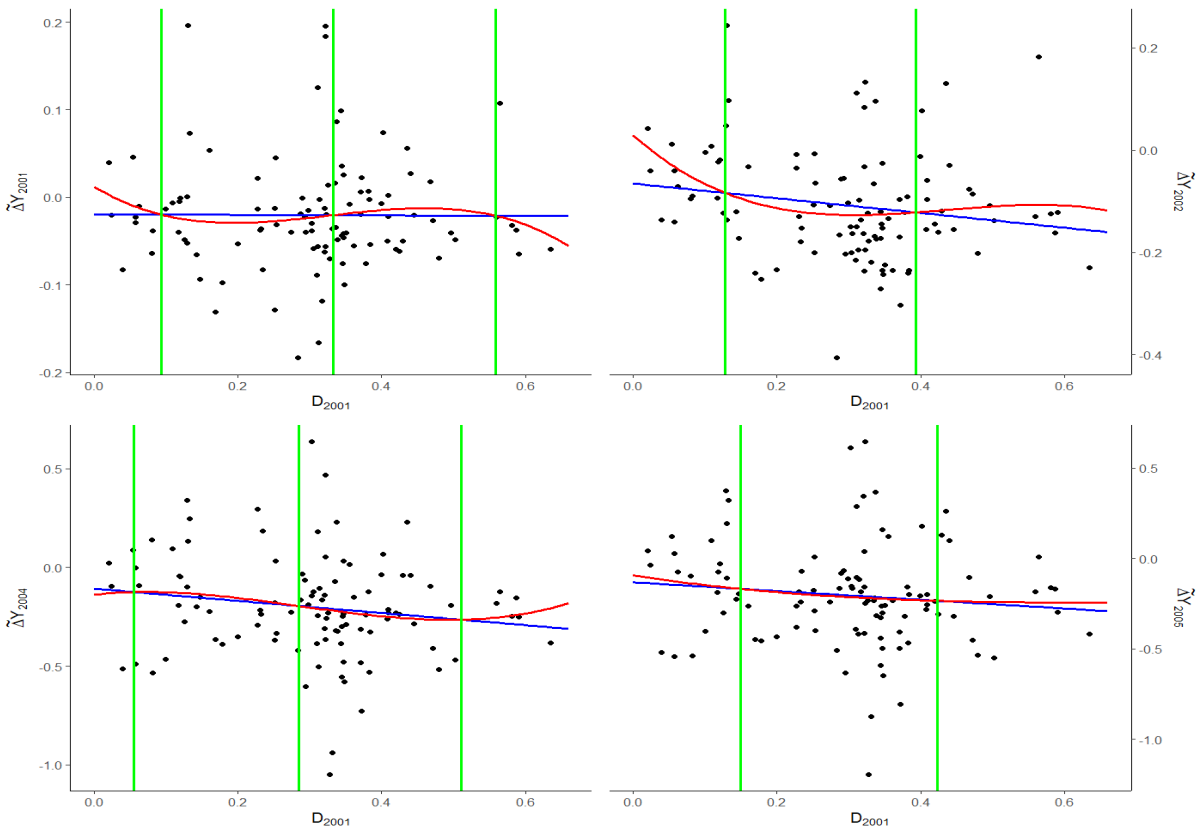


Figure 5: Neyman-Pearson classes comparing the linear (blue line) and cubic (red line) fit of $\mathbb{E}\left[\tilde{\Delta}Y_t | D_{2001}\right]$ for $t \in \{2001, 2002, 2004, 2005\}$, shown respectively in the northeast, northwest, southeast, and southwest panels.

	$\hat{\tau}_{\mathbf{AS}}$			
	2001	2002	2004	2005
H_0	-0.003	-0.143	-0.307	-0.241
	(0.046)	(0.076)	(0.186)	(0.207)
	[0.945]	[0.064]	[0.102]	[0.246]
$H_{1,2}$	-0.015	-0.384	-0.429	-0.381
	(0.088)	(0.144)	(0.356)	(0.396)
	[0.861]	[0.008]	[0.231]	[0.337]
$H_{1,3}$	-0.151	-0.563	-0.144	-0.405
	(0.158)	(0.259)	(0.643)	(0.715)
	[0.340]	[0.032]	[0.823]	[0.572]

Table 5: Estimated τ_{AS} for $t \in \{2001, 2002, 2004, 2005\}$ using the parametric specification in H_0 and $H_{1,K}$, $K \in \{2, 3\}$. Standard errors in parenthesis.

8.3 Observational Data under Unconfoundedness and Binary Treatment

Consider a setting where the researcher observes random draws $\{(Y_i, D_i, X_i')'\}_{i=1}^n$ from $(Y, D, X')'$, where $Y \in \mathbb{R}$ is the outcome of interest, $D \in \{0, 1\}$ is a binary treatment indicator, and $X \in \mathbb{R}^{d_x}$ represents individual characteristics. Let $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes for unit i under treatment and control, respectively, so that $Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$. The objective is to estimate the Average Treatment Effect on the Treated (ATT),

$$\tau_{\text{ATT}} = \mathbb{E}\left[Y(1) - Y(0)|D = 1\right].$$

This is done under the unconfoundedness and overlap framework (Rosenbaum and Rubin 1983), where the following assumptions hold:

Assumption 7

(a) $(Y_i(1), Y_i(0))$ *is independent of* D_i *conditional on* X_i.

(b) The propensity score, $p(X) = P(D = 1|X)$, *satisfies* $0 < p(X) < 1$ *a.s.*

Assumption 7(a) implies that, given X, treatment assignment is as if random, allowing identification of the ATT as $\tau_{\text{ATT}} = \mathbb{E}\left[Y|D = 1\right] - \mathbb{E}\left[m_0(X)|D = 1\right]$, where $m_0(X) = \mathbb{E}\left[Y|X, D = 0\right]$. Assumption 7(b) requires common support in the characteristics of treated and untreated populations, providing an alternative identification of the ATT (Rosenbaum 1987) as $\tau_{\text{ATT}} = \mathbb{E}\left[D\right]^{-1}\mathbb{E}\left[DY - (p(X)/(1 - p(X)))(1 - D)Y\right]$. Under Assumptions 7(a)-(b), the two identification strategies yields an expression for the ATT in terms of a *Neyman-Orthogonal* moment (Chernozhukov, Escanciano, Ichimura, Newey, and Robins 2022):

$$\tau_{\text{ATT}} = \frac{1}{P(D = 1)}\mathbb{E}\left[\left(D - \frac{p(X)(1 - D)}{1 - p(X)}\right)(Y - m_0(X))\right]. \tag{32}$$

Based on this identifying equation, a natural estimator of the ATT is the Augmented Inverse Probability Weighting (AIPW) estimator (Robins, Rotnitzky, and Zhao 1994),

$$\hat{\tau}_{\text{ATT}} = \frac{1}{n_1}\sum_{i=1}^{n}\left[D_i - \frac{\hat{p}(X_i)}{1 - \hat{p}(X_i)}(1 - D_i)\right](Y_i - \hat{m}_0(X_i)), \tag{33}$$

where $n_1 = \sum_{i=1}^{n} \mathbb{I}\{D_i = 1\}$, and $\hat{m}_0(x)$ and $\hat{p}(x)$ are estimates of $m_0(X)$ and $p(x)$, respectively.

The AIPW estimator is *doubly robust*, requiring that either the regression function model, $\hat{m}_0(x)$, or the propensity score model, $\hat{p}(x)$, be correctly specified for consistent estimation of the ATT. Thus, as long as one of the two models is correctly specified, the estimator will yield a consistent ATT estimate. When parametric models are assumed for both functions—i.e., $\hat{m}_0 = m_{0,\tilde{\theta}}$ and $\hat{p} = p_{\tilde{\beta}}$—the validity of the estimated causal effect is assessed by testing the hypothesis:

$$H_{0,U} : m_0 = m_{0,\theta_0} \text{ a.s. for some } \theta_0 \in \Theta \ \cup \ p = p_{\beta_0} \text{ a.s. for some } \beta_0 \in B.$$

A simple test for $H_{0,U}$ can be constructed using the Intersection-Union Test (IUT) (Casella and Berger 2024, Section 8.2.3), which combines two size-α test statistics for the correct model specification of m_θ and p_β as follows:

$$\hat{T}_U = \mathbb{I}\{\hat{t}_1 > z_{1,\alpha/2}\}\mathbb{I}\{\hat{t}_2 > z_{2,\alpha/2}\}, \tag{34}$$

where $\hat{t}_1$ and $\hat{t}_{2,\beta_0}$ are test statistics such that:

$$\lim_{n\to\infty} P(\hat{t}_1 > z_{1,\alpha/2}) = \alpha \quad \text{when } m = m_{\theta_0} \text{ a.s.,} \tag{35}$$

$$\lim_{n\to\infty} P(\hat{t}_2 > z_{2,\alpha/2}) = \alpha \quad \text{when } p = p_{\beta_0} \text{ a.s.} \tag{36}$$

for known critical values $z_{1,\alpha/2}$ and $z_{2,\alpha/2}$. The following proposition shows that the IUT is indeed a size α test whenever $\hat{t}_1$ and $\hat{t}_2$ have size α under their respective null hypothesis.

Proposition 4 *Let $H_{0,U}$, $\hat{T}_U$, $\hat{t}_1$, and $\hat{t}_2$, be as defined as above. Let $\mathcal{P}_0$ be the set of probability distributions such that $P(H_{0,U}) = 1$ for each $P \in \mathcal{P}_0$, then*

$$\lim_{n\longrightarrow\infty} \sup_{P\in\mathcal{P}_0} P(\hat{T} = 1) = \alpha.$$

Next, we apply the IUT test to assess the validity of causal estimates regarding the effects of attending Historically Black Colleges and Universities (HBCUs) on future labor outcomes for African American students, such as wages.

8.4 The Effects of Attending HBCU on African American Students

HBCUs play a crucial role in reinforcing identity, confidence, and self-esteem among students—especially Black students—by fostering a strong sense of belonging through higher Black student ratios (Price and Viceisza 2023). Determining whether HBCUs are more effective than Traditional White Institutions (TWIs) in supporting Black students has important policy implications. Despite their historic dependence on federal funding, HBCUs consistently face underfunding relative to TWIs, which impacts educational quality and limits student opportunities. Proposals to merge HBCUs with other institutions or reallocate resources to TWIs suggest potential paths toward a more efficient, cost-effective higher education system.

Following Fryer and Greenstone (2010), I examine the impact of HBCU attendance on wages for African American students using data from the National Longitudinal Survey of the High School Class of 1972 (NLS). The sample includes 624 African American students who pursued higher education post-high school and were tracked from 1972 to 1986; 260 attended an HBCU, and 364 attended a TWI. The outcome variable is the log of hourly wages in 1986. The set of characteristics, X, include family income, parental education, and indicators for student sex, one for private school attendance, and another for Southern residence, along with SAT, ACT, and high school GPA scores. The treatment variable, D, is a binary indicator for whether the first college attended was an HBCU.

The dataset contains many missing values, especially in pre-college standardized test scores. Unlike the original paper, which replaces missing values with zeros and groups covariates into blocks, I create a covariate representing each student's highest rank among the SAT and ACT tests and omit observations with missing data. After this adjustment, the sample reduces to 242 students, of whom 103 attended an HBCU as their first college.

I test the following specifications of $m_0(x)$ and $p(x)$,

$$H_{0,U} : m(D, X) = \theta_0 + \theta_1 D + \theta_2' X \ \cup \ p(X) = \Phi(\beta' X),$$

where $\Phi(x)$ is the CDF of the standard normal distribution, and $m(D, X) = \mathbb{E}[Y|D, X]$

so that $m_0(x) = m(0, x)$. The IUT test for $H_{0,U}$ is the product of individual tests.

When the probit model is correctly specified, the MLE is efficient, and $\text{Avar}\left(\boldsymbol{\Phi}_{\boldsymbol{\gamma}}(\tilde{\theta})\right) = \Sigma_{\boldsymbol{\gamma},0} - \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0}\mathcal{I}^{-1}\boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma},0}$ is positive definite, with $\boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} = \mathbb{E}\left[\phi(X'\theta_0)\mathbf{I}_{\boldsymbol{\gamma}}(X)X'\right]$, $\phi(x)$ as the standard normal density, and $\mathcal{I} = \mathbb{E}\left[\phi(X'\theta_0)^2 XX'/\Phi(X'\theta_0)(1-\Phi(X'\theta_0))\right]$ as the Fisher information matrix. Thus, $\text{Avar}^-\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ is full rank. Only the Wald test is reported here, as the iterative GMM estimation in (10) often fails to converge in the probit model specification.

I test $H_{0,U}$ using **FNP** partitions with both w_{poly} and w_{step}, as well as with **PNP** partitions specializing against,

$$H_{1,U} : m(D, X) = \theta_0 + \theta_1 D + \theta_2' X + \theta_3 \mathbb{I}\{\text{faminc} > q_{3,f}\} \cup$$

$$p(X) = \Phi\left(\beta_0' X + \beta_1 \mathbb{I}\{\text{score} > q_{3,s}\} + \beta_2 \mathbb{I}\{\text{faminc} \leq q_{3,s}\} \cdot \text{south}\right),$$

where $q_{j,s}$ and $q_{j,f}$ denote the $j-$quartile of family income and standardized test score, respectively. For **FNP** partitions, I use a polynomial of order $q = 2$ in w_{poly}, and a step function over each quartile of the fitted value in w_{step}. Each test, is performed for $L \in \{4, 6, 8\}$. Additional splits are done with **FS** for the regression specification, and **PC** for the propensity score specification.

The regression specification in $H_{0,U}$ is the standard short regression under unconfoundedness, while the alternative allows a differential effect for higher-income students. The alternative propensity score also accounts for HBCUs concentration in the South, where lower mobility costs may increase HBCU attendance, especially for low-income students, while higher-scoring students may prefer more prestigious TWIs.

Finally, I report the estimated ATT using the specification in $H_{0,U}$ and in $H_{1,U}$, along with semi-nonparametric estimates, where $m(x)$ and $p(x)$ are estimated using random forest. Following common practice, the ATT is estimated from the subsample with propensity score outside of the $[0.05, 0.95]$ interval (Crump, Hotz, Imbens, and Mitnik 2009).

The tests results are shown in Table 7. The null hypothesis of correct specification for the short regression cannot be rejected, regardless of the partitions used. Conversely, the propensity score specification is rejected with both **FNP** partitions using w_{poly} and **PNP** partitions, suggesting that $H_{1,U}$ offers a better fit to the data. The ATT point

estimate with the more comprensive specification in $H_{1,U}$ is one-third smaller than with $H_{0,U}$; however, in all three cases, including the nonparametric specification, the estimated effect remains stastically negligble.

	w_{poly}		w_{step}		**PNP**	
L	m_θ	p_β	m_θ	p_β	m_θ	p_β
4	0.272	0.044	0.558	0.558	0.802	0.019
6	0.833	0.009	0.877	0.877	0.465	0.068
8	0.725	0.035	0.627	0.627	0.876	0.027

Table 6: P-values from the Wald tests for H_0 using **FNP** partitions with w_{poly}, w_{step}, and **PNP** partitions specializing against $H_{1,U}$.

	$\hat{\tau}_{\mathbf{ATT}}$		
	$H_{0,U}$	$H_{1,U}$	NP
$\hat{\tau}$	0.069	0.045	0.032
s.e.	(0.069)	(0.067)	(0.069)
pval	[0.320]	[0.501]	[0.643]

Table 7: Estimated τ_{ATT} using the parametric specification in $H_{0,U}$, $H_{1,U}$, and the nonparametric fits. Standard errors in parenthesis and pvalue for the null of zero effect in bracket.

9 Conclusion

This article introduces a novel approach for validating the correct specification of conditional moment restriction (CMR) models, which are fundamental for identifying and estimating causal relationships. The proposed method harnesses the well-established chi-squared (χ^2) tests, commonly used in classical goodness-of-fit contexts, to goodness-of-fit checks for CMR specifications. Traditionally, CMR model checks have been adapted from tests for the cumulative density function (CDF) based on functionals of the standard empirical process (SEP), but these often exhibit limitations, especially when dealing with high-dimensional data and high-frequency alternatives. The introduced χ^2 tests, on the other hand, are distribution-free, do not necessitate complex bootstrapping or smoothing techniques, and offer flexibility in partitioning the data to favor specific alternative hypotheses. Thus, providing a valid complementary instruments to the existing model

checking procedures. Monte Carlo simulations and empirical evidence highlight the effectiveness of these χ^2 tests, especially in scenarios involving a large number of covariates. The empirical analysis illustrates the practical application of these tests in assessing the validity of estimated average treatment effects in difference-in-difference settings with continuous treatments and static binary treatment designs with many covariates. Two real-data examples are presented considering the impact of Permanent Normal Trade Relations on US employment and the effects of attending Historically Black Colleges and Universities on African American students' wages.

A Appendix A

A.1 Lemmas

I first state auxiliary lemmas for the propositions and theorems in the main text. Let $\rightsquigarrow$ denote weak convergence on $l^\infty(\mathbb{D})$ (see definition 13.3 in Van Der Vaart 1996, hereafter VW), where $l^\infty(\mathbb{D})$ is the space of all real-valued functions that are uniformly bounded on $\mathbb{D}$, and $\rightarrow_d$ denote convergence of real-valued random variables. Troughout, to highlight the dependency on the partition, denote as $\hat{\mathbf{\Phi}}_\theta(\boldsymbol{\gamma}) = \hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\theta)$ and $\hat{\mathbf{\Phi}}_0(\boldsymbol{\gamma}) = \hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\theta_0)$.

Lemma 1 *Under the null H_0,*

(a) if Assumption 1, 2, and 2' hold, then $\Sigma_{\boldsymbol{\gamma}}(\tilde{\theta}) = \Sigma_{\boldsymbol{\gamma},0} + o_p(1)$.

(b) if Assumption 1, 2, 3 hold, and $\text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ is p.d., then $\hat{W}_{\boldsymbol{\gamma}}(\tilde{\theta}) = \text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right) + o_p(1)$

Lemma 2 *Under the null hypothesis H_0, Assumptions 1, and 5,*

$$\hat{\mathbf{\Phi}}_0(\cdot) \rightsquigarrow \Phi_0(\cdot) \text{ as a process on } l^\infty(\mathbb{D}),$$

where $\Phi_0(\cdot)$ is an $\mathbb{R}^L$-valued Gaussian process with zero mean vector and covariance structure given by,

$$\mathbb{E}\left[\Phi_0(\boldsymbol{\gamma})\Phi_0(\tilde{\boldsymbol{\gamma}})\right] = \mathbb{E}\left[\varepsilon_{\theta_0}(Z)^2 \mathbf{I}_{\boldsymbol{\gamma}}(X)\mathbf{I}_{\tilde{\boldsymbol{\gamma}}}(X)'\right] \quad \forall \boldsymbol{\gamma}, \tilde{\boldsymbol{\gamma}} \in \mathbb{D}.$$

Lemma 3 *Under the null hypothesis H_0, and Assumptions 1-5, it holds that:*

(a) $\sup_{\boldsymbol{\gamma}\in\mathbb{D}} \left|\hat{\mathbf{\Phi}}_{\tilde{\theta}}(\boldsymbol{\gamma}) - (\hat{\mathbf{\Phi}}_0(\boldsymbol{\gamma}) - \hat{\boldsymbol{\mu}}^{*\prime}_{\boldsymbol{\gamma}}(\theta_0)\sqrt{n}(\hat{\theta} - \theta_0))\right| = o_p(1)$.

(b) $\hat{\boldsymbol{\mu}}^*_{\tilde{\boldsymbol{\gamma}}}(\theta_0) = \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} + o_p(1)$.

Lemma 4 *Under the null H_0, and Assumptions 4,5,*

(a) if Assumption 1, 2, and 2' hold, then $\Sigma_{\tilde{\boldsymbol{\gamma}}}(\tilde{\theta}) = \Sigma_{\boldsymbol{\gamma},0} + o_p(1)$.

(b) if Assumption 1, 2, 3 hold, and $\text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)$ is p.d., then $\hat{W}_{\tilde{\boldsymbol{\gamma}}}(\tilde{\theta}) = \text{Avar}\left(\hat{\mathbf{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right) + o_p(1)$

A.2 Proofs

For any class of functions $\mathcal{F}$, denote as $\{P_n f : f \in \mathcal{F}\}$ the empirical measure indexed by $\mathcal{F}$, such that $P_n f = n^{-1} \sum f(Z_i)$; alike, Pf denotes the population measure, $Pf = \int f(Z)dP$. We say that a class of functions is: i) Glivenko-Cantelli for P (hereafter, P-GC) whenever $\sup_{f \in \mathcal{F}} |P_n - P|f = o_p(1)$; ii) P-Donsker if $\{\sqrt{n}(P_n - P)f : f \in \mathcal{F}\}$ converge in distribution to a tight random element in the space $l^\infty(\mathcal{F})$. Throughout, we refer to both classes of sets with finite VC dimension and classes of functions with finite VC subgraph dimension as VC classes. These classes, having uniformly bounded covering numbers (Theorem 2.6.7 in VW), are Glivenko-Cantelli and Donsker (see Theorem 2.4.3 and 2.5.2 in VW) for any probability measure on the sample space, provided that they have integrable and square-integrable envelope function, respectively.

Proof of Lemma 1. For the first part of the Lemma, by the weak law of large numbers (WLLN) and a mean value theorem argument (MVT), suffices to show that

$$\frac{1}{n} \sum_{i=1}^{n} \left(\varepsilon^2_{\tilde{\theta}}(Z_i) - \varepsilon^2_{\theta_0}(Z_i) \right) \mathbb{I}_{\gamma_l}(X_i) = I + II + III = o_p(1)$$

for each $l \in 1, 2, ..., L$, where,

$$\begin{aligned} I &= (\tilde{\theta} - \theta_0)' \frac{1}{n} \sum_{i=1}^{n} \nabla m_{\bar{\theta}}(X_i) \nabla m_{\bar{\theta}}(X_i)' \mathbb{I}_{\gamma_l}(X_i)(\hat{\theta} - \theta_0), \\ II &= \frac{2}{n} \sum_{i=1}^{n} m_{\theta_0}(X_i) \nabla m_{\bar{\theta}}(X_i)' \mathbb{I}_{\gamma_l}(X_i)(\tilde{\theta} - \theta_0), \\ III &= \frac{2}{n} \sum_{i=1}^{n} Y_i (m_{\tilde{\theta}}(X_i) - m_{\theta_0}(X_i)) \mathbb{I}_{\gamma_l}(X_i), \end{aligned}$$

and $|\bar{\theta} - \theta_0| \leq |\tilde{\theta} - \theta_0|$. The triangle inequality, Assumption 2, and the consistency of $\tilde{\theta}$ show that, $|I| \leq d^2_\theta \left\| \tilde{\theta} - \theta_0 \right\|^2 n^{-1} \sum_{i=1}^{n} R(X_i)^2 = o_p(1)$, where $\|\cdot\|$ denotes the euclidean

norm. By a similar reasoning,

$$
\begin{aligned}
|II| &\leq d_\theta \left\|\tilde{\theta} - \theta_0\right\| \frac{2}{n} \sum_{i=1}^{n} m_{\theta_0}(X_i) R(X_i) \\
&\leq d_\theta \left\|\tilde{\theta} - \theta_0\right\| \left(\mathbb{E}\left[Y^2\right]\right)^{1/2} \left(\mathbb{E}\left[R(X)^2\right]\right)^{1/2} + o_p(1) = o_p(1)
\end{aligned}
$$

where the last inequality follows from the WLLN, the law of iterated expectation, and Cauchy-Schwarz inequality. Finally, after expanding again around θ_0 it is easy to see that $|III| \leq d_\theta \left\|\tilde{\theta} - \theta_0\right\| 2n^{-1} \sum_{i=1}^{n} Y_i R(X_i) \mathbb{I}_{\gamma l}(X_i) = o_p(1)$.

For the second part of the lemma, we need to show that $\hat{L}(\tilde{\theta}) = L_0 + o_p(1)$, $\hat{C}_{\boldsymbol{\gamma}}(\tilde{\theta}) = C_{\gamma,0} + o_p(1)$, and $\hat{\boldsymbol{\mu}}^*_{\boldsymbol{\gamma}}(\tilde{\theta}) = \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} + o_p(1)$. By the usual MVT argument and the law of large numbers,

$$
L_n = L_0 + I + II + II' + o_p(1)
$$

with,

$$
\|I\| = \left\| \frac{1}{n} \sum_{i=1}^{n} \nabla l_{\bar{\theta}}(Z_i) \left(\hat{\theta} - \theta_0\right) \left(\hat{\theta} - \theta_0\right)' \nabla l_{\bar{\theta}}(Z_i)' \right\| \leq d_\theta^4 \left\|\hat{\theta} - \theta_0\right\|^2 \frac{1}{n} \sum_{i=1}^{n} R_2^2(Z_i) = o_p(1),
$$

$$
\begin{aligned}
\|II\| &= \left\| \frac{1}{n} \sum_{i=1}^{n} \nabla l_{\bar{\theta}}(Z_i) \left(\hat{\theta} - \theta_0\right) l'_{\theta_0}(Z_i) \right\| \leq d_\theta^2 \left\|\hat{\theta} - \theta_0\right\| \frac{1}{n} \sum_{i=1}^{n} \|l_{\theta_0}(Z_i)\| R_2(Z_i) \\
&\leq d_\theta^2 \left\|\hat{\theta} - \theta_0\right\| \left(\frac{1}{n} \sum_{i=1}^{n} \|l_{\theta_0}(Z_i)\|^2 \right)^{1/2} \left(\frac{1}{n} \sum_{i=1}^{n} R_2^2(Z_i) \right)^{1/2} = o_p(1).
\end{aligned}
$$

Alike, we write $\hat{C}_{\boldsymbol{\gamma}}(\tilde{\theta})$ as,

$$
\hat{C}_{\boldsymbol{\gamma}}(\tilde{\theta}) = I - II - III + C_{\gamma,0} + o_p(1),
$$

where,

$$
\begin{aligned}
I &= \frac{1}{n}\sum_{i=1}^{n} l_{\theta_0}(Z_i)\nabla m_{\bar{\theta}}(X_i)'(\hat{\theta}-\theta_0)\mathbf{I}_{\boldsymbol{\gamma}}(X_i)' \\
II &= \frac{1}{n}\sum_{i=1}^{n} \nabla l_{\bar{\theta}}(Z_i)'(\hat{\theta}-\theta_0)\varepsilon_{\theta_0}(Z_i)\mathbf{I}_{\boldsymbol{\gamma}}(X_i)' \\
III &= \frac{1}{n}\sum_{i=1}^{n} \nabla l_{\bar{\theta}}(Z_i)(\hat{\theta}-\theta_0)(\hat{\theta}-\theta_0)'\nabla m_{\bar{\theta}}(X_i)\mathbf{I}_{\boldsymbol{\gamma}}(X_i)'
\end{aligned}
$$

By Assumptions 2, and 3,

$$
\begin{aligned}
\|I\| &\leq \left\|\hat{\theta}-\theta_0\right\| \frac{1}{n}\sum_{i=1}^{n} \|l_{\theta_0}(Z_i)\| \, \|\nabla m_{\bar{\theta}}(X_i)\| \, \|\mathbf{I}_{\boldsymbol{\gamma}}(X_i)\| \\
&\leq \sqrt{L}\left\|\hat{\theta}-\theta_0\right\| \left(\frac{1}{n}\sum_{i=1}^{n} \|l_{\theta_0}(Z_i)\|^2\right)^{1/2} \left(\frac{1}{n}\sum_{i=1}^{n} R^2(X_i)\right)^{1/2} = o_p(1),
\end{aligned}
$$

An analogous reasoning shows that $\|II\| = o_p(1)$, and,

$$
\begin{aligned}
\|III\| &\leq \left\|\hat{\theta}-\theta_0\right\|^2 \frac{1}{n}\sum_{i=1}^{n} \|\nabla l_{\bar{\theta}}(Z_i)\| \, \|\nabla m_{\bar{\theta}}(X_i)\| \, \|\mathbf{I}_{\boldsymbol{\gamma}}(X_i)\| \\
&\leq \sqrt{L} d_\theta^3 \left\|\hat{\theta}-\theta_0\right\|^2 \left(\frac{1}{n}\sum_{i=1}^{n} R^2(X_i)\right)^{1/2} \left(\frac{1}{n}\sum_{i=1}^{n} R_2^2(Z_i)\right)^{1/2} = o_p(1).
\end{aligned}
$$

Finally, $\hat{\boldsymbol{\mu}}^*_{\boldsymbol{\gamma}}(\tilde{\theta}) = \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} + o_p(1)$ follows from the proof of Lemma 3 below. ■

Proof of Lemma 2. By Lemma 2.6.17 in VW and Assumption 5, both $\mathbb{D}$ and $\{\mathbf{I}_{\boldsymbol{\gamma}}(x) : \gamma \in \mathbb{D}\}$ are VC classes. Therefore, $\mathcal{F} = \{\varepsilon_{\theta_0}(z)\mathbf{I}_{\boldsymbol{\gamma}}(x) : \gamma \in \mathbb{D}\}$ is a VC class (Lemma 2.6.18 in VW), with square integrable envelope function $F = |\varepsilon_{\theta_0}|$, and, hence, is P-Donsker. The convergence of the finite-dimensional distributions (fidis) of $\hat{\boldsymbol{\Phi}}_0(\cdot)$ to those of $\boldsymbol{\Phi}_0(\cdot)$, by the multivariate central limit theorem, characterize the limit process. ■

Proof of Lemma 3. By an MVT argument,

$$
\hat{\boldsymbol{\Phi}}_{\hat{\theta}}(\boldsymbol{\gamma}) = \hat{\boldsymbol{\Phi}}_0(\boldsymbol{\gamma}) - I'\sqrt{n}(\hat{\theta}-\theta_0) - \hat{\boldsymbol{\mu}}^{*\prime}_{\boldsymbol{\gamma}}(\theta_0)\sqrt{n}(\hat{\theta}-\theta_0)
$$

where,

$$I = \hat{\boldsymbol{\mu}}^*_{\boldsymbol{\gamma}}(\bar{\theta}) - \hat{\boldsymbol{\mu}}^*_{\boldsymbol{\gamma}}(\theta_0) = \frac{1}{n}\sum_{i=1}^{n}\big(\nabla m_{\bar{\theta}}(X_i) - \nabla m_{\theta_0}(X_i)\big)\mathbf{I}_{\boldsymbol{\gamma}}(X_i)'$$

and $|\bar{\theta} - \theta_0| \leq |\tilde{\theta} - \theta_0|$. The class $\{\nabla m_\theta(x) : \theta \in \Theta\}$ is a collection of continuous mapping, $\theta \to \nabla m_\theta$, over the compact metric space Θ with integrable envelope function $R(\cdot)$ and, therefore, is P-GC (e.g., Example 19.8 in Van der Vaart 2000). Thus,

$$\begin{aligned}\sup_{\boldsymbol{\gamma}\in\mathbb{D}} \|I\| &\leq \sup_{\gamma\in\mathbb{D}} \frac{1}{n}\sum_{i=1}^{n} \|\nabla m_{\bar{\theta}}(X_i) - \nabla m_{\theta_0}(X_i)\| \, \|\mathbf{I}_{\boldsymbol{\gamma}}(X_i)\| \\ &\leq \sqrt{L}\frac{1}{n}\sum_{i=1}^{n} \|\nabla m_{\bar{\theta}}(X_i) - \nabla m_{\theta_0}(X_i)\| = o_p(1).\end{aligned}$$

where the last equality follows from an application of the uniform law of large numbers (e.g., Davidson 1994, Theorem 21.6). For the second part of the Lemma is sufficient to prove that,

$$|II| = \left|\frac{1}{n}\sum_{i=1}^{n} \nabla^{(j)} m_{\theta_0}(X_i)\big(\mathbf{I}_{\tilde{\boldsymbol{\gamma}}}(X_i) - \mathbf{I}_{\boldsymbol{\gamma}}(X_i)\big)\right| = o_p(1),$$

for each $j \in \{1, .., d_\theta\}$. To see this is true, notice that by Assumption 5, $\mathbb{D}\Delta\mathbb{D} = \{\boldsymbol{\gamma}_1 \Delta \boldsymbol{\gamma}_2 : \boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2 \in \mathbb{D}\}$ is a class of subsets of unions of VC classes, and hence is VC. Therefore, $\{|\nabla^{(j)} m_{\theta_0}(x)|\mathbf{I}_{\tilde{\boldsymbol{\gamma}}}(x) : \tilde{\boldsymbol{\gamma}} \in \mathbb{D}\Delta\mathbb{D}\}$ is also VC with integrable envelope $R(\cdot)$ and, hence, P-GC. Thus, for each $j \in \{1, .., d_\theta\}$,

$$\begin{aligned}|II| &\leq \frac{1}{n}\sum_{i=1}^{n} |\nabla^{(j)} m_{\theta_0}(X_i)|\mathbf{I}_{\hat{\boldsymbol{\gamma}}\Delta\boldsymbol{\gamma}}(X_i) \\ &\leq \sup_{\tilde{\gamma}\in\mathbb{D}\Delta\mathbb{D}} (P_n - P)|\nabla^{(j)} m_{\theta_0}|\mathbf{I}_{\tilde{\gamma}} + \mathbb{E}\left[|\nabla^{(j)} m_{\theta_0}(X)|\mathbf{I}_{\hat{\boldsymbol{\gamma}}\Delta\boldsymbol{\gamma}}(X)\right] \\ &= o_p(1) + \mu_R(\hat{\boldsymbol{\gamma}}\Delta\boldsymbol{\gamma}) = o_p(1)\end{aligned}$$

where $\mu_R(\hat{\boldsymbol{\gamma}}\Delta\boldsymbol{\gamma}) = \big(\mu_R(\hat{\gamma}_1\Delta\gamma_1), ..., \mu_R(\hat{\gamma}_L\Delta\gamma_L)\big)'$, and $\mu_R(A) = \int_A R(X)dP$ is a (signed) measure absolutely continuous with respect to P. The last equality follows from Assumption 4. ■

Proof of Lemma 4. For each element on the main diagonal of $\hat{\Sigma}_{\hat{\boldsymbol{\gamma}}}(\tilde{\theta}) - \hat{\Sigma}_{\boldsymbol{\gamma}}(\tilde{\theta})$ write,

$$\frac{1}{n}\sum_{i=1}^{n} \varepsilon_{\hat{\theta}}(Z_i)^2(\mathbb{I}_{\hat{\gamma}_l}(X_i) - \mathbb{I}_{\gamma_l}(X_i)) = I + II + III$$

where

$$\begin{aligned}
I &= \frac{1}{n}\sum_{i=1}^{n} \varepsilon_{\theta_0}(Z_i)^2(\mathbb{I}_{\hat{\gamma}_l}(X_i) - \mathbb{I}_{\gamma_l}(X_i)) \\
II &= (\hat{\theta} - \theta_0)'\frac{1}{n}\sum_{i=1}^{n} \nabla m_{\bar{\theta}}(X_i)\nabla m_{\bar{\theta}}(X_i)'(\mathbb{I}_{\hat{\gamma}_l}(X_i) - \mathbb{I}_{\gamma_l}(X_i))(\hat{\theta} - \theta_0) \\
III &= -(\hat{\theta} - \theta_0)'\frac{2}{n}\sum_{i=1}^{n} \varepsilon_{\theta_0}(Z_i)\nabla m_{\bar{\theta}}(X_i)(\mathbb{I}_{\hat{\gamma}_l}(X_i) - \mathbb{I}_{\gamma_l}(X_i))
\end{aligned}$$

The class $\{\varepsilon_{\theta_0}(z)^2\mathbb{I}_{\tilde{\gamma}_l} : \tilde{\gamma} \in \mathbb{C}\Delta\mathbb{C}\}$ is VC with integrable envelope function $\varepsilon_{\theta_0}^2$ and, hence, is P-GC. Therefore, $|I| \leq \mu_\sigma(\hat{\gamma}_l\Delta\gamma_l) + o_p(1) = o_p(1)$, by Assumption 4. Also, $|II| \leq \sqrt{L}d_\theta^2 \left\|\hat{\theta} - \theta_0\right\|^2 n^{-1}\sum_{i=1}^{n} R(X_i)^2 = o_p(1)$, by Assumptions 1-2 and the consistency of $\hat{\theta}$, and $|III| \leq \sqrt{L}d_\theta \left\|\hat{\theta} - \theta_0\right\| n^{-1}\sum_{i=1}^{n} \varepsilon_{\theta_0}R(X_i) = o_p(1)$ by Cauchy-Schwarz inequality. Thus, $\hat{\Sigma}_{\hat{\boldsymbol{\gamma}}}(\tilde{\theta}) = \hat{\Sigma}_{\boldsymbol{\gamma}}(\tilde{\theta}) + o_p(1)$, and the first part of the lemma follows from Lemma 1(a).

For the second part of the lemma, notice that by Lemma 3(b) (and the proof of the first part of Lemma 3), $\hat{\boldsymbol{\mu}}^*_{\hat{\boldsymbol{\gamma}}}(\hat{\theta}) = \boldsymbol{\mu}^*_{\boldsymbol{\gamma},0} + o_p(1)$, and for each element of $C_n(\hat{\gamma}) - C_n(\gamma_0)$ it holds that,

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n} \varepsilon_{\hat{\theta}}(Z_i)l_{\hat{\theta},j}(\mathbb{I}_{\hat{\gamma}_l}(X_i) - \mathbb{I}_{\gamma_l}(X_i)) &\leq \left(\frac{1}{n}\sum_{i=1}^{n} \varepsilon_{\hat{\theta}}(Z_i)^2\mathbb{I}_{\hat{\gamma}_l\Delta\gamma_l}(X_i)\right)^{1/2}\left(\frac{1}{n}\sum_{i=1}^{n} l_{\hat{\theta},j}\right)^{1/2} \\
&= o_p(1)O_p(1),
\end{aligned}$$

where $l_{\theta,j}$ denotes the j-th component of l_θ and the last equality follows from the first part of this proof and Lemma 1. ■

Proof of Theorem 1. The consistency of the grouped GMM estimator follows from $\sup_{\theta\in\Theta} |Q_n(\theta) - Q_0(\theta)| = o_p(1)$, where $Q_0(\theta) = \mathbb{E}\left[\varepsilon_\theta(Z)\mathbf{I}_{\boldsymbol{\gamma}}(X)\right]'(\Sigma_{\boldsymbol{\gamma},0})^{-1}\mathbb{E}\left[\varepsilon_\theta(Z)\mathbf{I}_{\boldsymbol{\gamma}}(X)\right]$ and $Q_n(\theta) = n^{-1}\hat{\chi}^2_{\hat{\boldsymbol{\gamma}},\tilde{\theta}}(\theta)$ (see Theorem 2.1 in Newey and McFadden 1994, for instance).

To see that the condition holds, notice that,

$$\sup_{\theta\in\Theta}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_\theta(Z_i)\mathbf{I}_{\hat{\gamma}}(X_i)-\mathbb{E}\left[\varepsilon_\theta(Z)\mathbf{I}_{\boldsymbol{\gamma}}(X)\right]\right|\leq\sup_{\theta\in\Theta}|I|+\sup_{\theta\in\Theta}|II|+\sup_{\theta\in\Theta}|III|$$

where

$$\begin{aligned}
I &= (P_n-P)\varepsilon_\theta\mathbf{I}_{\boldsymbol{\gamma}},\\
II &= \frac{1}{n}\sum_{i=1}^{n}\varepsilon_{\theta_0}(Z_i)(\mathbf{I}_{\hat{\gamma}}(X_i)-\mathbf{I}_{\boldsymbol{\gamma}}(X_i)),\\
III &= \frac{1}{n}\sum_{i=1}^{n}\left[\nabla m_{\bar{\theta}}(Z_i)(\mathbf{I}_{\hat{\gamma}}(X_i)-\mathbf{I}_{\boldsymbol{\gamma}}(X_i))'\right]'(\theta-\theta_0).
\end{aligned}$$

The mapping $\theta\to\varepsilon_\theta$ is continuous over the compact Θ, with

$$\mathbb{E}\left[\sup_{\theta\in\Theta}\varepsilon_\theta(Z)\right]\leq\mathbb{E}\left[\sup_{\bar{\theta},\theta\in\Theta}\varepsilon_{\theta_0}(Z)-\nabla m_{\bar{\theta}}(X)'(\theta-\theta_0)\right]\leq d_\theta\mathbb{E}\left[R(Z)\right]D<\infty,$$

by Assumption 2 and the compactness of Θ, where D denotes the diameter of Θ. Therefore, both $\{\varepsilon_\theta(z):\theta\in\Theta\}$ and $\{\varepsilon_\theta(z)\mathbf{I}_{\boldsymbol{\gamma}}(X):\theta\in\Theta\}$ are P-GC classes (see Corollary 8.6 in Giné and Zinn 1984, for instance) and, hence, $\sup_{\theta\in\Theta}|I|=o_p(1)$. Then, by Cauchy-Schwarz inequality and Assumptions 1 and 4,

$$\sup_{\theta\in\Theta}|II|\leq\left(\frac{1}{n}\sum_{i=1}^{n}\varepsilon_{\theta_0}^2(Z_i)\right)^{1/2}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{I}_{\hat{\gamma}\Delta\boldsymbol{\gamma}}(X_i)\right)^{1/2}=o_p(1).$$

Finally,

$$\begin{aligned}
\sup_{\theta\in\Theta}\|III\| &\leq \sup_{\theta\in\Theta}\frac{1}{n}\sum_{i=1}^{n}\|\mathbf{I}_{\hat{\gamma}\Delta\boldsymbol{\gamma}}(X_i)\|\,\|\nabla m_{\bar{\theta}}(X_i)\|\,\|\theta-\theta_0\|\\
&\leq d_\theta D\frac{1}{n}\sum_{i=1}^{n}\|\mathbf{I}_{\hat{\gamma}\Delta\boldsymbol{\gamma}}(X_i)\|\,R(X_i)=o_p(1),
\end{aligned}$$

where the second inequality follows from Assumption 2 and the compactness of Θ. This result, together with the consistency of $\hat{\Sigma}_{\hat{\gamma}}(\tilde{\theta})$ implies that $\sup_{\theta\in\Theta}|Q_n(\theta)-Q_0(\theta)|=o_p(1)$ and, therefore, $\hat{\theta}_{\hat{\gamma}}=\theta_0+o_p(1)$. For the asymptotic normality: by Assumptions 2 and

1(c), the first-order conditions of the minimization problem are satisfied with probability approaching one, $\hat{\boldsymbol{\mu}}^*_{\hat{\gamma}}(\hat{\theta}_{\boldsymbol{\gamma}})\hat{\Sigma}_{\hat{\gamma}}(\tilde{\theta})^{-1}n^{-1/2}\hat{\boldsymbol{\Phi}}_{\hat{\gamma}}(\hat{\theta}_{\boldsymbol{\gamma}}) = 0$. Expanding $\hat{\boldsymbol{\Phi}}_{\hat{\gamma}}(\hat{\theta}_{\boldsymbol{\gamma}})$ around θ_0 and solving gives the Bahadur representation,

$$\sqrt{n}(\hat{\theta}_{\boldsymbol{\gamma}} - \theta_0) = -\left[\hat{\boldsymbol{\mu}}^*_{\hat{\gamma}}(\hat{\theta}_{\boldsymbol{\gamma}})\hat{\Sigma}_{\hat{\gamma}}(\tilde{\theta})^{-1}\hat{\boldsymbol{\mu}}^{*\prime}_{\hat{\gamma}}(\bar{\theta})\right]^{-1}\hat{\boldsymbol{\mu}}^*_{\hat{\gamma}}(\hat{\theta}_{\boldsymbol{\gamma}})\hat{\Sigma}_{\hat{\gamma}}(\tilde{\theta})^{-1}\hat{\boldsymbol{\Phi}}_0(\hat{\boldsymbol{\gamma}})$$

where $|\bar{\theta} - \theta_0| \leq |\hat{\theta}_{\hat{\gamma}} - \theta_0|$.

From the proof of Lemma 3, it follows that $\hat{\boldsymbol{\mu}}^*_{\hat{\gamma}}(\theta) - \hat{\boldsymbol{\mu}}^*_{\hat{\gamma}}(\theta_0) = o_p(1)$ for any $\theta \underset{p}{\rightarrow} \theta_0$. Thus by consistency of $\hat{\theta}_{\hat{\gamma}}$ and Lemma 1(b), it follows that both $\hat{\boldsymbol{\mu}}^*_{\hat{\gamma}}(\hat{\theta}_{\boldsymbol{\gamma}})$ and $\hat{\boldsymbol{\mu}}^*_{\hat{\gamma}}(\bar{\theta})$ converge in probability to $\boldsymbol{\mu}^*_{\boldsymbol{\gamma},0}$. Finally, by Assumptions 4-5, and the uniform continuity of the sample paths of $\Phi_0(\cdot)$,

$$\Phi_0(\hat{\boldsymbol{\gamma}}) \xrightarrow{d} N\left(0, \Sigma_{\boldsymbol{\gamma},0}\right).$$

The result follows by Slutsky's lemma. ■

Proof of Theorem 2. By Lemma 3, 4, and Assumption 3 (or Theorem 1 and Assumption 2' for the $\hat{\chi}^2$ tests) both test statistics are asymptotically equivalent to the following quadratic form,

$$q(\hat{\theta}, W, \hat{\gamma}) = \left(\Phi_0(\hat{\boldsymbol{\gamma}}) - \sqrt{n}\boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma},0}\bar{l}_{\theta_0}\right)' W^{-1}\left(\Phi_0(\hat{\boldsymbol{\gamma}}) - \sqrt{n}\boldsymbol{\mu}^*_{\boldsymbol{\gamma},0}\bar{l}_{\theta_0}\right), \tag{A1}$$

where $\bar{l}_{\theta_0} = n^{-1}\sum_{i=1}^n l_{\theta_0}(Z_i)$ and W^{-1} is the probability limit of the weighting matrix W_n^{-1}. In particular, the couple $(\hat{\theta}, W_n^{-1})$ is equal to $(\hat{\theta}_{\hat{\gamma}}, \hat{\Sigma}_{\hat{\gamma}}(\tilde{\theta}))$ in the $\hat{\chi}^2$ test and $\left(\tilde{\theta}, \widehat{\text{Avar}^-}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right)\right)$ in the Wald test. The functional

$$\phi(z, w, \gamma) = \left(z(\gamma) - \boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma},0}w\right)' W^{-1}\left(z(\gamma) - \boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma},0}w\right),$$

mapping $(\hat{\boldsymbol{\Phi}}_0(\cdot), \sqrt{n}\bar{l}_{\theta_0}, \hat{\boldsymbol{\gamma}})$ into $q(\hat{\theta}, W, \hat{\boldsymbol{\gamma}})$ is continuous with respect to the product topology on $l^\infty(\mathbb{D}) \times \mathbb{R}^L \times \mathbb{D}$ (see Lemma 4 in Andrews 1988a). Thus, Theorem 2 follows by establishing the limit null distribution of $(\Phi_0(\hat{\boldsymbol{\gamma}}) - \boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma},0}\sqrt{n}\bar{l}_{\theta_0})$ and an application of the continuous mapping theorem (e.g., Theorem 1.3.6 in VW). Lemma 2, Assumptions 1, 4, 5, and the central limit theorem imply that $(\Phi_0(\cdot), \sqrt{n}\bar{l}_{\theta_0}, \hat{\gamma})$ is a uniformly tight process

on $\mathbb{D}$ with fidis converging weakly to those of $(\Phi_0(\cdot), l_0, \gamma_0)$, where $l_0 \stackrel{d}{=} N(0, L_0)$ and $\mathbb{E}\left[l_0 \Phi_0(\boldsymbol{\gamma})\right] = \mathbb{E}\left[l_{\theta_0}(Z)\varepsilon_{\theta_0}(Z)\mathbf{I}_{\boldsymbol{\gamma}}(X)'\right]$. Thus,

$$(\hat{\boldsymbol{\Phi}}_0(\cdot), \sqrt{n}\bar{l}, \hat{\boldsymbol{\gamma}}) \rightsquigarrow (\Phi_0(\cdot), l_0, \gamma_0) \text{ on } l^\infty(\mathbb{D}),$$

and by the continuous mapping theorem,

$$q(\hat{\theta}, W, \hat{\gamma}) \xrightarrow{d} \tilde{Y}' W^{-1} \tilde{Y},$$

where $\tilde{Y} \stackrel{d}{=} N(0, \Sigma_{\tilde{Y}})$ and $\Sigma_{\tilde{Y}} = \text{Avar}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\hat{\theta})\right)$. In the Wald test, where W^{-1} is a generalized inverse of $\Sigma_{\tilde{Y}}$, $\tilde{Y}' W^{-1} \tilde{Y} \stackrel{d}{=} \chi^2_{r(\text{Avar}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\tilde{\theta})\right))}$ by Theorem 7.3(i) in Rao and Mitra (1972). In the $\hat{\chi}^2$ test, $W = \Sigma_{\boldsymbol{\gamma},0}$, and $\Sigma_{\tilde{Y}} = \text{Avar}\left(\hat{\boldsymbol{\Phi}}_{\boldsymbol{\gamma}}(\hat{\theta}_{\hat{\gamma}})\right) = \Sigma_{\boldsymbol{\gamma},0} - \boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma}}(\boldsymbol{\mu}^*_{\boldsymbol{\gamma}} \Sigma^{-1}_{\boldsymbol{\gamma},0} \boldsymbol{\mu}^{*\prime}_{\boldsymbol{\gamma}}) \boldsymbol{\mu}^*_{\boldsymbol{\gamma}}$. Since $\text{Var}\left(\Sigma^{-1/2}_{\boldsymbol{\gamma},0} \tilde{Y}\right) = \Sigma^{-1/2}_{\boldsymbol{\gamma},0} (\Sigma_{\tilde{Y}}) \Sigma^{-1/2}_{\boldsymbol{\gamma},0}$ is idempotent with rank equal to $L - d_\theta$, it follows that $\tilde{Y}' W^{-1} \tilde{Y} = (\Sigma^{-1/2}_{\boldsymbol{\gamma},0} \tilde{Y})' (\Sigma^{-1/2}_{\boldsymbol{\gamma},0} \tilde{Y}) \stackrel{d}{=} \chi^2_{L-d_\theta}$, proving the theorem. ■

Proof of Proposition 1. By an MVT argument, for fixed x, one can rewrite the difference between the two models fit as,

$$m_{\theta_0}(x) + \Delta m_{\bar{\theta}}(x)(\tilde{\theta} - \theta_0) = m_{1,\theta_1^*}(x) + \Delta m_{1,\bar{\theta}^*}(x)(\tilde{\theta}^* - \theta_1^*).$$

where $|\bar{\theta} - \theta_0| \leq |\tilde{\theta} - \theta_0|$ and $|\bar{\theta}^* - \theta_1^*| \leq |\tilde{\theta}^* - \theta_1^*|$. Then using conditions (a) and (b) of Proposition 2, we obtain, that the splitting points, x_0, solve,

$$\Delta m_{\bar{\theta}}(x_0) C n^{-1/2} \sum_{i=1}^n g(X_i, \varepsilon_{\theta_0}) - \Delta m_{1,\bar{\theta}^*}(x_0) D n^{-1/2} \sum_{i=1}^n g(X_i, \varepsilon_{\theta_0}) + o_p(1) = 0,$$

and, thus,

$$\Delta m_{\bar{\theta}}(x_0) C - \Delta m_{1,\bar{\theta}^*}(x_0) D = o_p(1).$$

■

Proof of Proposition 2. Let

$$d_{\boldsymbol{\gamma}}(I_L) = \delta(\boldsymbol{\gamma})' \delta(\boldsymbol{\gamma}) = \sum_{l=1}^L \delta_l^2 = \sum_{l=1}^L \mathbb{E}\left[h(X) \mathbb{I}_{\gamma_l}(X)\right]^2.$$

If $h(x) > 0$ ($h(x) \leq 0$) for all $x \in \mathcal{X}$, then $\delta_l \geq 0$ ($\delta_l \leq 0$) for all $l = 1, 2, ..., L$, and, since $(\delta_l + \delta_f)^2 \geq (\delta_l^2 + \delta_f^2)$ for each δ_l, δ_f such that $sign(\delta_l) = sign(\delta_f)$, it follows that,

$$\sum_{l=1}^{L} \delta_l^2 \leq \left(\sum_{l=1}^{L} \delta_l \right)^2 .$$

Thus, a unique cells maximize the drifts norm.

Consider, instead, the case $h(x) = 0$ for some $x \in \mathcal{X}$, then for any finite split of $\mathcal{X}$ with $L > 1$, we have that

$$\delta_l^2 + \delta_f^2 = (\delta_l^+ + \delta_l^-)^2 + (\delta_f^+ + \delta_f^-)^2 \leq (\delta_l^+ + \delta_f^+)^2 + (\delta_l^- + \delta_f^-)^2 = \delta_l^{*2} + \delta_f^{*2},$$

where δ_l^+ and δ_l^- denote the positive and negative part of δ_l, respectively. Therefore, any partition is dominated by a partition with the same number of cells obtained by merging the positive and negative part of each δ_l into a new cell. Since the δ_l^* are either non-negative or non-positive, by merging them in two cells containing only the positive and negative δ_l^* we obtain a partition with two cells that dominates any partition with $L > 1$ cells. Of course, since

$$(\delta_l^+ + \delta_l^-)^2 \leq (\delta_l^+)^2 + (\delta_l^-)^2,$$

the two-cells partition dominates the one-cell partition as well. Thus, the two-cells partition into positive and negative values maximizes the drifts norm. ■

Proof Proposition 3. Assume that for an integer K, there are functions $(g_1(\cdot), ..., g_K(\cdot))$ such that,

$$\mathbb{E}\left[TE_2 | D_2\right] = \sum_{k=1}^{K} \delta_k g_k(D_2) \;\; a.s. \;\; \text{for some} \;\; (\delta_1, ..., \delta_K) \in \mathbb{R}^K.$$

with $(1, D_2 g_1(D_2), ..., D_2 g_K(D_2))$ being a linearly independent vector. Since $\Delta Y = \Delta Y(0) + D_2 TE_2$, it follows that,

$$\mathbb{E}\left[\Delta Y | D_2\right] = \mathbb{E}\left[\Delta Y(0) | D_2\right] + D_2 \mathbb{E}\left[TE_2 | D_2\right] = \mu_0 + D_2 \sum_{k=1}^{K} \delta_k g_k(D_2),$$

and $(\mu_0, \delta_1, ..., \delta_K) = (\beta_0, \beta_1, ..., \beta_K)$ by the properties of linear projections (see, e.g.,

Wooldridge 2010 Property LP.1). For the reverse implication, let $\mathbb{E}[\Delta Y|D_2] = \beta_0 + D_2 \sum_{k=1}^{K} \beta_k g_k(D_2)$. Since $0 \in \mathcal{D}_2$, $\mathcal{D}_2$ being the support of D_2, it holds that,

(a) $\mathbb{E}[\Delta Y|D_2 = 0] = \mathbb{E}[\Delta Y(0)|D_2 = 0]$ if and only if $\mu_0 = \beta_0$,

(b) $\mathbb{E}[\Delta Y|D_2 > 0] = \mathbb{E}[\Delta Y(0)|D_2 > 0] + D_2 \mathbb{E}[TE_2|D_2 > 0]$ if and only if $\mathbb{E}[TE_2|D_2 > 0] = \sum_{k=1}^{K} \beta_k g_k(D_2)$.

Finally, by assumption, $\mathbb{E}[TE_2|D_2 = 0] = 0 = \sum_{k=1}^{K} \beta_k g_k(0)$. ■

Proof Proposition 4. Notice that under $H_{0,U}$,

$$\lim_{n\to\infty} \sup_{P\in\mathcal{P}_0} P(\hat{T} = 1) \leq \lim_{n\to\infty} \sup_{P\in\mathcal{P}_0} P(\hat{t}_1 > z_{\alpha/2}) = \alpha.$$

Furthermore, by Bonferroni inequality,

$$\lim_{n\to\infty} \sup_{P\in\mathcal{P}_0} P(\hat{T} = 1) \geq \lim_{n\to\infty} \sup_{P\in\mathcal{P}_0} \big(P(\hat{t}_1 > z_{\alpha/2}) + P(\hat{t}_2 > z_{\alpha/2})\big) - 1 = \alpha + 1 - 1 = \alpha.$$

Therefore, the IUT test has asymptotic size equal to α. Notice also, that when both $H_{0,1}$ and $H_{0,2}$ hold true, the test has a rejection probability weakly below the nominal level α.

■

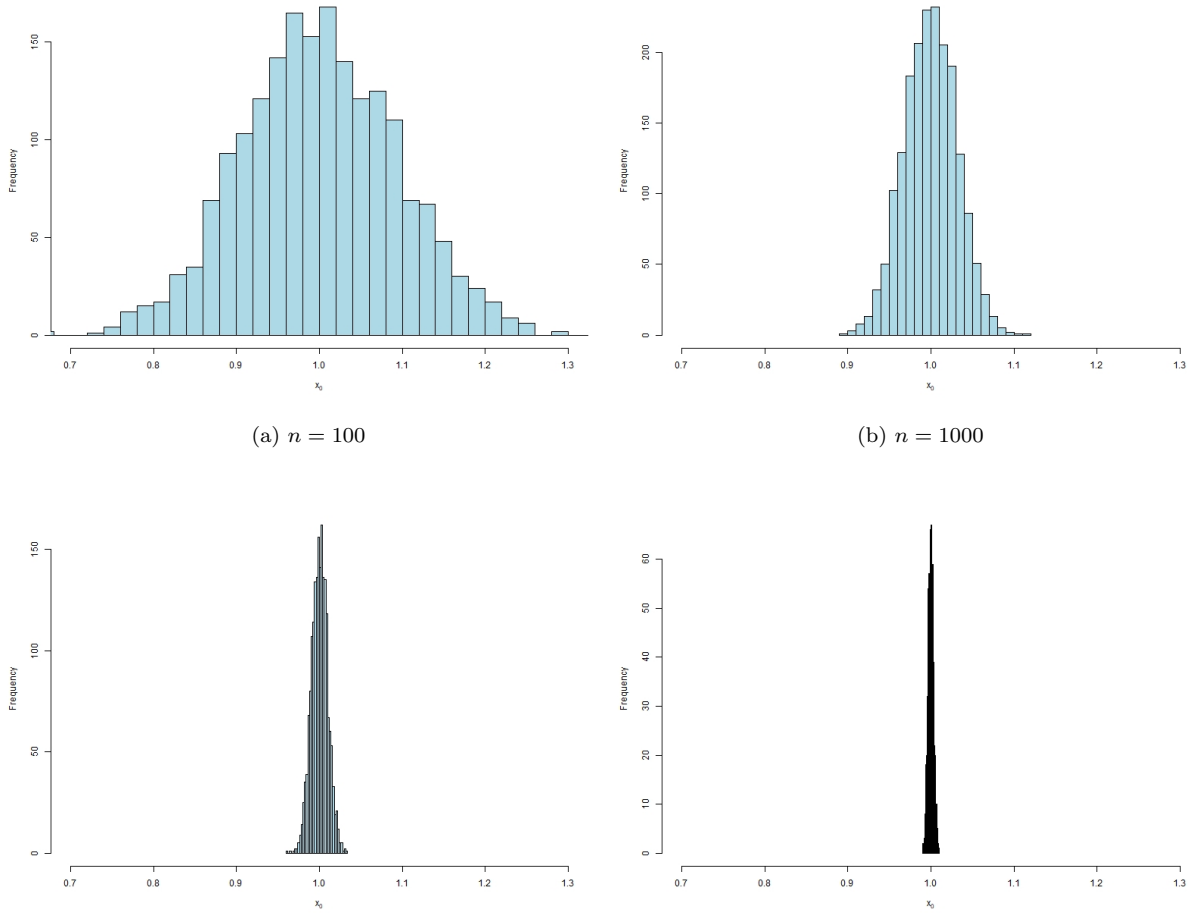
B Appendix B

B.1 Convergence of NP Cells: Simulations

I present Monte Carlo evidence demonstrating NP partitions converging to fixed cells in $\mathbb{D}$. In Figure 1, we generate data as $Y = 1 + \varepsilon$, where $\varepsilon \sim N(0, 1)$. Under the null hypothesis, $H_0 : m(X) = c$ a.s., with c as a constant. The alternative model, $H_1 : m(X) = c + \beta X$ a.s., includes a constant c and slope β, where X is generated as $X \sim N(1, 1)$. In Figure 2, we create a linear model, $Y = 1 + X + \varepsilon$, with $\varepsilon \sim N(0, 1)$. The null hypothesis is $H_0 : m(X) = c + \beta X$ a.s., with c and β as constants. The alternative model, $H_1 : m(X) = c + \beta X + \gamma X^2$ a.s., introduces a curvature term γ. I consider sample sizes of $n = 100$, $n = 1000$, $n = 10000$, and $n = 10000$, to compute the splitting points

using the NP partition algorithm over $R = 1000$ replications. The results, depicted in the figures below, align with theoretical findings, illustrating the convergence of splitting points to fixed points in $\mathcal{X}$.

Figure 6: Convergence of x_0 for the constant model under linear alternative.



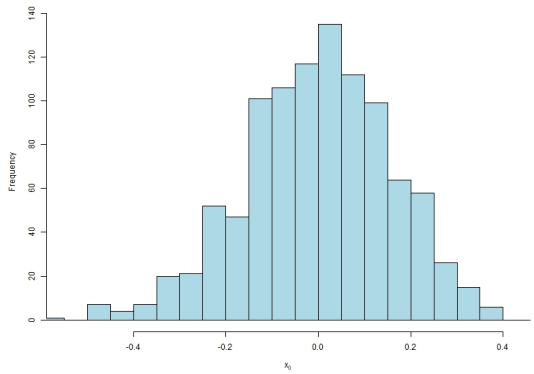
(a) $n = 100$

(b) $n = 1000$

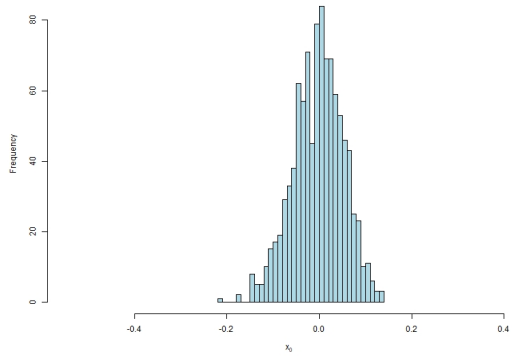
(c) $n = 10,000$

(d) $n = 100,000$

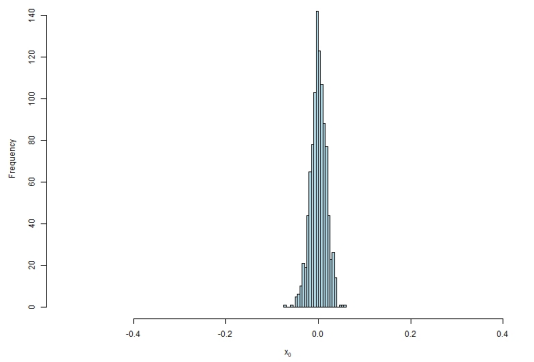
Figure 7: Convergence of x_0 for the linear model under quadratic alternative. The first four graphs and the last four refer to the two roots of the quadratic equation.



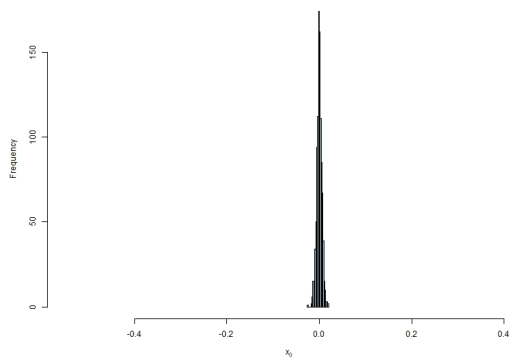
(a) $n = 100$



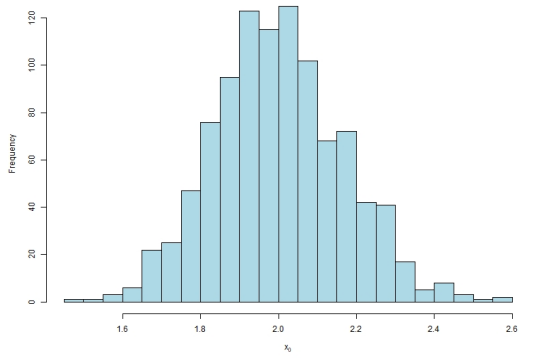
(b) $n = 1000$



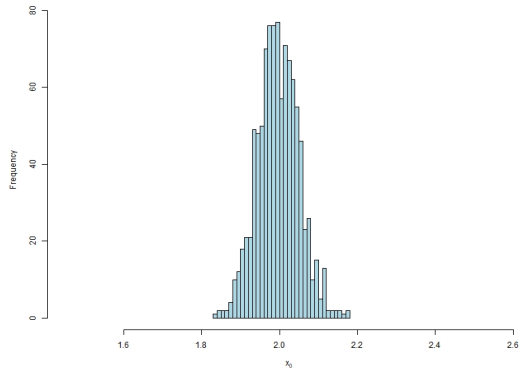
(c) $n = 10,000$



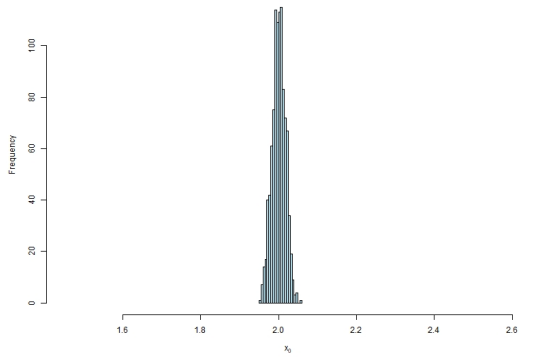
(d) $n = 100,000$



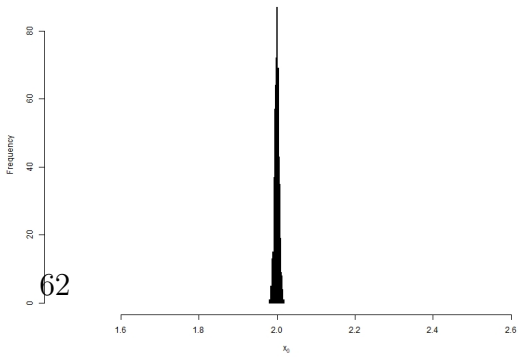
(e) $n = 100$



(f) $n = 1000$



(g) $n = 10,000$



(h) $n = 100,000$

C Appendix C

In this section of the appendix, I report the simulations results under the same DGP as in Section 7 of the main text for the χ^2 tests, where the additional splits on the preliminary two-cell partition is done using the first principal component of X (**PCS**).

Table A1: First Block: χ^2 tests with **PCS** Partition

				PCS					
				w_{poly}		w_{step}		**PNP**	
n	DGP	L	L^*	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$
50	1	4	7	4.9	3.5	5.0	3.5	4.3	3.5
		6	9	3.8	2.6	3.8	2.6	3.8	2.6
		8	11	2.7	1.4	2.7	1.4	2.7	1.4
50	1-E	4	7	3.7	2.7	3.8	2.7	3.6	2.7
		6	9	2.5	2.0	2.5	2.0	2.5	2.0
		8	11	1.7	0.9	1.7	0.9	1.7	0.9
50	2	4	7	5.7	4.7	5.7	4.7	22.4	4.7
		6	9	4.2	2.9	4.2	2.9	4.2	2.9
		8	11	3.0	2.2	3.0	2.2	3.0	2.2
50	3	4	7	31.9	30.3	30.3	30.3	33.3	30.3
		6	9	28.3	20.6	28.3	20.6	28.3	20.6
		8	11	18.2	14.6	18.2	14.6	18.2	14.6
50	4	4	7	48.0	45.4	47.8	45.4	91.5	45.4
		6	9	48.5	40.5	48.5	40.5	48.5	40.5
		8	11	46.8	32.8	46.8	32.8	46.8	32.8
100	1	4	7	5.0	5.0	5.2	5.0	5.2	4.9
		6	9	5.1	4.3	5.3	4.3	4.7	4.3
		8	11	4.0	3.1	4.0	3.1	3.6	3.1
100	1-E	4	7	4.6	5.0	4.9	5.0	4.7	4.3
		6	9	4.5	3.8	4.1	3.9	4.5	3.8
		8	11	3.8	3.6	3.8	3.6	3.1	3.6
100	2	4	7	5.1	9.8	5.0	15.9	48.0	39.2
		6	9	8.3	6.4	6.1	7.0	36.6	24.4
		8	11	5.2	5.6	5.3	5.6	29.3	5.6
100	3	4	7	54.8	75.7	45.4	70.9	78.9	74.0
		6	9	66.2	61.9	49.9	61.9	74.5	61.9
		8	11	62.4	55.4	61.3	55.4	62.3	55.4
100	4	4	7	44.0	74.3	37.9	51.0	99.9	98.1
		6	9	89.7	92.1	70.0	87.3	99.6	97.8
		8	11	95.9	93.6	95.8	93.6	99.8	93.6
300	1	4	7	4.1	4.5	4.5	5.5	5.2	4.5
		6	9	5.2	4.3	5.3	4.9	5.0	4.7
		8	11	4.5	4.6	4.3	5.2	4.4	3.9
300	1-E	4	7	5.6	5.2	5.1	6.3	4.6	5.4
		6	9	6.0	5.7	5.1	6.0	5.9	5.3
		8	11	5.3	5.6	5.2	7.3	5.5	5.8
300	2	4	7	5.0	29.6	5.2	56.4	95.1	87.6
		6	9	27.8	38.0	21.4	58.3	91.5	86.4
		8	11	30.6	36.4	26.7	58.0	87.1	82.3
300	3	4	7	98.3	99.9	92.5	99.8	99.9	99.9
		6	9	99.7	99.8	96.6	99.8	99.9	99.9
		8	11	99.8	99.9	96.6	99.7	99.8	99.8
300	4	4	7	92.1	98.9	77.1	79.1	100	99.9
		6	9	99.8	99.3	91.3	93.9	100	99.9
		8	11	99.9	99.9	99.6	97.9	100	100

Percentage of rejections at the nominal level $\alpha = 0.05$ of the $\hat{\chi}^2$ and the Wald test under different partitioning methods. n, DGP, L, and L^* denote the sample size, the DGP, and the number of cells used for Wald and J tests, respectively.

Table A2: Second Block: Size ($b = 0$), χ^2 tests with **PCS** Partition

					PCS					
					w_{poly}		w_{step}		PNP	
n	d_x	a	L	L^*	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$
100	5	0	4	9	5.5	5.4	5.3	5.0	5.3	5.2
			6	11	4.9	4.4	5.6	4.4	4.2	4.4
			8	13	4.3	3.5	4.1	3.5	4.0	3.5
100	5	3	4	9	5.8	4.7	5.3	5.0	4.2	4.3
			6	11	5.4	3.8	5.4	3.8	3.8	3.8
			8	13	3.8	3.3	3.7	3.3	3.1	3.3
100	10	0	8	18	5.0	4.7	5.7	4.7	5.2	4.7
			10	20	4.7	5.0	4.7	5.0	4.7	5.0
			12	22	4.4	3.7	4.4	3.7	4.4	3.7
100	10	3	8	18	4.8	4.0	4.7	4.0	4.2	4.0
			10	20	3.7	4.7	3.7	4.7	3.7	4.7
			12	22	3.7	2.2	3.7	2.2	3.7	2.2
200	5	0	4	9	5.2	6.1	4.8	5.6	3.7	4.7
			6	11	4.8	6.2	5.2	4.6	4.6	5.3
			8	13	4.6	4.6	5.2	5.2	4.6	4.9
200	5	3	4	9	4.7	6.1	5.8	5.9	4.5	4.8
			6	11	5.2	5.0	5.3	4.7	4.9	3.9
			8	13	4.5	5.0	4.3	5.2	3.8	3.8
200	10	0	8	18	5.0	5.2	5.4	5.2	4.7	5.3
			10	20	4.4	4.8	4.8	4.8	5.3	4.8
			12	22	5.2	4.5	4.8	4.5	4.3	4.5
200	10	3	8	18	4.8	5.2	5.1	5.2	4.5	5.2
			10	20	4.2	5.4	4.4	5.4	4.8	5.4
			12	22	4.5	4.4	4.2	4.4	4.3	4.4
500	5	0	4	9	5.6	6.3	5.7	6.2	4.9	5.3
			6	11	5.2	5.7	5.6	6.0	4.5	4.5
			8	13	5.3	5.7	5.6	6.1	4.5	4.5
500	5	3	4	9	5.2	6.4	5.2	6.0	5.2	5.3
			6	11	5.4	5.8	5.4	6.0	4.7	4.8
			8	13	4.9	5.7	4.1	6.5	4.4	4.6
500	10	0	8	18	5.5	5.7	5.8	5.6	4.6	4.9
			10	20	5.4	5.8	5.9	5.8	4.6	4.4
			12	22	4.9	5.4	5.5	4.9	4.9	4.3
500	10	3	8	18	5.3	5.2	5.3	5.0	5.0	5.0
			10	20	5.0	5.9	5.6	5.6	4.3	4.3
			12	22	5.0	5.0	4.5	4.5	4.9	4.1
1000	5	0	4	9	6.2	6.9	5.9	6.4	5.0	5.3
			6	11	5.4	6.0	5.3	7.0	4.6	4.3
			8	13	5.1	5.4	6.1	5.8	5.4	4.9
1000	5	3	4	9	5.6	7.2	5.7	6.5	5.3	5.0
			6	11	5.2	6.1	4.6	5.6	4.8	4.5
			8	13	5.5	6.3	5.6	5.8	4.9	4.8
1000	10	0	8	18	5.9	4.9	5.4	5.6	6.0	4.3
			10	20	5.4	5.4	5.7	5.4	5.7	4.2
			12	22	5.7	5.2	4.7	5.9	5.2	4.7
1000	10	3	8	18	5.9	6.1	4.8	6.3	5.2	5.6
			10	20	5.1	5.8	5.3	7.0	4.9	4.8
			12	22	4.9	4.9	6.1	4.8	5.0	5.2

Percentage of rejections at the nominal level $\alpha = 0.05$ of the $\hat{\chi}^2$ and the Wald test under different partitioning methods. n, d_x, a, L, and L^* denote the sample size, the number of covariates, the degree of heteroskedasticity, and the number of cells of the Wald and J test, respectively. The last four columns report the rejection rates of the Kolmogorov-Smirnov tests of Stute (1997) (KS1) and Stute and Zhu (2002) (KS2), the Cramer-Von-Mises test of Escanciano (2006) (CvM1), and the T-test on $b = 0$.

Table A3: Second Block: Power ($b = 0.5$), χ^2 tests with **PCS** Partition

					PCS					
					w_{poly}		w_{step}		**PNP**	
n	d_x	a	L	L^*	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$	$\hat{\mathcal{W}}$	$\hat{\chi}^2$
100	5	10	4	9	38.9	5.7	44.5	10.8	42.4	13.8
			6	11	26.8	5.7	37.5	5.7	36.6	5.7
			8	13	15.6	4.7	25.6	4.7	28.2	4.7
100	5	50	4	9	6.1	5.3	6.7	5.2	73.7	49.1
			6	11	6.4	4.4	5.6	4.4	67.2	4.4
			8	13	3.9	3.6	3.7	3.6	58.5	3.6
100	10	10	8	18	6.3	4.6	7.0	4.6	23.4	4.6
			10	20	5.4	4.4	5.4	4.4	5.4	4.4
			12	22	4.3	3.2	4.3	3.2	4.3	3.2
100	10	50	8	18	4.9	4.0	5.2	4.0	54.8	4.0
			10	20	4.1	4.0	4.1	4.0	4.1	4.0
			12	22	3.5	3.4	3.5	3.4	3.5	3.4
200	5	10	4	9	72.1	37.6	76.9	41.0	79.6	56.7
			6	11	69.1	38.0	72.9	40.1	77.2	54.7
			8	13	61.8	16.6	66.8	35.9	70.7	52.2
200	5	50	4	9	8.5	9.0	8.7	8.2	97.1	86.9
			6	11	9.7	8.1	8.1	7.0	96.2	87.8
			8	13	7.4	7.4	9.1	5.4	94.4	85.3
200	10	10	8	18	58.0	5.3	66.5	5.3	82.5	5.9
			10	20	53.6	5.6	58.5	5.6	79.0	5.6
			12	22	43.8	5.2	36.1	5.2	75.3	5.2
200	10	50	8	18	5.8	5.2	5.3	5.0	93.0	72.4
			10	20	5.2	4.9	5.0	4.9	90.6	4.9
			12	22	5.1	3.8	4.6	3.8	86.8	3.8
500	5	10	4	9	99.1	74.1	99.5	77.4	99.7	93.4
			6	11	99.0	77.9	99.5	80.6	99.6	94.3
			8	13	98.1	78.0	99.0	81.0	99.5	95.2
500	5	50	4	9	22.7	23.3	21.8	21.0	100	98.9
			6	11	30.7	24.9	22.8	19.7	100	99.6
			8	13	23.2	27.4	28.7	20.0	100	99.8
500	10	10	8	18	99.1	83.4	99.6	84.4	100	97.2
			10	20	98.3	84.4	99.4	85.1	99.9	97.5
			12	22	97.9	85.1	99.1	86.0	99.8	97.9
500	10	50	8	18	8.1	7.4	7.8	7.1	100	98.5
			10	20	7.1	7.6	7.6	6.6	100	98.8
			12	22	6.7	6.1	8.1	6.3	100	99.3
1000	5	10	4	9	100	92.4	100	94.1	100	99.3
			6	11	100	95.6	100	96.7	100	99.9
			8	13	100	96.7	100	98.0	100	99.9
1000	5	50	4	9	45.9	44.8	45.3	41.5	100	99.7
			6	11	62.3	49.8	50.1	42.2	100	99.9
			8	13	53.9	47.4	58.9	42.1	100	100
1000	10	10	8	18	100	98.5	100	98.2	100	99.9
			10	20	100	98.7	100	98.6	100	99.9
			12	22	100	99.3	100	99.2	100	100
1000	10	50	8	18	20.4	13.9	15.0	12.9	100	99.9
			10	20	18.3	14.3	19.3	14.4	100	100
			12	22	18.5	13.1	19.1	14.1	100	100

Percentage of rejections at the nominal level $\alpha = 0.05$ of the $\hat{\chi}^2$ and the Wald test under different partitioning methods. n, d_x, a, L, and L^* denote the sample size, the number of covariates, the degree of heteroskedasticity, and the number of cells of the Wald and J test, respectively. The last four columns report the rejection rates of the Kolmogorov-Smirnov tests of Stute (1997) (KS1) and Stute and Zhu (2002) (KS2), the Cramer-Von-Mises test of Escanciano (2006) (CvM1), and the T-test on $b = 0$.

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